भारत सरकार
Government of India
जल शक्ति मंत्रालय,
Ministry of Jal Shakti,
जल संसाधन, नदी विकास और गंगा संरक्षण विभाग,
Department of Water Resources,
River Development and
Ganga Rejuvenation

केंद्रीय भूमि जल बोर्ड
Central Ground Water Board

वार्षिक रिपोर्ट

ANNUAL REPORT

2023-24

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Central Ground Water Board

FARIDABAD
FEBRUARY 2025

डॉ. सुनील कुमार अम्बष्ट
अध्यक्ष
Dr. Sunil Kumar Ambast
Chairman

भारत सरकार
जल शक्ति मंत्रालय
जल संसाधन,
नदी विकास और गंगा संरक्षण विभाग
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River Development & Ganga Rejuvention
Central Ground Water Board

MESSAGE

Groundwater is a vital resource for agriculture, drinking, and industry, playing a central role in the nation's economic and social development. With a vision of sustainable development and management of India's groundwater resources, the Central Ground Water Board (CGWB) undertakes a range of activities related to the exploration, assessment, monitoring, and responsible management of these critical resources.

I am pleased to present the CGWB Annual Report for 2023-24. This report offers a comprehensive overview of the Board's initiatives, achievements, and key milestones over the past year.

More than just a summary of activities, the Annual Report 2023-24 provides valuable insights into the challenges, progress, and future directions of groundwater management in India. I trust that this report will serve as a crucial resource for all stakeholders, underscoring the importance of sustainable groundwater management for the country's long-term prosperity.

I would like to extend my appreciation to the entire CGWB team for their tireless efforts in achieving the goals set for 2023-24.

(Dr. Sunil Kumar Ambast)
Chairman, CGWB

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डॉ. ए. असोकन
सदस्य
Dr. A. Asokan
Member

भारत सरकार
जल शक्ति मंत्रालय
जल संसाधन, नदी विकास
और गंगा संरक्षण विभाग
केंद्रीय भूमि जल बोर्ड

Government of India
Ministry of Jal Shakti
Department of Water Resources,
River Development and Ganga Rejuvenation
Central Ground Water Board

MESSAGE

Central Ground Water Board (CGWB) remains steadfast in its commitment to the sustainable development and management of India's groundwater resources. As water demands continue to grow, we recognize the urgent need for scientific approaches to ensure the judicious use and long-term sustainability of this vital resource. Our initiatives encompass a wide range of activities, including groundwater exploration, monitoring water levels and quality, identifying regions that require focused attention, and preparing plans for sustainable development of ground water. In addition, CGWB is dedicated to raising awareness about the looming water crisis and fostering engagement with stakeholders across all levels.

The Annual Report of CGWB highlights our efforts and key achievements during the year 2023-24, offering valuable insights into the state of groundwater management in India.

I extend my sincere appreciation to all CGWB officers for their outstanding contributions and accomplishments. I also commend our team for their diligent efforts in compiling and presenting this comprehensive report.

Dr. A. Asokan
Member

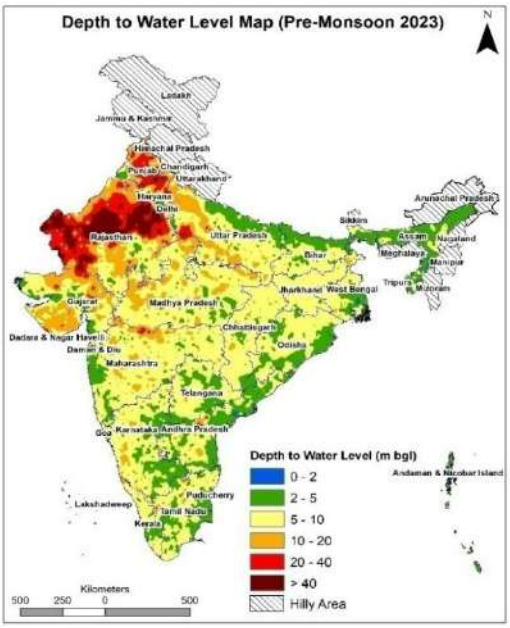
वार्षिक रिपोर्ट 2023-24

सारांश

केंद्रीय भूमि जल बोर्ड (सीजीडब्ल्यूबी) भूजल के क्षेत्र में एक वैज्ञानिक संगठन है। इसका मुख्यालय फरीदाबाद, हरियाणा में स्थित है। यह बोर्ड केंद्रीय भूजल प्राधिकरण (सीजीडब्ल्यूए) के रूप में भी कार्य करता है। इसका कार्यालय नई दिल्ली में स्थित है। सीजीडब्ल्यूबी की मुख्य भूमिका भारत के भूजल संसाधनों के वैज्ञानिक और सतत विकास एवं प्रबंधन के लिए वैज्ञानिक इनपुट प्रदान करना और इससे संबंधित विभिन्न गतिविधियों की मॉनीटरिंग एवं कार्यान्वयन करना है। बोर्ड की अधिकांश गतिविधियों का निष्पादन 'भूजल प्रबंधन और विनियमन (जीडब्ल्यूएमआर) स्कीम' नामक केंद्रीय क्षेत्र की योजना के एक भाग के रूप में की जाती हैं। उपरोक्त के अतिरिक्त, सीजीडब्ल्यूबी राजीव गांधी राष्ट्रीय भूमि जल प्रशिक्षण एवं अनुसंधान संस्थान (आरजीएनजीडब्ल्यूटीआरआई) के तत्वावधान में प्रशिक्षण और क्षमता निर्माण गतिविधियों का कार्यान्वयन भी करता है। सीजीडब्ल्यूबी राष्ट्रीय जल विज्ञान परियोजना (एनएचपी) और जल शक्ति मंत्रालय के अन्य कार्यक्रमों से संबंधित गतिविधियों का निष्पादन भी करता है। वर्ष 2023-24 में सीजीडब्ल्यूबी की गतिविधियों और उपलब्धियों का संक्षिप्त विवरण निम्नलिखित है।

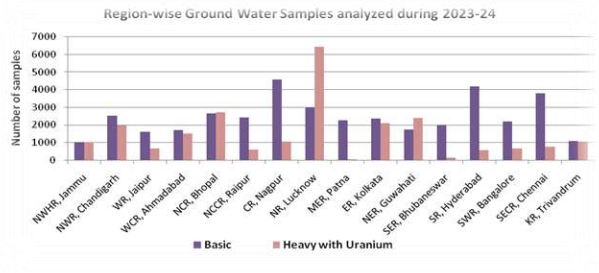
क . भूजल प्रबंधन और विनियमन योजना

1. **भूजल स्तर की मॉनीटरिंग :** सीजीडब्ल्यूबी के देश भर में 27163 भूजल प्रेक्षण कूपों (डग वेल : 16516, पीजोमीटर: 9552, हैंड पंप: 913, स्प्रिंग: 182) का एक नेटवर्क है। देश के मानसून पूर्व 2023 के भूजल स्तर के आंकड़ों से यह ज्ञात होता हैं कि देश में जल स्तर की सामान्य गहराई 5 से 10 मीटर बीजीएल तक पाई गई है। असम, उत्तर प्रदेश, बिहार, ओडिशा के तटीय भागों, आंध्र प्रदेश, गुजरात, केरल, तमिलनाडु, तेलंगाना और महाराष्ट्र के कुछ हिस्सों में सामान्यतः 2 से 5 मीटर बीजीएल की सीमा में भूजल स्तर पाया गया है। देश के अधिकांश राज्यों में जलस्तर 5 से 10 मीटर बीजीएल की रेंज में है। उत्तर-पश्चिमी और पश्चिमी राज्यों के प्रमुख हिस्सों में, विशेष रूप से दिल्ली, हरियाणा, पंजाब और राजस्थान राज्यों में, जल का स्तर समान्यतः अधिक गहरा है । भूजल स्तर की गहराई लगभग 20 से 40 मीटर बीजीएल के मध्य है ।



चित्र:1 जलस्तर की गहराई (मानसून-पूर्व 2023)

2. भूजल गुणवत्ता की मॉनीटरिंग : केन्द्रीय भूमि जल बोर्ड के अंतर्गत भूजल नमूनों का रासायनिक विश्लेषण करने के लिए 16 क्षेत्रीय रासायनिक प्रयोगशालाएं हैं। वार्षिक कार्य योजना 2023-24 में, सीजीडब्ल्यूबी द्वारा लगभग 62603 भूजल नमूनों का विश्लेषण किया गया है, जिसमें मूल पैरामीटर के लिए 39017, यूरेनियम सहित भारी धातुओं के लिए 23586 भूजल नमूने शामिल हैं।



चित्र:2 : भूजल नमूनों का किया गया क्षेत्रवार आकलन

3. भूभौतिकीय अन्वेषण:

सतही भूभौतिकीय अन्वेषण लिथोलॉजी से संबंधित उपसतही जानकारी को रेखांकित करने, भूजल के विघमान होने, इसके वितरण और गति को प्रभावित करने में समर्थन प्रदान करती है।

वार्षिक कार्य योजना 2023-24 के दौरान, सीजीडब्ल्यूबी द्वारा 2753 वर्टिकल इलेक्ट्रिकल साउंडिंग (VES), 3125 ट्रांसिएंट इलेक्ट्रो-मैग्नेटिक (TEM) सर्वेक्षण और नेक्यूम 2.0 के तहत 35 बोरहोल लॉगिंग, अल्पकालिक जल आपूर्ति जांच और अन्य संबंधित अध्ययन किए गए हैं। नई दिल्ली जंगपुरा, प्रगति मैदान में जल भराव के मुद्दों के समाधान हेतु एक उन्नत प्रौद्योगिकी (2डी इमेजिंग) का उपयोग करते हुए एक विशेष अध्ययन भी किया गया था।

4 समन्वेषी ड्रिलिंग : सीजीडब्ल्यूबी द्वारा अपने 76 ड्रिलिंग रिगों (23 डायरेक्ट रोटरी, 51 डाउन द होल और 02 डुअल रोटरी) के समृद्ध संसाधन के माध्यम से भूजल अन्वेषण किया जा रहा है। वर्ष 2023-24 के दौरान, सीजीडब्ल्यूबी द्वारा 730 कूपों(ईडब्ल्यू-223, ओडब्ल्यू -73, पीजेड-434) के अपने इन-हाउस लक्ष्य की तुलना में 800 अन्वेषण कुपों (ईडब्ल्यू -288, ओडब्ल्यू - 85, पीजेड - 427) का निर्माण किया गया है। इसमें 162 उच्च उत्पादकता कुप शामिल है ।

5. भूजल संसाधन आकलन: देश के समस्त डॉयनेमिक भूजल संसाधनों (वर्ष 2023 की स्थिति के अनुसार) का आकलन सीजीडब्ल्यूबी और राज्य भूजल विभागों द्वारा संयुक्त रूप से किया गया है। माननीय जल शक्ति मंत्री द्वारा दिनांक 1 दिसंबर, 2023 को " डॉयनेमिक भूजल संसाधनों का राष्ट्रीय संकलन, 2023" पर रिपोर्ट जारी की गई।

चित्र 3 :माननीय जलशक्ति मंत्री डॉयनेमिक भूजल संसाधन 2023 पर राष्ट्रीय संकलन जारी करते हुए

वर्ष 2023 के आकलन के अनुसार कुल वार्षिक भूजल पुनर्भरण 449 बीसीएम है और वार्षिक

निष्कर्षण योग्य भूजल संसाधन 407 बीसीएम है। सभी उपयोगों के लिए वार्षिक भूजल निष्कर्षण 241 बीसीएम है। समग्र रूप से देश में भूजल निष्कर्षण का स्तर 59% है। देश में कुल 6553 आकलन इकाइयों (ब्लॉक/मंडल/तालुका/तहसील/घाटी/जिला) में से 17 राज्यों/संघ राज्य क्षेत्रों में 736 इकाईयाँ अतिदोहित हैं जहां वार्षिक भूजल निष्कर्षण वार्षिक भूजल संसाधन से अधिक है। 199 इकाइयां 'गंभीर हैं; 'अर्ध-गंभीर' के रूप में 698 इकाइयाँ; 4793 इकाई सुरक्षित और 127 इकाई लवणीय के रूप में मौजूद है।

6. **नेक्यूम 2.0**: राष्ट्रीय जलभृत मैपिंग और प्रबंधन पहल (नेक्यूम) का कार्यान्वयन सीजीडब्ल्यूबी द्वारा किया जा रहा है। मार्च 2023 तक, नेक्यूम के तहत देश के कुल मैपिंग योग्य 33 लाख वर्ग किमी क्षेत्र में से 25 लाख वर्ग किमी क्षेत्र को शामिल कर लिया गया ।

नेक्यूम 2.0 के तहत, इस वर्ष, 44,000 वर्ग किमी के क्षेत्र को शामिल करते हुए विभिन्न क्षेत्रों में 76 अध्ययन किए गए थे। मुख्यालय, फरीदाबाद में प्रस्तुत किए जाने के बाद इन 76 अध्ययनों की प्रारंभिक रिपोर्टों को संबंधित राज्य सरकार के साथ साझा किया गया है।

सीजीडब्ल्यूबी के विभिन्न क्षेत्रों में मध्यावधि समीक्षा कार्यशालाएं आयोजित की गई थीं। ये समीक्षा कार्यशालाएं 4 स्थानों नामतः मुख्यालय फरीदाबाद (शहरी समूहों के लिए), दक्षिण पश्चिम क्षेत्र बेंगलुरु (ओसीएस और निम्न उपलब्धता के लिए), उत्तरांचल क्षेत्र देहरादून में (स्प्रिंगशेड और ऑटोफ्लो के लिए), और पश्चिमी क्षेत्र जयपुर (निम्न जल गुणवत्ता, औद्योगिक क्लस्टर, खनन क्षेत्र और तटीय अध्ययन के लिए) में आयोजित की गईं। वर्ष 2023-24 में आरंभ किए गए 76 अध्ययनों में से एक वर्ष की अवधि के 68 अध्ययन पूरे हो चुके हैं। दो वर्ष की अवधि वाले शेष 8 अध्ययनों को वर्ष 2024-25 में पूरा करने का लक्ष्य रखा गया है।

7. **कृत्रिम पुनर्भरण और भूजल संरक्षण**: राजस्थान में प्रदर्शनात्मक कृत्रिम पुनर्भरण परियोजनाएं कार्यान्वित की जा रही हैं। वर्ष 2023-24 के दौरान 107 संरचनाओं का निर्माण किया गया है।

इसके अतिरिक्त, दो बड़े बांधों (i) बस्तवा माता (ब्लॉक बालेसर) में कंक्रीट ग्रेविटी बांध और (ii) राजस्थान के जोधपुर जिले में इंद्रोका (ब्लॉक मंडोर) में मृदा सहित ज़ोनयुक्त मिट्टी के बांध का निर्माण किया गया।

चित्र 4 : माननीय जल शक्ति मंत्री इंदरोका में बांध का उदघाटन करते हुए

राजस्थान के जोधपुर जिले में इंद्ररोका (ब्लॉक मंडोर) में मृदा सहित ज़ोनयुक्त मिट्टी के बांध का निर्माण पूरा हो गया है और माननीय जल शक्ति मंत्री द्वारा दिनांक 16.03.2024 को इसका उद्घाटन किया गया है (चित्र 6)।

8. **सीजीडब्ल्यूए द्वारा भूजल निष्कर्षण का विनियमन और नियंत्रण**: सीजीडब्ल्यूए द्वारा संशोधित दिशानिर्देशों के अनुसार भूजल निष्कर्षण के लिए अनापत्ति प्रमाण पत्र (एनओसी) प्रदान करने के

लिए उद्योगों, अवसंरचनात्मक इकाइयों और खनन परियोजनाओं से प्राप्त आवेदनों की समीक्षा की गई। वर्ष 2023-2024 के दौरान, कुल 2054 नए एनओसी जारी किए गए और 1833 एनओसी का नवीकरण किया गया। इसके अतिरिक्त, नवीकरण के लिए 147 मामलों और 6966 नए आवेदनों को छूट प्रदान की गई थी । वर्ष 2023-2024 में कुल मिलाकर 11,000 एनओसी और छूट जारी की गई थी।

9. **राज्य सरकारों को तकनीकी समर्थन**: सरकारी विभागों को तकनीकी समर्थन प्रदान करने के लिए 160 अल्पकालिक जल आपूर्ति अन्वेषण किए गए ।

10. **रिपोर्ट और प्रकाशन**: तकनीकी रिपोर्ट सीजीडब्ल्यूबी द्वारा किए गए अध्ययनों के सबसे महत्वपूर्ण आउटपुट हैं। वर्ष 2023-24 के दौरान सीजीडब्ल्यूबी द्वारा प्रकाशित प्रमुख रिपोर्ट निम्नलिखित हैं:
    - डॉयनेमिक भूजल संसाधन आकलन 2023 पर राष्ट्रीय संकलन
    - भूजल संसाधन आकलन का 36 राज्य/संघ राज्य क्षेत्रवार संकलन
    - भूजल वार्षिकी इंडिया 2023
    - राज्यों और संघ राज्य क्षेत्रों को शामिल करते हुए 23 भूजल वार्षिकी ।
    - भारत के उथले जलभृतों में भूजल गुणवत्ता 2023
    - अल्पावधि जल आपूर्ति जांच की 160 रिपोर्टें।
    - भूजल संवाद - 4 खंड।

उपर्युक्त के अतिरिक्त, वर्तमान वर्ष में नेक्यूम अध्ययनों की लगभग 80 रिपोर्टें जारी की गई थीं और सभी नेक्यूम अध्ययनों की संचयी रिपोर्टें जारी की गई थीं। इसके अतिरिक्त, विभिन्न क्षेत्रों में भूजल अन्वेषण, भूभौतिकीय अध्ययन आदि पर भी रिपोर्ट प्रकाशित की गई।

11. **आउटरीच गतिविधियाँ**: जल संरक्षण, भूजल प्रबंधन के विभिन्न पहलुओं और नेक्यूम अध्ययन के निष्कर्षों पर जमीनी स्तर पर लोगों में जागरूकता सृजन के लिए सीजीडब्ल्यूबी द्वारा जन संवाद कार्यक्रम (पीआईपी) आयोजित किए गए। वर्ष 2023-24 के दौरान कुल 146 ऐसे कार्यक्रम आयोजित किए गए जिनमें 15,887 लोगों ने भाग लिया ।

ख. सार्वजनिक निवेश बोर्ड (पीआईबी) द्वारा राष्ट्रीय जलभृत मैपिंग और प्रबंधन (नेक्यूम) के लिए आँकड़े तैयार करने हेतु अवसंरचनात्मक परियोजना को मंजूरी

1. **पीआईबी परियोजना**: जीडब्ल्यूएमआर योजना के तहत चल रही गतिविधियों के अतिरिक्त, वर्ष 2022-2026 की अवधि के लिए जीडब्ल्यूम और आर योजना के एक भाग के रूप में कार्यान्वित किए जाने वाले राष्ट्रीय जलभृत मैपिंग और प्रबंधन (नेक्यूम) के लिए आँकड़े तैयार करने हेतु अवसंरचना के निर्माण के लिए सार्वजनिक निवेश बोर्ड (पीआईबी) द्वारा एक परियोजना को मंजूरी दी गई है। निविदा और अन्य संबंधित गतिविधियों की प्रक्रिया जारी है।

- टेलीमेट्री प्रणाली सहित 7000 डिजिटल वाटर लेवल रिकॉर्डर (डीडब्ल्यूएलआर) की संस्थापना के साथ 7000 पीजोमीटर का निर्माण ।
- उत्तर-पश्चिम भारत के शुष्क भागों में लगभग 3 लाख वर्ग किमी क्षेत्र में उच्च रिज़ॉल्यूशन मैपिंग के लिए हेली-बोर्न सर्वेक्षण ।
- कूपों के निर्माण के माध्यम से जलभृत मैपिंग के लिए डेटा सृजन (11 राज्यों में 1135 कूप) ।

ग. प्रशिक्षण और क्षमता निर्माण

1. **राष्ट्रीय भूजल प्रशिक्षण और अनुसंधान संस्थान (आरजीएनजीडब्ल्यूटीआरआई):** राजीव गांधी राष्ट्रीय भूजल प्रशिक्षण और अनुसंधान संस्थान (आरजीएनजीडब्ल्यूटीआरआई) केन्द्रीय भूमि जल बोर्ड का प्रशिक्षण स्कन्ध है। वर्ष 2012 (XIIवीं योजना) से आरजीएनजीडब्ल्यूटीआरआई त्रिस्तरीय प्रशिक्षण प्रणाली (टियर I-राष्ट्रीय स्तर, टियर II-राज्य/जिला टियर-III: ब्लॉक स्तर) का कार्यान्वयन कर रहा है। इसके अतिरिक्त, आरजीएनजीडब्ल्यूटीआरआई जल ऑडिटर्स , भूजल प्रोफेसनल्स को मान्यता आदि प्रदान करने हेतु प्रशिक्षण भी आयोजित करता है। वर्ष 2023-24 में आरजीएनजीडब्ल्यूटीआरआई की प्रमुख उपलब्धियां निम्नलिखित हैं:

चित्र 5 : आरजीएनजीडब्ल्यूटीआरआई को प्रदान किया गया मान्यता प्रमाण पत्र

- आरजीएनजीडब्ल्यूटीआरआई द्वारा वार्षिक कार्य योजना 2023-24 में 2796 महिला प्रतिभागियों सहित 7523 प्रतिभागियों को शामिल करते हुए 120 प्रशिक्षणों के लक्ष्य की तुलना में 128 प्रशिक्षण (टियर I, टियर II, टियर III) पूर्ण किए गए हैं।
- आरजीएनजीडब्ल्यूटीआरआई द्वारा अटल भूजल योजना पर छह ई-लर्निंग मॉड्यूल तैयार किए गए हैं और उन्हें गॉट कर्मयोगी पर अपलोड किया गया है।
- आरजीएनजीडब्ल्यूटीआरआई द्वारा 2 जल ऑडिटर प्रमाणन प्रशिक्षण पूरा कर लिए गए है।
- राष्ट्रीय शिक्षा और प्रशिक्षण प्रत्यायन बोर्ड (एनएबीईटी) द्वारा दिनांक 22.11.23 को किए गए मूल्यांकन के आधार पर आरजीएनजीडब्ल्यूटीआरआई को क्षमता निर्माण आयोग के राष्ट्रीय मानकों के तहत अतिउत्तम, के रूप में मान्यता दी गई है।

2. **स्रोत स्थायित्वता पर प्रशिक्षण**: भूजल आधारित पेयजल स्रोतों की स्थायित्वता के लिए योजना और उपायों हेतु परिचालन दिशानिर्देशों का एक खंड विकसित करने तथा राज्य सरकार की एजेंसियों को प्रशिक्षण प्रदान करने के उद्देश्य से प्रशिक्षण कार्यक्रम (प्रत्येक 3 दिन की अवधि) आयोजित किए गए थे। प्रशिक्षण कार्यक्रम जून 2023 से मार्च 2024 तक

आयोजित किए गए। 29 राज्यों और संघ राज्य क्षेत्रों में लगभग 1300 प्रतिभागियों के लिए कुल 49 प्रशिक्षण आयोजित किए गए ।

घ . राष्ट्रीय जल विज्ञान परियोजना (एनएचपी)

1. **राष्ट्रीय जल विज्ञान परियोजना (एनएचपी)**: राष्ट्रीय जल विज्ञान परियोजना (एनएचपी) जल विज्ञान परियोजना (एचपी)-I और एचपी-II की क्रमबद्ध योजना है । यह योजना जल संसाधन, नदी विकास और गंगा संरक्षण विभाग, जल शक्ति मंत्रालय, भारत सरकार की केंद्रीय क्षेत्र की एक योजना है, जिसका उद्देश्य बाढ़ बेसिन स्तर संसाधन आकलन/योजना के लिए जल संसाधन पेशेवरों और प्रबंधन संस्थानों की क्षमता को सुदृढ़ करने के लिए जल संसाधन सूचना की मात्रा, गुणवत्ता और सुगम्यता में सुधार करना है।

एनएचपी की कार्यान्वयन एजेंसियों में से एक के रूप में केंद्रीय भूमि जल बोर्ड (सीजीडब्ल्यूबी) द्वारा भूजल गुणवत्ता जांच के साथ 60 डिजिटल जल स्तर रिकार्डर की संस्थापना और देश भर में 5200 डिजिटल जल स्तर रिकार्डर की संस्थापना का लक्ष्य रखा गया था। यद्यपि यह गतिविधि पहले ही पूरी हो गई थी, वर्ष 2023-24 के अंत में 5200 डीडब्ल्यूएलआर के लक्ष्य की तुलना में, 5190 (संचयी) डीडब्ल्यूएलआर की संस्थापना पूरी हो गई थी।

एनएचपी के अंतर्गत, सीजीडब्ल्यूबी द्वारा सितंबर 2023 में आईएचई डेल्फ़्ट, नीदरलैंड में भूजल मॉडलिंग पर 3 सप्ताह की अवधि का एक उद्देश्य आधारित प्रशिक्षण का आयोजन किया गया । इस प्रशिक्षण कार्यक्रम में भारत के कुल 25 अधिकारियों ने भाग लिया, जिनमे सीजीडब्ल्यूबी से 15, एनएचपी-एनपीएमयू से 02 और राज्य सरकार के विभागों से 08 अधिकारी शामिल थे ।

चित्र 6 : आईएचई डेल्फ्ट, नीदरलैंड में भारतीय प्रशिक्षु अधिकारी

ड़. प्रधानमंत्री कृषि सिंचाई योजना - भूजल (पीएमकेएसआई-एचकेकेपी-जीडब्ल्यू)

1. **पीएमकेएसवाई-एचकेकेपी-जीडब्ल्यू**: "प्रधानमंत्री कृषि सिंचाई योजना - हर खेत को पानी - भूजल" स्कीम का कार्यान्वयन भूजल का विकास करते हुए सिंचाई क्षेत्र का विस्तार करने के उद्देश्य से व्यवहार्य क्षेत्रों में किया गया था। यह योजना वर्तमान में कार्यरत नहीं है, परंतु वर्ष 2023-24 में चल रही गतिविधियां जारी रहीं। वर्ष 2023-24 के दौरान, इस योजना के तहत 84 कूपों का निर्माण किया गया, 1058.3 हेक्टेयर कमान क्षेत्र का सृजन किया गया और 624 छोटे और सीमांत किसान इस योजना से लाभान्वित हुए हैं। मार्च 2024 तक, परियोजना के तहत 29779 कूपों का निर्माण किया गया है, जिससे 88,547 हेक्टेयर का कमान क्षेत्र का सृजन किया गया एवं इससे 67,909 छोटे और सीमांत किसान लाभान्वित हुए है।

च . जल शक्ति अभियान (जेएसए)

1. **जल शक्ति अभियान (जेएसए):** जल शक्ति अभियान (जेएसए) की शुरुआत जल शक्ति मंत्रालय द्वारा वर्ष 2019 में हुई थी। जल शक्ति अभियान- कैच द रेन (जेएसए: सीटीआर) अभियान 2023 का विषय "पेयजल स्रोतों की स्थायित्वता" था। इस अभियान में सीजीडब्ल्यूबी के परामर्श से एनडब्ल्यूएम द्वारा चिन्हित 150 जिलों को शामिल किया गया और सीजीडब्ल्यूबी के 150 अधिकारियों ने इस अभियान के अंतर्गत केंद्रीय टीमों में तकनीकी अधिकारियों के रूप में कार्य किया।

ज . राजभाषा

1. **राजभाषा हिन्दी**: केंद्रीय भूमिजल बोर्ड और इसके सभी अधीनस्थ क्षेत्रीय, प्रभागीय और राज्य एकक कार्यालयों में वर्ष 2023-2024 के दौरान राजभाषा हिन्दी के कार्यान्वयन और प्रचार-प्रसार के सक्रिय प्रयास जारी रखे गए। माननीय संसदीय राजभाषा समिति की दूसरी उपसमिति द्वारा केंद्रीय भूमि जल बोर्ड के विभिन्न अधीनस्थ क्षेत्रीय / प्रभागीय / राज्य एकक कार्यालयों का राजभाषायी निरीक्षण किया गया। समिति द्वारा कार्यालयों में राजभाषा के प्रचार-प्रसार और इसके कार्यान्वयन की दिशा में किए जा रहे कार्यों पर संतोष प्रकट किया गया।

चित्र 7: माननीय संसदीय राजभाषा समिति द्वारा केन्द्रीय भूमिजल बोर्ड वाराणसी कार्यालय का राजभाषायी निरीक्षण

ज . बजट और व्यय

1. **आवंटन और व्यय**: वर्ष 2023-24 के दौरान निधि के आवंटन और व्यय का एक संक्षिप्त सारांश निम्नलिखित है-

तालिका 1: वर्ष 2023-24 के दौरान सीजीडब्ल्यूबी की विभिन्न योजनाओं/परियोजनाओं के तहत आवंटन और व्यय

लाख रुपए

	जीडब्लूएमएवं आर योजना	मुख्यालय स्थापना	आरजीएनजीडब्ल्यूटीआरआई-गैर योजना	राष्ट्रीय जल विज्ञान परियोजना (एनएचपी)	कुल
आवंटन	28000	30400	300	932.12	59632.12
व्यय	20422	29028.63	242.44	860.14	50553.21

Annual General Report 2023-24

Summary

Central Ground Water Board (CGWB) is a scientific organisation in the field of groundwater with its headquarters at Faridabad, Haryana. The Board also functions as Central Ground Water Authority (CGWA) with its office at New Delhi. Role of CGWB is to provide scientific inputs, and monitor and implement various activities for the Scientific and Sustainable development and management of India's Ground Water Resources. Most of the activities of the Board are undertaken as a part of the Central Sector Scheme titled 'Ground Water Management and Regulation (GWMR) scheme'. In addition to above, CGWB also implements training and capacity building activities under the aegis of Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI). CGWB also carries out activities related to National Hydrology Project (NHP) and other programmes of Ministry of Jal Shakti. A brief account of activities and achievements of CGWB in the year 2023-24 is given below.

A. Ground Water Management and Regulation Scheme

1. **Ground Water Level Monitoring:** CGWB has a network of 27163 (Dug Wells: 16516, Piezometers: 9552, Hand Pump: 913, spring: 182) groundwater observation wells throughout the Country. Groundwater level data of pre-monsoon 2023 for the country reveals that the general depth to water level of the country ranges from 5 to 10 m bgl. Groundwater level in the range of 2 to 5 m bgl is prominently seen in Assam, Uttar Pradesh, Bihar, Coastal parts of Odisha, few pockets in Andhra Pradesh, Gujarat, Kerala, Tamil Nadu, Telangana and Maharashtra. Major part of the country shows water level in the range 5 to 10 m bgl, in almost all states of the country. In major parts of north-western and western states, especially in the states of Delhi, Haryana, Punjab and Rajasthan, depth to water level is generally deeper and ranges from about 20 to more than 40 m bgl.

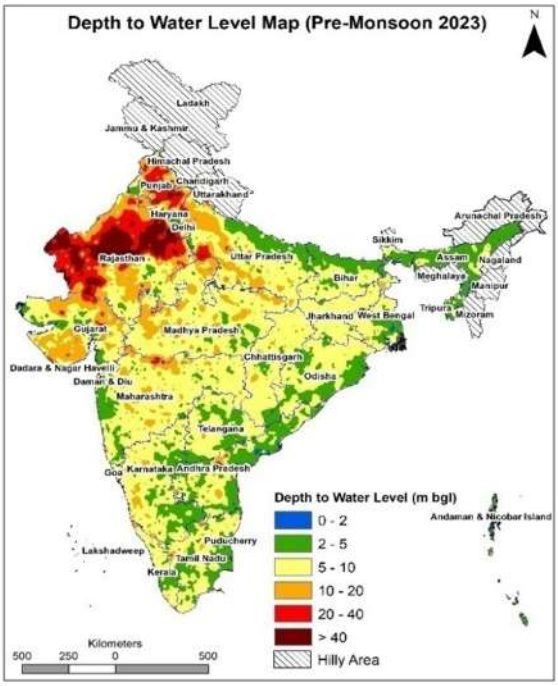


Figure 1: Depth to Water Level (Premonsoon 2023)

2. **Ground Water Quality Monitoring:** Central Ground Water Board has 16 Regional Chemical Laboratories to carry out chemical analysis of ground water samples. In AAP 2023-24, nearly 62603 groundwater samples have been analyzed by CGWB which included 39017 for Basic parameters, 23586 for Heavy Metals with Uranium

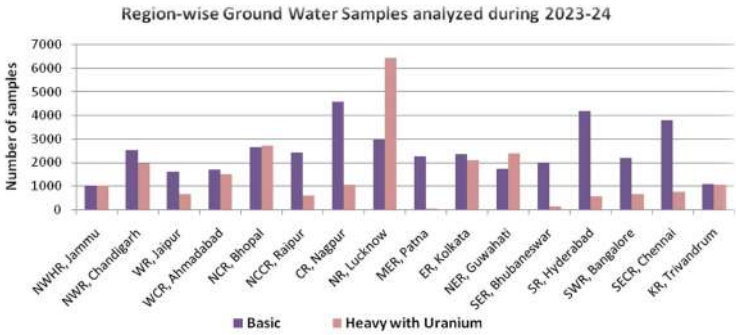


Figure 2: Region wise groundwater samples analysed

3. **Geophysical Investigations**: Surface geophysical investigations aid in delineating subsurface information regarding lithology, influencing the occurrence, distribution, and movement of groundwater.

   During the AAP 2023-24, CGWB has carried out 2753 Vertical Electrical Sounding (VES), 3125 Transient Electro-magnetic (TEM) survey and 35 borehole logging in NAQUIM 2.0, short term water supply investigations and other related studies. A special study was also carried out using advanced technology (2D Imaging) for water logging issues from Pragati maidan to Jungpura, New Delhi.

4. **Exploratory Drilling**: Groundwater Exploration is being carried out by CGWB through its fleet of 76 drilling rigs (23 Direct Rotary, 51 Down the Hole and 02 Dual Rotary). During the year 2023-24, CGWB has constructed 800 exploration Wells (EW-288, OW- 85, PZ- 427), under its in-house target of 730 wells (EW-223, OW-73, PZ-434). This included 162 high-yielding wells.

5. **Ground Water Resource Assessment**: The Dynamic Ground Water Resources (as in 2023) of the entire country have been assessed jointly by CGWB and State Ground Water Departments. The Report on "National Compilation of Dynamic Ground Water Resources, 2023" was released by Honourable Minister of Jal Shakti on 1st December, 2023.

Figure 3: Hon'ble Minister of Jal Shakti releasing the National Compilation on Dynamic Ground Water Resources 2023

   As per the 2023 assessment total Annual Ground Water Recharge is 449 BCM and the Annual Extractable Ground Water Resource is 407 BCM. The Annual Ground Water

Extraction for all uses is 241 BCM. The Stage of Ground Water Extraction is 59% for the country as a whole. Out of the total 6553 assessment units (Blocks/ Mandals/ Talukas/ Tehsil/ Valley/ District) in the country, 736 units in 17 States/UTs are 'Over-exploited' where the Annual Ground Water Extraction is more than Annual Extractable Ground Water Resource. 199 units are 'Critical'; 698 units as 'Semi-critical'; 4793 units as 'Safe' and 127 units as 'Saline'.

6. **NAQUIM 2.0**: The National Aquifer Mapping and Management initiative (NAQUIM) is being implemented by CGWB. Up until March 2023, this initiative covered the complete mappable area of 25 lakh km^2 out of the 33 lakh km^2 of the country.
   Under NAQUIM 2.0, this year, 76 studies were undertaken in different prority areas, covering an area of 44,000 sq. km.. The inception reports of these 76 studies have been shared with the concerned State Government after presentation at CHQ, Faridabad.
   Mid-term review workshops were organized in different regions of CGWB. These review workshops were held at 4 places, namely CHQ Faridabad (for Urban Agglomerates), at SWR Bengaluru (for OCS & Poor Availability), at UR Dehradun (for Springshed & Autoflow), and at WR Jaipur (for Poor Water Quality, Industrial Cluster, Mining Area and Coastal studies). Out of the 76 studies initiated in 2023-24, 68 studies of one year duration are complete. The remaining 8 studies, which are of two year duration are targeted to be completed in 2024-25.

7. **Artificial Recharge & Ground water Conservation**: Demonstrative Artificial Recharge Projects are being implemented in Rajasthan. During the year 2023-24, 107 structures have been constructed.
   In addition to this, two large dams were consrructed (i) Concrete Gravity Dam at Bastawa Mata (Block Balesar) and (ii) Zoned Earth Fill Dam with clay Core at Indroka (Block Mandore) in Jodhpur district, Rajasthan.
   The construction of Zoned Earth Fill Dam with Clay Core at Indroka (Block Mandore) in Jodhpur district, Rajasthan has been completed and inaugurated by the Hon'ble Minister of Jal Shakti on 16.03.2024 (Fig. 6).

Figure 4: Hon'ble Minister of Jal Shakti inaugurating the dam at Indroka

8. **Regulation and Control of Ground Water Extraction by CGWA**: Applications from industries, infrastructure units, and mining projects were reviewed by CGWA in order to provide No Objection Certificate (NoC) for ground water withdrawal as per revised guidelines. During 2023–2024, a total of 2054 new NOCs were issued and 1833 NOCs were renewed. Furthermore, exemptions were granted for 147 cases for renewal and 6966 new applications. In all, 11,000 NoCs and exemptions were issued in 2023–2024.

9. **Technical Assistance to State Governments**: 160 Short Term Water Supply investigations

10. **Reports and Publications**: Technical reports are the most important outputs of studies carried out by CGWB. Major reports published by CGWB during 2023-24 are as follows:
    - National Compilation on Dynamic Ground Water Resource Assessment 2023
    - 36 State/UT wise compilations of Ground Water Resource Assessment
    - Ground Water Year Book India 2023
    - 23 Ground Water Year Books covering states and UTs.
    - Ground Water Quality in Shallow Aquifers of India 2023
    - 160 reports of Short Term Water Supply Investigations.
    - Bhujal Samvad – 4 volumes.

    In addition to above, nearly 80 reports of NAQUIM studies were issued in the current year and cumulatively reports of all the NAQUIM studies were issued. Further, reports on groundwater exploration, geophysical studies etc. were also brought out in different regions.

11. **Out-reach Activities**: Public Interaction programs (PIPs) were organised by CGWB to sensitize people at grassroots level on various aspects of water conservation, ground water management and findings of the NAQUIM study. During 2023-24 a total of 146 such programmes were conducted in which 15,887 people have participated.

B. Public Investment Board (PIB) approved project for data generation under National Aquifer Mapping and Management (NAQUlM)

1. **PIB Project**: In addition to the ongoing activities under GWMR scheme, a Project has been approved by the Public Investment Board (PIB) for creating infrastructure for data generation for National Aquifer Mapping and Management (NAQUlM) to be implemented as a part of the GWM&R Scheme for the period 2022-2026. Major Activities being taken up as a part of the project include the following. Process for tendering and other related activities are in process.
    - Construction of 7000 piezometers along with installation of 7000 Digital Water Level Recorders (DWLRs) with telemetry system
    - Heli-borne surveys for high resolution mapping in ~ 3 lakh sq km area in arid parts of NW India.
    - Data Generation for Aquifer Mapping through construction of Wells (1135 wells in 11 States)

C. Training and Capacity Building

1. **National Ground Water Training and Research Institute (RGNGWTRI)**: Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI) is the training wing of CGWB. Since 2012 (XII Plan) RGNGWTRI is implementing a three-tiered training system (Tier I- National Level, Tier II-State/ District Tier- III: Block Level). In addition to this RGNGWTRI also carries out training for water auditors, accreditation of groundwater professionals etc. Major achievements of RGNGWTRI in 2023-24 are listed below:

   - RGNGWTRI has completed 128 Trainings (TierI, Tier II, TierIII) against a target of 120 trainings with 7523 participants including 2796 female participants in AAP 2023-24.
   - RGNGWTRI has created Six e-Learning modules on Atal Bhujal Yojana and uploaded them on iGOT Karmayogi

Figure 5: Certificate of recognition presented to RGNGWTRI

   - RGNGWTRI has completed 2 Water Auditor Certification trainings
   - RGNGWTRI has been accredited as अतिउत्तम under the Capacity Building Commission's National Standards as assessed by the National Accreditation Board of Education and Training (NABET) on 22.11.23.

2. **Training on Source Sustainability**: Training programmes (3 day duration each) were organized with the objective to develop a set of operational guidelines and impart training to state government agencies for planning and interventions for sustainability of groundwater based drinking water sources. The Training programmes were conducted from June 2023 to March 2024. Total 49 trainings have been conducted in 29 States and UTs with about 1300 participants.

D. National Hydrology Project (NHP)

1. **National Hydrology Project (NHP)**: National Hydrology Project (NHP) is a continuation of Hydrology Project (HP)- I and HP- II, a central sector scheme of the Department of Water Resources, RD&GR, Ministry of Jal Shakti, Government of India with an aim to improve the extent, quality, accessibility of water resources information, decision support system for floods basin level resource assessment/ planning strengthen the capacity of targeted water resources professionals and management institutions in India.

Central Ground Water Board (CGWB), as one of the implementing agencies of NHP had a target of installation of 60 Digital Water Level Recorders with groundwater quality probes and installation of 5200 Digital Water Level Recorders across the country. While the former activity was completed earlier, at the end of 2023-24 against a target of 5200 DWLRs, installation of 5190 (cumulative) DWLR were completed.

E. Pradhan Mantri Krishi Sinchayee Yojna – Ground Water (PMKSY-HKKP-GW)

1. **PMKSY-HKKP-GW**: The "Pradhan Mantri Krishi Sinchayi Yojana - Har Khet Ko Pani - Ground Water" scheme was implemented with the aim of expanding irrigation coverage by developing groundwater in feasible areas. The scheme is currently not in operation, but the ongoing activities continued in 2023-24. During the year 2023-24, 84 wells have been constructed, 10583 Ha command area have been created and 624 small & marginal farmers have benefited under the scheme. As of March 2024, so far (since inception) under the project 29779 wells have constructed, creating command area of 88,547 ha benefiting 67,909 small and marginal farmers.

F. Jal Shakti Abhiyan (JSA)

1. **Jal Shakti Abhiyan (JSA)**: Jal Shakti Abhiyan (JSA) was launched by Ministry of Jal Shakti in 2019. Jals Shakti Abhiyan- Catch The Rain (JSA: CTR) campaign 2023 had the theme "Source Sustainability for Drinking Water". The campaign covered 150 districts identified by NWM in consultation with CGWB and 150 officers of CGWB served as technical officers in the central teams for the campaign.

G. राजभाषा

1. **राजभाषा हिन्दी**: केंद्रीय भूमिजल बोर्ड और इसके सभी अधीनस्थ क्षेत्रीय, प्रभागीय और राज्य एकक कार्यालयों में वर्ष 2023-2024 के दौरान राजभाषा हिन्दी के कार्यान्वयन और प्रचार-प्रसार के सक्रिय प्रयास जारी रखे गए। माननीय संसदीय राजभाषा समिति की दूसरी उपसमिति द्वारा केंद्रीय भूमि जल बोर्डके विभिन्न अधीनस्थ क्षेत्रीय / प्रभागीय / राज्य एकक कार्यालयों का राजभाषायी निरीक्षण किया गया। समिति द्वारा कार्यालयों में राजभाषा के प्रचार-प्रसार और इसके कार्यान्वयन की दिशा में किए जा रहे कार्यों पर संतोष प्रकट किया गया।

चित्र 6:माननीय संसदीय राजभाषा समिति द्वारा केन्द्रीय भूमि जल बोर्ड वाराणसी कार्यालय का राजभाषायी निरीक्षण

H. Budget and Expenditure

1. **Allocation and Expenditure**: A brief summary of fund allocation and expenditure during the year 2023-24 is given below:

Table 1: Allocation and Expenditure under various schemes/projects of CGWB during 2023-24 (Rs are in lakhs)

	GWM&R Scheme	HQ-Establishment	RGNGWTRI-Non Scheme	National Hydrology Project (NHP)	Total
Allocation	28000	30400	300	932.12	59632.12
Expenditure	20422	29028.63	242.44	860.14	50553.21

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1. Overview of Central Ground Water Board

1.1 Introduction

Central Ground Water Board (CGWB) is an organisation under Department of Water Resources, River Development & Ganga Rejuvination, Ministry of Jal Shakti, Government of India and it is the national apex organization for all aspects of ground water surveys, exploration, development and management.

Scientific studies related to groundwater (hydrogeological studies) started in the mid-19th century with the establishment of the Geological Survey of India (GSI) in 1851. In 1954 a dedicated organization named Exploratory Tubewell Organisation (ETO) was set up for exploring ground water. In 1970, ETO was renamed as Central Ground Water Board (CGWB). In 1972, the ground water wing of Geological Survey of India (GSI) was merged with CGWB.

Central Ground Water Board is also discharging the functions as Central Ground Water Authority (CGWA) to regulate and control the development and management of ground water in the country since 1997. In the same year (1997), the Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI) was established as a part of CGWB.

1.2 Vision

Sustainable Development and Managemement of Ground Water Resources of the Country.

1.3 Mandate

CGWB's mandate is "To develop and disseminate technologies, monitor and implement National policies for the scientific and sustainable development and management of India's Ground Water resources, including its exploration, assessment, conservation, augmentation, protection from pollution and distribution, based on principles of economic and ecological efficiency and equity"

1.4 Organizational Set up

Central Ground Water Board (CGWB) operates as a subordinate office of Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti Government of India. CGWB is a multi-disciplinary scientific organization consisting of hydrogeologists, geophysicists, chemists, hydrologists, hydrometeorologists, engineers, and other streams.

The Board is headed by the Chairman and has five Members. The extended Board of CGWB consists of five permanent members representing Central Water Commission, Ministry of Jal

Shakti, Ministry of Environment, Forest& Climate Change, Oil and Natural Gas Commission (ONGC).

Total sanctioned strength of the Board is 4017, which includes 472 Group A officers, 985 Group B officers and 2560 Group C employees.

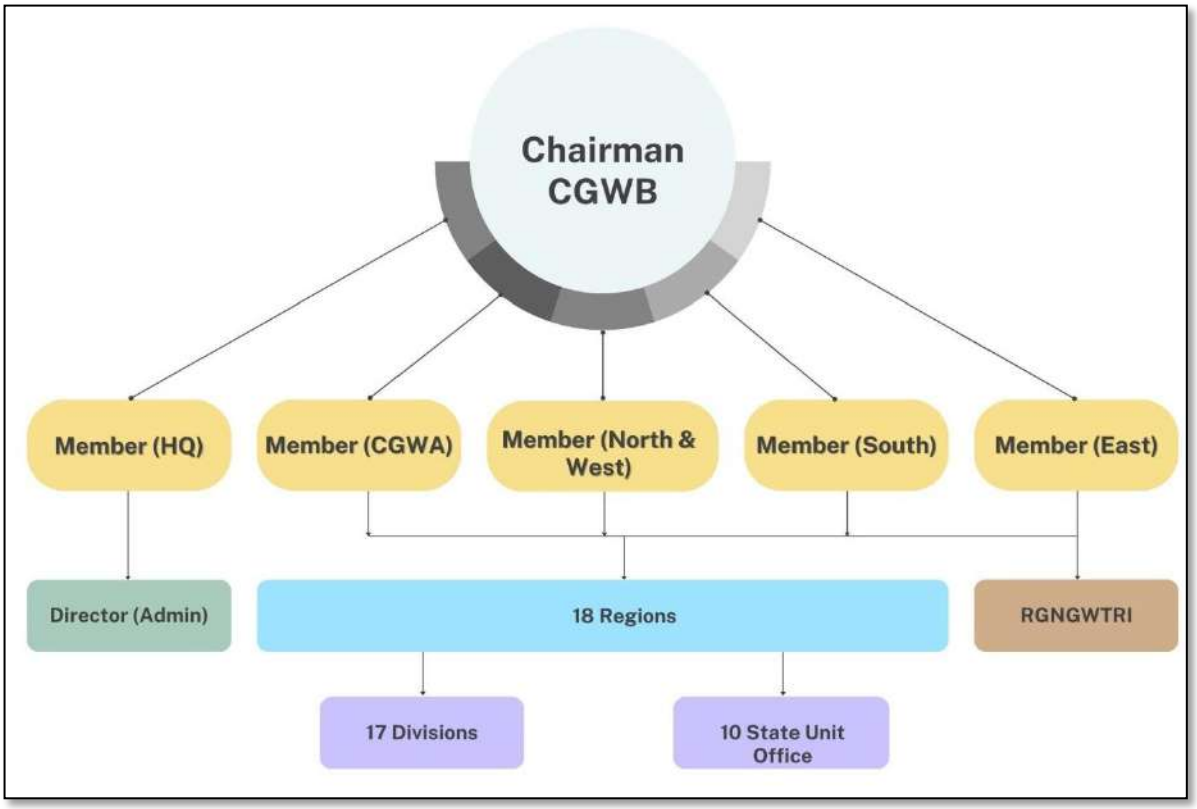


Fig. 1. Organizational setup of CGWB

1.5 Offices of Central Ground Water Board

CGWB has its headquarters at Bhujal Bhawan, Faridabad, Haryana; the Central Ground Water Authority (CGWA) operates from its office in New Delhi. Field activities are performed through 18 Regional Offices, 10 State Unit Offices and 17 Divisional Offices located in various part of the Country. The Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI) located in Raipur, Chhattisgarh, serves as the training arm of CGWB.

Table 1: Regions & Regional Offices of CGWB

Sl No.	Name of Office	Location	Jurisdiction
Central Headquarters (CHQ)			
1	CGWB, CHQ	Faridabad	Entire Country
Central Ground Water Authority (CGWA)			
2	CGWA	New Delhi	Entire Country

Rajiv Gandhi National Ground Water Training and Research Institute			
3	RGNGWTRI	Raipur	Entire Country
North West Himalayan Region (NWHR)			
4	North West Himalayan Region	Jammu	U.T. of Jammu & Kashmir and Ladakh
5	Division-VII	Jammu	U.T. of Jammu & Kashmir and Ladakh
North Himalayan Region (NHR)			
6	North Himalayan Region	Dharamshala	Himachal Pradesh
7	Division-XVII	Dharamshala	Himachal Pradesh
North Western Region (NWR)			
8	North Western Region	Chandigarh	Punjab, Haryana, Delhi & U.T. of Chandigarh
9	State Unit Office (SUO)	Delhi	Delhi
10	Division-II	Ambala	Punjab, Haryana, NCT of Delhi & U.T. of Chandigarh
Uttarakhand Region (UR)			
11	Uttarakhand Region	Dehradun	Uttarakhand
Northern Region (NR)			
12	Northern Region	Lucknow	Uttar Pradesh
13	Division-III	Varanasi	Parts of Uttar Pradesh
14	Division-XVI	Bareilly	Part of Uttar Pradesh & Uttarkhand
15	State Unit Office	Prayagraj	Part of Uttar Pradesh
Western Region (WR)			
16	Western Region	Jaipur	Parts of Rajasthan
17	Division-XI	Jodhpur	Rajasthan
18	State Unit Office	Jodhpur	Part Rajasthan
Mid-Eastern Region (MER)			
19	Mid-Eastern Region	Patna	Bihar, Jharkhand
20	Division-V	Ranchi	Bihar and Jharkhand
21	State Unit Office	Ranchi	Jharkhand
Eastern Region (ER)			
22	Eastern Region	Kolkata	West Bengal, Sikkim, U.T. of Andaman & Nicobar Islands
23	Division-XV	Kolkata	West Bengal, Sikkim, U.T. of Andaman & Nicobar Islands
North Eastern Region (NER)			
24	North Eastern Region	Guwahati	Assam, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Meghalaya & Tripura

25	Division-VII	Guwahati	Assam, Arunachal Pradesh, Nagaland, Manipur, Mizoram, Meghalaya & Tripura
26	State Unit Office	Shillong	Meghalaya
27	State Unit Office	Agartala	Tripura
28	State Unit Office	Itanagar	Arunachal Pradesh
West Central Region (WCR)			
29	West Central Region	Ahmedabad	Gujarat, U.T of Daman & Diu and U.T. of Dadra & Nagar Haveli
30	Division-I	Ahmedabad	Gujarat, Daman & Diu and & U.T. of Dadra & Nagar Haveli
North Central Region (NCR)			
31	North Central Region	Bhopal	Madhya Pradesh
32	Division-XII	Bhopal	Madhya Pradesh
North Central Chhattisgarh Region (NCCR)			
33	North Central Chhattisgarh Region	Raipur	Chhattisgarh
34	Division-XIII	Raipur	Chhattisgarh
South Eastern Region (SER)			
35	South Eastern Region	Bhubaneswar	Odisha
36	Division-X	Bhubaneswar	Odisha
Central Region (CR)			
37	Central Region	Nagar	Maharashtra
38	Division-VI	Nagpur	Maharashtra
39	State Unit Office	Pune	Parts of Maharashtra
Southern Region (SR)			
40	Southern Region	Hyderabad	Andhra Pradesh & Telangana
41	Division-IX	Hyderabad	Andhra Pradesh & Telangana
42	State Unit Office	Visakhapatnam	Parts of Andhra Pradesh
South Western Region (SWR)			
43	South Western Region	Bengaluru	Karnataka & Goa
44	Division-XIV	Bengaluru	Karnataka, Goa and Parts of Kerala
45	State Unit Office	Belagavi	Parts of Karnataka and Goa
South East Coastal Region (SECR)			
46	South East Coastal Region	Chennai	Tamil Nadu and U.T. of Puducherry
47	Division-IV	Chennai	Tamil Nadu, U.T. of Puducherry and Parts of Kerala
Kerala Region (KR)			

48	Kerala Region	Thiruvananthapuram	Kerala and U.T. of Lakshadweep

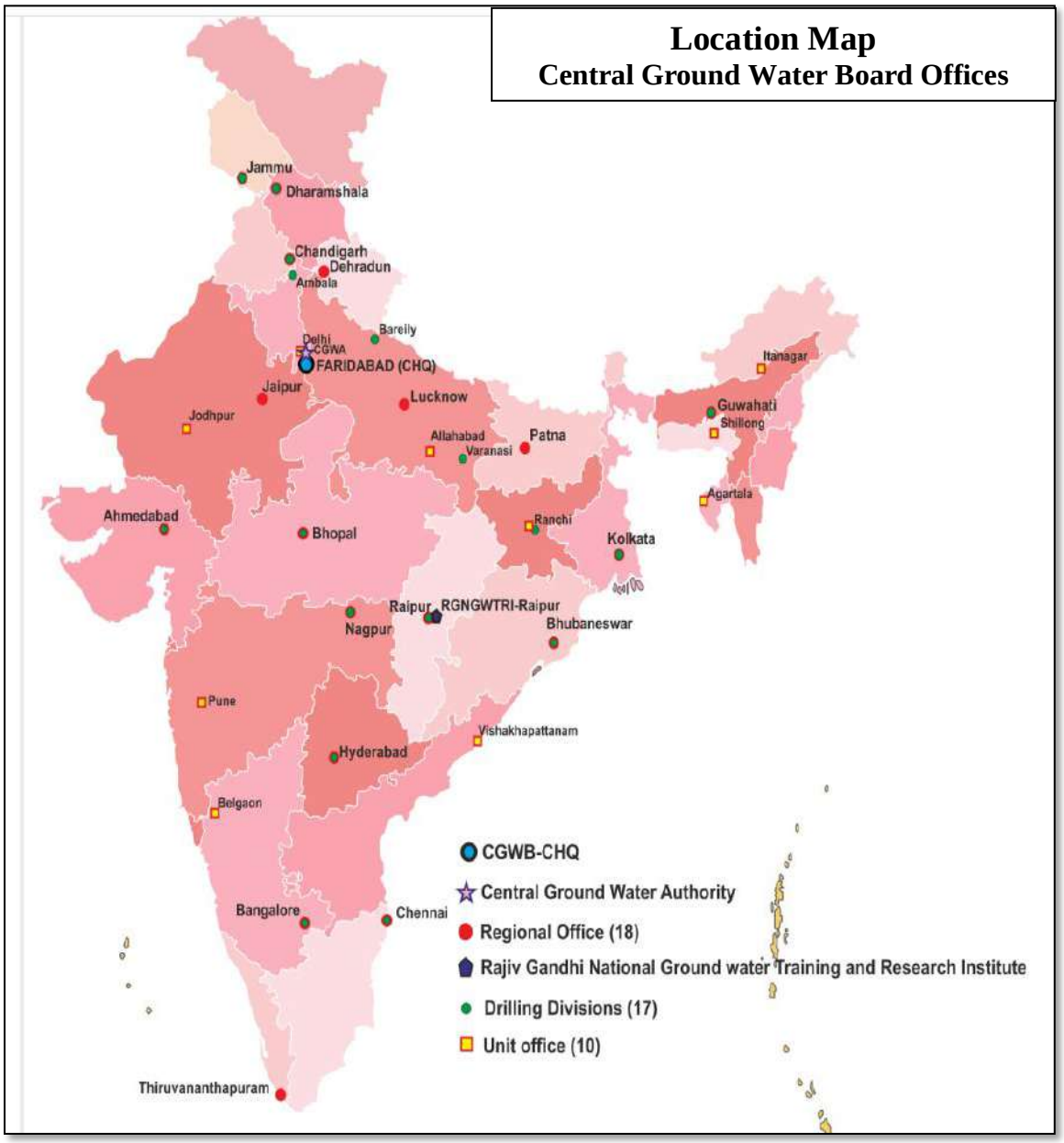


Fig 2. Location and Jurisdiction of the CGWB offices

1.6 Major Activities of Central Ground Water Board

Activities of CGWB involve scientific studies for monitoring assessment and management of ground water resources of the country.

Central Sector Scheme 'Ground Water Management and Regulation (GWMR)' is the major scheme that CGWB is implementing. Major activities under the scheme include Aquifer mapping, Exploratory drilling, Monitoring of water level and quality, Resource assessment, Aquifer Rejuvenation, Ground water regulation through CGWA, Outreach activities etc. Rajiv

Gandhi National Ground Water Training and Research Instittue implements a three tiered training programme for training and capacity building of groundwater professionals and other stakeholders. In addition to above, CGWB also is an implementing agency of National Hydrology Project (NHP). CGWB also implements specific components of other schemes of DoWR, RD & GR like the Ground Water component of the PMKSY – HKKP scheme. Details of the activities and achievements of CGWB in the year 2023-24 are given in the subsequent chapters.

1.7 Aquifer systems of India

Aquifer System is an outcome of the extensive hydrogeological studies carried out by Central Ground Water Board (CGWB) include Hydrogeological information collected by CGWB through its survey, investigations and exploration were collated, analyzed and integrated to define the extent and characteristics of aquifer systems.

Based on the hydrogeological characteristics, the entire country has been classified into 14 Principal Aquifer Systems and 42 major aquifers.

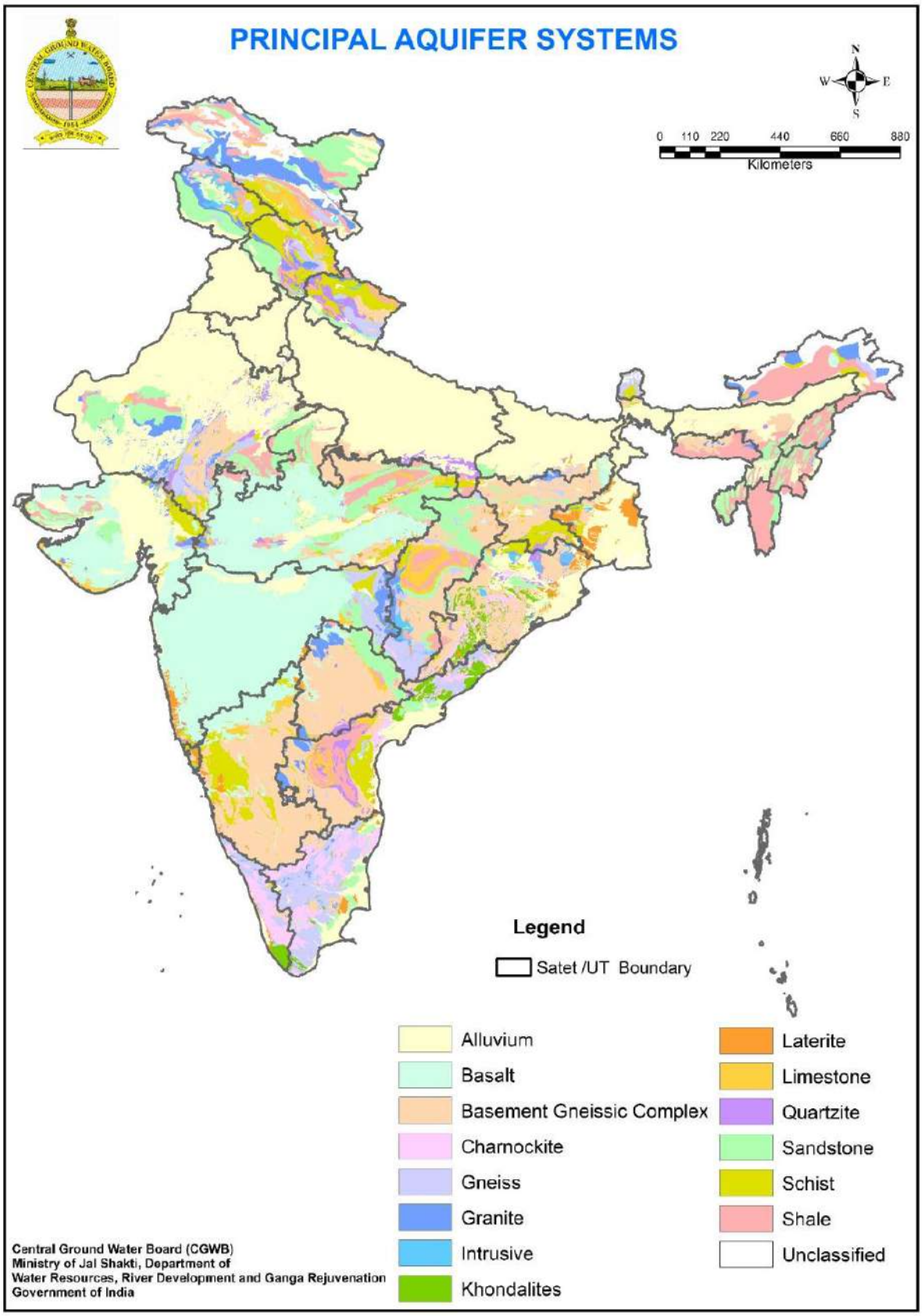


Fig 3: Principal Aquifers of India

2. Schemes, Projects, and Activities

2.1 Ground Water Management and Regulation (GWMR) Scheme

2.1.1 Introduction

Ground Water Management and Regulation scheme is a continuing Central Sector Scheme, which is being implemented since 2007-08 by Central Ground Water Board (CGWB). One of the major activities under the scheme is National Aquifer Mapping & Management (NAQUIM) Programme, under which it was targeted to cover ~25 lakh sq km of mappable area and it has been covered by 31st March 2023. In addition to above other activities like ground water level & quality monitoring, assessment of dynamic ground water resources, regulation and control of ground water withdrawal, demonstrative recharge projects etc. have also been implemented.
To ensure continuity of the activities, the scheme has been approved for implementation till 31st March 2026 and includes activities like monitoring, assessment and regulation of groundwater resources and strengthening of infrastructure for technological up gradation.

2.1.2 Major Activities under the Scheme

- Aquifer Mapping
- Interventions for Aquifer Rejuvenation and Springshed Mapping
- Construction of Piezometers
- Monitoring of ground water levels
- Monitoring of ground water quality
- Assessment of ground water resources
- Regulation and control of ground water extraction
- Outreach Activities

2.2 Public Investment Board

2.2.1 Introduction

In addition to the above ongoing activities under GWMR scheme, a project has been approved by the Public Investment Board (PIB) for data generation in National Aquifer Mapping and Management (NAQUlM). The project will be implemented as a part of the GWM&R Scheme during the period 2022-2026.

2.2.2 Major Activities envisaged under this project

- Construction of 7000 piezometers along with installation of 7000 Digital Water Level Recorders (DWLRs) with telemetry system

- Heli-borne surveys for high resolution mapping in ~ 3 lakh sq km area in arid parts of NW India.
- Data Generation for Aquifer Mapping through construction of Wells (1135 wells in 11 States)

2.3 Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI) - non-scheme

2.3.1 Introduction

Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI) is the training wing of Central Ground Water Board, Ministry of Jal Shakti, Govt. of India and functions as a `Centre of Excellence' with the national role of capacity building of CGWB,groundwater professionals from Central and State Government Organizations, Public Sector Undertakings, Non-Governmental Organizations, Academic institutions, and other stakeholders through three-tier training -Tier I (National Level), Tier II (State Level) and Tier III (Block level) trainings and is located in Sector 23, Naya Raipur, Chhattisgarh.

The institute was established at Raipur in the year 1997, for training and capacity building of groundwater professionals, and sub-professionals in core and applied fields of groundwater sciences. Due to varied hydrogeological settings in the country and excessive demands of groundwater, it has become imperative to train professionals on capacity building for the sustainable development of groundwater. The institute endeavors: (i) to be a center of excellence in Training, Research and Development in the groundwater sector; (ii) to provide training to groundwater professionals and sub-professionals in various fields of groundwater; (iii) to train NGOs, Panchayati Raj Institutions and other stakeholders in groundwater; (iv) to train various stakeholders for taking up groundwater monitoring and data collection work for various national groundwater schemes under the Participatory Ground Water Management Program (v) to undertake Research and Development works in Ground Water Sector.

2.4 National Hydrology Project (NHP)

2.4.1 Introduction

National Hydrology Project (NHP**)** is a continuation of Hydrology Project (HP)- I and HP- II, a central sector scheme of the Department of Water Resources, RD&GR, Ministry of Jal Shakti, Government of India with an aim to improve the extent, quality, accessibility of water resources information, decision support system for floods basin level resource assessment/ planning strengthen the capacity of targeted water resources professionals and management institutions in India, and a total outlay of Rs. 3679.7674 Crore (Rs. 3,640 Crore for NHP and Rs. 39.7674 Crore for the establishment of the National Water Informatics Centre (NWIC), a repository of Nation-wide Water Resources Data. The project spans over a period of 08 years starting from 2016-17 to 2023-24, which has been extended for a period of one-and-half years from April 2024 to Sep 2025.

2.5 Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)

2.5.1 Introduction

Pradhan Mantri Krishi Sinchayee Yojana- Har Khet Ko Paani- Ground Water (PMKSY-HKKP-GW) scheme, launched by Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti, envisages to provide irrigation facility for small and marginal farmers in areas having sufficient potential for future development of ground water.

The scheme provides financial assistance to States for assured ground water irrigation to small and marginal farmers with priority to SC/ST and women farmers. The funding pattern is in the ratio of 90:10 (C:S) in case of NE/Hilly areas and 60:40 (C:S) in case of other areas. The scheme is applicable only in areas having stage of ground water development less than 60%, average rainfall more than 750 mm rainfall and having shallow ground water levels (less than 15 m below ground level).

Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti has sanctioned 13 projects under PMKSY-HKKP-GW scheme since 2019 in 10 States namely Arunachal Pradesh, Assam, Gujarat, Nagaland, Manipur, Mizoram, Tripura, Tamil Nadu, Uttar Pradesh, and Uttarakhand. All the 13 projects have been successfully completed. Under these thirteen projects, construction of 29779 wells was targeted with the creation of 88679 Ha of command area benefitting 67930 small & marginal farmers.

During the period, an amount of Rs.29.71 crore has been released (01.01.2023 to 31.03.2024) to the projects in the States of Assam, Arunachal Pradesh, Gujarat, Manipur, Mizoram, Nagaland, Tripura, Tamil Nadu, Uttar Pradesh and Uttarakhand towards central assistance and 550 wells have been constructed creating additional command area of 11305 Ha, benefitting 1309 small & marginal farmers.

3. Ground Water Management and Regulation Scheme

3.1 Ground Water Level Monitoring

3.1.1 Introduction

Groundwater regime monitoring is one of the key activities of Central Ground Water Board (CGWB). Monitoring of groundwater regime is an effort to obtain information on groundwater level and chemical quality through representative sampling. The primary objective of groundwater monitoring is to record the response of various natural and anthropogenic stress on the groundwater regime which impacts the recharge and discharge parameters with reference to geology, climate, physiography, land use pattern and hydrologic characteristics on a regional scale. The key applications of the groundwater data acquired during regime monitoring in the country are:

- Volumetric estimation of groundwater resources.
- To assess the impact of groundwater recharge and draft on long term basis.
- To categorise area into overexploited, critical, semi-critical and safe based on long term water level trend analysis.
- In deciding the depth of water well drilling as well as depth of lowering the pumps.
- Help in design, implement and monitor the effectiveness of groundwater management, protection and conservation program
- To identify groundwater quality affected areas for taking remedial measures.
- To plan groundwater recharge interventions and study the impact in time and space.

CGWB currently has a network of 27163 (Dug Wells: 16516, Piezometers: 9552, Hand Pump: 913, spring: 182) groundwater observation wells throughout the Country. The state wise breakup is given below in table 1.

The groundwater regime monitoring started in the year 1969 by Central Ground Water Board. The groundwater levels are measured manually four times a year during the months of January, March/April/ May, August and November coinciding with crop season and onset of monsoon as well as to capture deepest and shallowest water level in the hydrological year.

Table 1: State/UT -Wise Status of Groundwater Monitoring Wells

Sl No	Name of the State/UT	Dug Well	Piezometer	Handpump	Spring	Total Number of Monitoring Stations (March 2024)
1	Andhra Pradesh	676	797	0	0	1473
2	Arunachal Pradesh	26	6	0	0	32
3	Assam	355	80	8	0	443
4	Bihar	796	120	0	0	916
5	Chhattisgarh	1044	269	5	0	1318
6	Goa	83	52	0	0	135
7	Gujarat	789	504	0	0	1293
8	Haryana	478	819	0	0	1297
9	Himachal Pradesh	138	55	0	24	217
10	Jharkhand	460	122	0	0	582
11	Karnataka	1360	931	0	0	2291
12	Kerala	1330	296	0	24	1650

13	Madhya Pradesh	1386	485	0	0	1871
14	Maharashtra	1779	296	0	0	2075
15	Manipur	4	0	0	2	6
16	Meghalaya	67	13	0	19	99
17	Mizoram	3	0	0	0	3
18	Nagaland	99	1	0	28	128
19	Odisha	1507	277	0	0	1784
20	Punjab	175	995	0	0	1170
21	Rajasthan	642	660	0	0	1302
22	Sikkim	0	4	0	0	4
23	Tamil Nadu	743	694	0	0	1437
24	Telangana	273	1008	0	0	1281
25	Tripura	97	18	0	0	115
26	Uttar Pradesh	978	486	0	0	1464
27	Uttarakhand	39	12	149	79	279
28	West Bengal	695	286	751	0	1732
	UTs					
1	Andaman & Nicobar	111	2	0	0	113
2	Chandigarh	1	22	0	0	23
3	Daman & Diu and Dadra & Nagar Haveli	38	5	0	0	43
4	Delhi	22	113	0	0	135
5	Jammu and Kashmir	313	106	0	6	425
6	Ladakh	0	0	0	0	0
7	Puducherry	9	18	0	0	27
	Total	**16516**	**9552**	**913**	**182**	**27163**

The database thus generated forms the basis for planning the groundwater development and management programme. This data is used for assessment of groundwater resources and changes in the regime consequent to various development and management activities.

3.2 Ground Water Quality Monitoring

3.2.1 Introduction

Central Ground Water Board has 16 Regional Chemical Laboratories to carry out chemical analysis of ground water samples. The Chemical laboratories are well equipped with sophisticated instruments like Inductively Coupled Plasma Mass-Spectrometer (ICP-MS), Atomic Absorption Spectrophotometer (AAS) etc. as detailed below in table 1. Ground water sampling for quality monitoring is done twice a year during the pre-monsoon and post-monsoon period. In addition to it ground water samples are also collected and analyzed as a part of other scientific studies. CGWB has also taken up sampling and analysis of ground water samples for Uranium content across the country.

Table 1: Status of Instruments in Regional Chemical Laboratories of CGWB

S. No.	Instrument/ Equipment	No. of Instruments/Equipment in 16 Regional Chemical Laboratory of CGWB
1	pH Meter	1-2 in each Laboratory
2	EC Meter	1-2 in each Laboratory
3	Flame- Photometer	1-2 in each Laboratory
4	UV/Visible-Spectro-photometer	1 in each Laboratory
5	Atomic Absorption Spectro-photometer	1 in each Laboratory (2 each at Lucknow & Kolkata)
6	Ion Chromatograph	1 in each Laboratory
7	Ion meter	1 in each Laboratory
8	ICP-MS	1 each in Lucknow, Chandigarh, Guwahati & Hyderabad laboratory
9	TOC	1 each in Lucknow, Chandigarh, Kolkata, Guwahati & Chennai laboratory
10	LSC	1 each in Lucknow, Chandigarh, Bengaluru, Nagpur laboratory
11	Ultra-Water Purification System	1 in each Laboratory
12.	Fluorimeter	1 in each Laboratory

In the AAP 2023-24, a total of 62603 groundwater samples have been analyzed by CGWB which included 39017 for Basic parameters, 23586 for Heavy Metals with Uranium. The details of water samples analyzed by different chemical laboratories during 2023 - 24 are presented in table 2 and its graphical presentation is shown in figure 1.

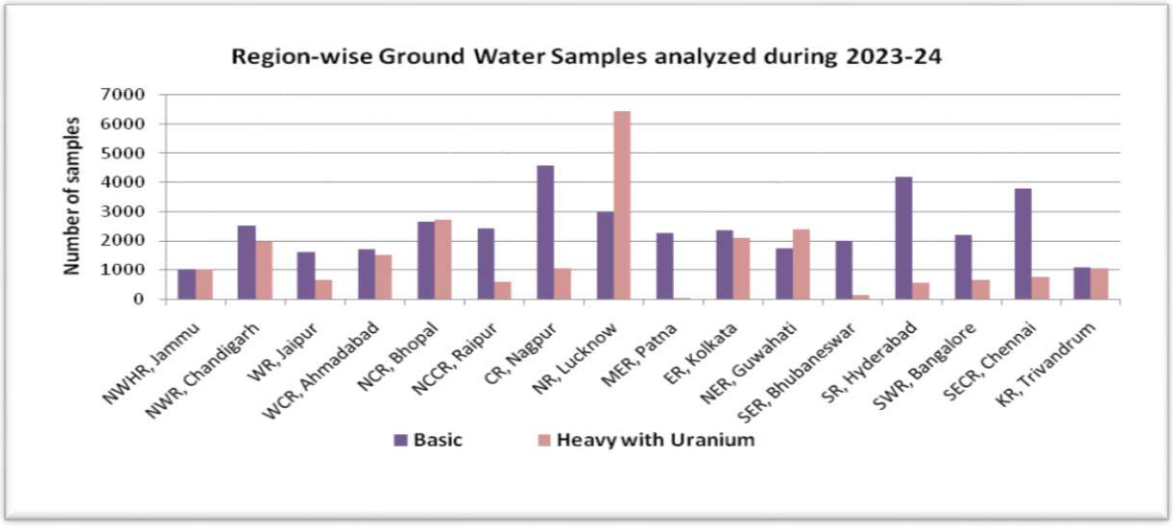


Fig. 1: Ground Water Samples analyzed during 2023-24

Table 2: Region wise Ground Water Samples analyzed during 2023-24

S. No.	Regional Labs	Number of Analyzed Samples								
		NHS Basic	NAQUIM 2.0 Basic	Other Basic	Total Basic	NHS Heavy Metals	NAQUIM 2.0 Heavy Metals	Other Heavy Metals	Total Heavy Metals	Total
1	NWHR, Jammu	315	371	325	1011	268	369	373	1010	2021
2	NWR, Chandigarh	1829	478	214	2521	1358	601	11	1970	4491
3	WR, Jaipur	1008	343	255	1606	0	400	261	661	2267
4	WCR, Ahmedabad	796	850	71	1717	649	848	2	1499	3216
5	NCR, Bhopal	1122	888	626	2636	1064	720	936	2720	5356
6	NCCR, Raipur	970	646	806	2422	214	347	31	592	3014
7	CR, Nagpur	2876	1492	197	4565	0	994	40	1034	5599
8	NR, Lucknow	1746	881	349	2976	2342	1617	2481	6440	9416
9	MER, Patna	1360	725	158	2243	0	0	44	44	2287
10	ER, Kolkata	1216	962	162	2340	1098	962	38	2098	4438
11	NER, Guwahati	829	731	167	1727	1095	945	338	2378	4105
12	SER, Bhubaneswar	1100	775	116	1991	0	114	31	145	2136
13	SR, Hyderabad	2237	1844	98	4179	0	552	0	552	4731

14	SWR, Bangalore	917	748	525	2190	171	354	141	666	2856
15	SECR, Chennai	1523	1936	351	3810	0	624	119	743	4553
16	KR, Trivandrum	364	565	154	1083	364	565	105	1034	2117
	Total	**20208**	**14235**	**4574**	**39017**	**8623**	**10012**	**4951**	**23586**	**62603**

3.2.2 Regional Chemical Laboratories

16 Regional Chemical Laboratories – NWHR Jammu, NWR Chandigarh, WR Jaipur, WCR Ahmadabad, NCR Bhopal, NCCR Raipur, CR Nagpur, NR Lucknow, MER Patna, ER Kolkata, NER Guwahati, SER Bhubaneswar, SR Hyderabad, SWR Bangalore, SECR Chennai and KR Trivandrum.

3.2.3 NABL accredited laboratories

11 NABL Accredited Laboratories - Chandigarh, Jaipur, Bhopal, Raipur, Nagpur, Lucknow, Kolkata, Guwahati, Bhubaneswar, Hyderabad and Chennai.

3.3 Geophysical Studies

3.3.1 Introduction

Geophysical investigations utilize non-invasive methodologies to yield indirect insights regarding the characteristics and arrangement of aquifers. These techniques allow researchers to gather valuable data without disturbing the subsurface environment, thereby preserving the integrity of the aquifer systems being studied. By employing various geophysical methods, scientists can effectively analyze the physical properties of aquifers, such as their porosity, permeability, and saturation levels. This approach not only enhances the understanding of groundwater resources but also aids in the management and protection of these vital water supplies, ensuring sustainable usage for future generations.

3.3.2 Surface and Sub-Surface Geophysical Studies

Geophysical studies employ non-invasive techniques, offering indirect insights into the disposition and characteristics of aquifers. The Central Ground Water Board (CGWB) possesses an in-house facility for various geophysical studies. Both surface and subsurface (well logging) geophysical techniques are extensively utilized in the quest for groundwater and the proper construction of water wells. The results of geophysical studies are amalgamated with hydrogeological investigations to establish a robust foundation.

Surface geophysical investigations play a crucial role in groundwater exploration, aquifer mapping, water supply investigations, recharge site selection, and specialized studies. These

investigations aid in delineating subsurface information regarding lithology, influencing the occurrence, distribution, and movement of groundwater. CGWB initiated the regular application of geophysical techniques for groundwater investigations in the seventies. Traditional methods such as Electrical Resistivity surveys, employed in both soft and hard rock formations, are standard in CGWB. They help delineate groundwater-bearing zones, pinpoint sites for borehole construction, and contribute to proposals for artificial recharge structures. Geophysical surveys also aid in delineating bedrock topography and sandy horizons of non-perennial channels. In recent times, Transient Electromagnetic techniques (TEM) have been introduced to identify sub-surface layer parameters like resistivity and thickness, similar to Electrical Resistivity surveys. TEM surveys, despite providing comparable information, require less time than conventional Electrical Resistivity surveys. CGWB is currently conducting advanced 2-D Imaging surveys using a Multi Electrodes Resistivity Meter for special studies and aquifer mapping.

Sub-surface borehole geophysics plays a crucial role in groundwater exploration, providing valuable insights into lithology, fractures, and water quality (Fresh/Saline). Its primary purpose is to delineate the subsurface disposition of aquifers. In drilled pilot boreholes, borehole-geophysical logging is employed to determine the character and thickness of various lithological and hydrogeological units. Different parts of India host saline or brackish water-bearing aquifers, often interspersed with fresh water-bearing aquifers. This information is vital for the proper placement of casing and screens in water-supply wells, ensuring effective water sourcing while addressing challenges associated with ground-water salinity.

A Region wise list of existing Geophysical Insturments is given below in table 1

Table 1: Region wise details of Geophysical Instruments in CGWB

Sl. No	Region	Geophysical Logger	Resistivity Meter	EM
1	ER-Kolkata	Uptron (18Y) Logger u-1	1.CRM Auto –C (7 y) 2. Res Meter(2 y)	TEM – 1 (New)
2	KR- Trivandrum	Nil	1. Terrameter (21 Y) 2. CRM-500 (15Y**)**	TEM – 1 (New)
3	NCCR- Raipur	Nil	SSR-MP-ATS (adv) (7Y)	TEM – 1 (New)
4	NCR - Bhopal	Nil	1. DDR-4 MP (20Y) 2. SSR-MP1 (18Y)	
5	WCR- Ahmadabad	Uptron (25 y) Logger	1.SSR-1 (7Y) 2.DDR-4 (25Y)	
6	NWHR- Jammu	Uptron (22 y) Logger	1.DDR-4 (18 Y) 2. Terrameter (18Y)	
7	NHR- Dharmshala	Nil		
8	UR - Dehradun	Nil	1.Terrameter SAS 300 (21Y)	
9	NER-Guwahati	Uptron (>27Y) Logger (+Gamma)	1. SSR-MP1 (13Y) 2. Terrameter (18 Y)	

10	CR- Nagpur	1.Uptron (27Y) Logger (+ Gamma), 2. Digital (25Y)Portable Logger (+ Gamma, Flow meter &Temp.) 3.Terrameter spot Logger (part of Terrameter)	1.Terrameter (19Y) 2.CRM – 20 (15Y)	
11	MER - Patna	Uptron (> 20 Y) Logger	1. CRM 500 – 40 W (Ranchi) (15Y) 2. CRM-Auto C – 40 W (5Y)	
12	SER - Bhubaneswar	Uptron (> 15 y) Logger	1. CRM 20 (5Y)	
13	NWR - Chandigarh	Uptron (27Y) Logger (+Gamma)	1. CRM – 500 (20Y)	
14	SR - Hyderabad	1.Upron (3no and 25 y) Logger (+Gamma)	1. DDR- 4–80 W (21Y) 2. SSR- MP1–80W (7Y) 3. DDR- 3 – 80 W (7Y) 4. Terrameter with manual logger (22Y) 5. Terrameter LS (6Y) 6. ERT System Terrameter LS–2D (6)	
15	SWR- Bangalore	Uptron (20Y) Logger(+Gamma)	1. SSR-MP-ATS - 100W – 2 no. (2Y) 2. Res Meter (5Y)	
16	SECR-Chennai	Uptron (19 Y) Logger	1. DDR – 3= (7Y) 2. CRM – 500- (15Y) 3. Adv WTS – 2 No (5Y)	
17	WR - Jaipur	Uptron (27Y) Logger (+Gamma), u-13	1. DDR- 4 – 80 W (19 Y) 2. CRM-Auto-C – 40 W (7 Y)	
18	NR - Lucknow	Uptron (21Y) Logger(+Gamma), u-14	1.Adv. Terrameter LS – 2D (6.5Y) 2.Adv Syscal R2 – 250 W (25 Y) 3. ERT system 4.Terrameter LS – 2D	
	Total Logger (17), two spot logger with resistivity meter	**1.Uptron - 14, (+8 gamma) (8 working)** **2.OYO potable-1 (+gamma, flow & Temp). (Not working)** **3.SAS log 200 – 2 (part of Terrameter), One is working**	**1. 33 working, 8 Repairable** **2. Imaging System- 2** **3. TEM- 9**	

During the AAP 2023-24, CGWB through its field officers carried out 2753 Vertical Electrical Sounding (VES), 3125 Transient Electro-magnetic (TEM) survey and 35 boreholes logging in NAQUIM 2.0, short term water supply investigations. A special study was also carried out using advanced technology (2D Imaging) for water logging issues from Pragati maidan to Jungpura, New Delhi.

Fig. 1: Geophysical Survey conducted at Delhi Golf Club, New Delhi

Table 2: Region wise progress of Gephysical Activities

Sr. No.	State/UT	Region	Annual Target (2023-24)	Progress as on 31.03.2024
1	Uttar Pradesh	NR	72	87
2	Delhi	SUO	30	21
3	Uttarakhand	UR	20	51
4	Andaman and Nicobar Islands	ER	25	25
5	West Bengal	ER	75	78
22	Bihar	MER	30	35
7	Jharkhand	MER	35	38
8	Chhattisgarh	NCCR	50	84
9	Assam	NER	50	51
10	Arunachal Pradesh	NER	20	18
11	Meghalaya	NER	10	11
12	Manipur	NER	0	0
13	Mizoram	NER	0	1
14	Nagaland	NER	0	0
15	Tripura	NER	20	21
16	Odisha	SER	200	201
17	Madhya Pradesh	NCR	230	198

18	Jammu and Kashmir	NWHR	50	51
19	Ladakh	NWHR	25	25
20	Haryana	NWR	260	280
21	Punjab	NWR	140	115
22	Gujarat	WCR	50	63
23	Rajasthan	WR	75	82
24	Maharashtra	CR	212	214
25	Kerala	KR	100	100
26	Lakshadweep	KR	0	22
27	Tamilnadu	SECR	250	290
28	Andhra Pradesh	SR	140	151
29	Telangana	SR	250	282
30	Goa	SWR	10	10
31	Karnataka	SWR	140	148
	Total		**2597**	**2753**

Table 3: Region wise achievements of TEM

	State/UT	Region	Annual Target (2023-24)	Achievement as on 31.03.2024
1	Uttar Pradesh	NR	350	408
2	West Bengal	ER	300	300
3	Bihar	MER	126	131
4	Jharkhand	MER	35	35
5	Chhattisgarh	NCCR	200	222
6	Rajasthan	WR	300	314
7	Kerala	KR	250	250
8	Tamilnadu	SECR	625	709
9	Andhra Pradesh	SR	125	137
10	Telangana	SR	150	158
11	Goa	SWR	20	21
12	Karnataka	SWR	430	440
	Total		**2911**	**3125**

3.3.3 High resolution aquifer mapping and management in arid areas of India:

Central Ground Water Board (CGWB) has initiated high resolution mapping of aquifers using state of art technology heliborne geophysical survey in parts of the arid areas spread over the states of Rajasthan, Gujarat and Haryana. The study is aimed at establishing aquifer geometry,

demarcation of de-saturated and saturated aquifers, fresh/saline aquifers, identification of paleochannels network, identification of potential sites for ground water withdrawal and identification of sites for water conservation structures, MAR sites etc. The phase-I heliborne survey project of an area of 97165 sq.km and 40313line km flight line has been completed covering 92 Blocks in Rajasthan, Gujarat and Haryana states. In Rajasthan, 47 Blocks spanning in parts of 8 districts were addressed, with 1 Block each in Bikaner, Sikar, and Jaisalmer districts, 7 Blocks in Churu district, 9 blocks in Ganganagar district, 8 blocks in Jalore district, 10 blocks in Pali district, and 10 blocks in Jodhpur district. In Gujarat, the areas covered included 11 blocks in Rajkot, 6 blocks in Jamnagar, 5 blocks in Morbi, 10 blocks in Surendranagar, and 4 blocks in Devbhumi Dwarka districts, while in Haryana, 5 blocks in Kurukshetra and 4 blocks in Yamunanagar districts were surveyed. The deliverables such as granular zones, saturated, de-saturated, saline/fresh aquifers, suitable sites for drilling, expected water level, suitable zones for artificial recharge and type of recharge structures at Gram Panchayat level are being extracted along the flight line from the heliborne survey results. Gram panchayat level deliverable extracted for 13 blocks out of 92 blocks covering 529 Gram panchayats up to March 2023. The remaining 79 blocks covering 4141 Gram panchayats is being proposed to be completed by March 2025. CGWB has proposed airborne geophysical survey of 2.71 lakh sq. km. area in phase-II under PIB project in arid region of northwest India in parts of Rajasthan, Punjab, Haryana, Gujarat and Himachal Pradesh having water stressed area.

3.4 Exploratory Drilling

3.4.1 Introduction

Ground Water Exploration aided by drilling is one of the major activities of the Board. It is aimed at delineation of aquifers in different hydrogeological setups and determination of their hydraulic parameters. The exploratory drilling operations have enabled demarcation of aquifers both in lateral and vertical extensions and evaluation of various aquifer parameters, designing of suitable structures and assessment of their yield potential in various hydrogeological settings. These studies have helped in identifying areas worthy for future ground water development. Ground Water exploration contributes to a large extent in guiding the States to implement ground water development schemes.

Groundwater exploration is being carried out by the Board through its fleet of 76 drilling rigs (23 Direct Rotary, 51 Down the Hole and 02 Dual Rotary). The region/division/state wise deployment of rigs during AAP 2023-24 is mentioned in table 1 and the division-wise details of pumping test units is given in table 2.

During the year 2023-24, Central Ground Water Board under its in-house Ground Water Exploration programme constructed 800 wells (EW- 288, OW- 85, Pz- 427) (table 3). Out of 288 exploratory wells, 235 wells were constructed in hard rock, 46 wells in alluvium and 7

wells in bouldary formations including 29 exploratory wells constructed in Tribal areas of the country.

The details of State/ Division / Region wise distribution of boreholes drilled/ completed during the year is presented in Table 4(a) & 4(b).

The Board since inception has drilled 46562 bore holes (including 9023 bore holes through outsourcing), as on 31.03.2024, to identify worthy areas for ground water development in the country. The statement showing State-wise distribution of boreholes drilled/completed till March, 2024 in the country is presented in table 5.

Table 1: Region/Division/State wise Deployment of Rigs during AAP 2023-24

Region	Division	State	Number Of Rigs			
			DR	DTH	DUR	T
NWHR,Jammu	VIII,Jammu	Jammu & Kashmir	0	2	1	**3**
NWR, Chandigarh	II,Ambala	Punjab	1	0	0	**1**
		Haryana	2	0	0	**2**
WR,Jaipur	XI,Jodhpur	Rajasthan	2	3	0	**5**
WCR,Ahmedabad	I,Ahmedabad	Gujarat	4	2	0	**6**
NCR,Bhopal	XII,Bhopal	Madhya Pradesh	0	4	0	**4**
NCCR,Raipur	XIII,Raipur	Chhattisgarh	0	4	0	**4**
CR,Nagpur	VI,Nagpur	Maharashtra	0	5	0	**5**
NR,Lucknow	III,Varanasi	Eastern U.P	3	2	0	**5**
	XVI,Bareilly	Western U.P	3	0	0	**3**
MER,Patna	V,Ranchi	Bihar	1	1	0	**2**
		Jharkhand	0	3	0	**3**
ER,Kolkatta	XV,Kolkata	West Bengal	3	2	0	**5**
NER,Guwahati	VII,Guwahati	Assam	2	1	0	**3**
		Meghalaya	0	1	0	**1**
SER,Bhubaneswar	X,Bhubaneswar	Orissa	1	4	0	**5**
SR,Hyderabad	IX, Hyderabad	Andhra Pradesh	0	1	0	**1**
		Telangana	0	4	0	**4**
SWR,Bangalore	XIV,Bangalore	Karnataka	0	4	0	**4**
SECR,Chennai	IV,Chennai	Tamil Nadu	1	4	0	**5**
KR,Trivendrum	IV,Chennai	Kerala	0	1	0	**1**
	XIV, Bangalore		0	1	0	**1**
UR,Dehradun	XVI,Bareilly	Uttarakhand	0	0	1	**1**
NHR,Dharamshala	XVII,Dharamshala	Himachal Pradesh	0	2	0	**2**
Total			**23**	**51**	**2**	**76**

Abbreviation :

DR : Direct Rotary **DUR :** Dual Rotary **DTH :** Down The Hole

Table 2: Status of Division wise Pumping Units of CGWB

Division	No. of Pumping Units
I.Ahmedabad	**1**
II.Ambala	**1**
III.Varanasi	**2**
IV.Chennai	**1**
V.Ranchi	**3**
VI.Nagpur	**2**
VII.Guwahati	**2**
VIII.Jammu	**1**
IX.Hyderabad	**1**
X.Bhubneshwar	**2**
XI.Jodhpur	**2**
XII.Bhopal	**1**
XIII.Raipur	**1**
XIV. Bengaluru	**2**
XV.Kolkata	**2**
XVI.Bareilly	**2**
XVII. Dharamshala	**0**

Table 3: State wise Wells constructed by CGWB during 2023-24 (till 31th Mar., 2024)

State	Tentative Target 2023-24				Cumulative Achiev2023-24 (Since April 2023)			
	EW	OW	PZ	T	EW	OW	PZ	T
West Bengal	5	5	18	**28**	9	7	12	**28**
Bihar	12	0	9	**21**	7	3	8	**18**
Jharkhand	7	0	6	**13**	2	4	11	**17**
Assam	8	0	8	**16**	7	0	6	**13**
Arunachal Pradesh	0	0	0	**0**	0	0	0	**0**
Meghalaya	3	0	3	**6**	2	0	2	**4**
Manipur	0	0	0	**0**	0	0	0	**0**
Nagaland	0	0	0	**0**	0	0	0	**0**
Tripura	0	0	0	**0**	0	0	0	**0**
Odisha	24	10	16	**50**	45	8	2	**55**
Chhatisgarh	7	3	35	**45**	8	8	30	**46**
Uttar Pradesh	24	0	30	**54**	13	12	31	**56**
Uttarakhand	2	0	0	**2**	1	0	0	**1**

New Delhi	0	0	0	**0**	0	0	0	**0**
Punjab	3	3	4	**10**	3	3	1	**7**
Haryana	3	3	12	**18**	4	4	6	**14**
Chandigarh	0	0	0	**0**	0	0	0	**0**
Jammu & Kashmir	0	0	15	**15**	0	0	15	**15**
Himachal Pradesh	3	3	8	**14**	0	0	14	**14**
Rajasthan	4	2	44	**50**	5	2	44	**51**
Gujarat	9	6	29	**44**	8	3	33	**44**
Daman & Div	0	0	0	**0**	0	0	0	**0**
Madhya Pradesh	18	6	26	**50**	23	2	25	**50**
Maharashtra	22	6	50	**78**	24	6	49	**79**
Dadra & Nagar Haveli	0	0	0	**0**	0	0	0	**0**
Andhra Pradesh	14	6	8	**28**	26	3	11	**40**
Telangana	14	6	28	**48**	41	10	51	**102**
Tamil Nadu	33	10	25	**68**	46	3	31	**80**
Puducherry (UTP)	0	0	0	**0**	0	0	0	**0**
Karnataka	8	4	36	**48**	11	4	30	**45**
Kerala	0	0	24	**24**	3	3	15	**21**
Total	**223**	**73**	**434**	**730**	**288**	**85**	**427**	**800**

Table 4(a) Division- wise wells constructed by CGWB during 2023-24

Division	Target 2023-24				Cumulative Achievement 2023-24 (since April 2023)				Achievement (%)
	EW	OW	PZ	T	EW	OW	PZ	T	
I.Ahmedabad	9	6	29	**44**	8	3	33	**44**	**100.00%**
II.Ambala	6	6	16	**28**	7	7	7	**21**	**75.00%**
III.Varanasi	12	0	27	**39**	6	6	28	**40**	**102.56%**
IV.Chennai	33	10	37	**80**	46	3	43	**92**	**115.00%**
V.Ranchi	19	0	15	**34**	9	7	19	**35**	**102.94%**
VI.Nagpur	22	6	50	**78**	24	6	49	**79**	**101.28%**
VII.Guwahati	11	0	11	**22**	9	0	8	**17**	**77.27%**
VIII.Jammu	0	0	15	**15**	0	0	15	**15**	**100.00%**
IX.Hyderabad	28	12	36	**76**	67	13	62	**142**	**186.84%**
X. Bhubneshwar	24	10	16	**50**	45	8	2	**55**	**110.00%**
XI.Jodhpur	4	2	44	**50**	5	2	44	**51**	**102.00%**
XII.Bhopal	18	6	26	**50**	23	2	25	**50**	**100.00%**
XIII.Raipur	7	3	35	**45**	8	8	30	**46**	**102.22%**
XIV. Bangalore	8	4	48	**60**	14	7	33	**54**	**90.00%**
XV.Kolkata	5	5	18	**28**	9	7	12	**28**	**100.00%**
XVI.Bareilly	14	0	3	**17**	8	6	3	**17**	**100.00%**
XVII. Dharamshala	3	3	8	**14**	0	0	14	**14**	**100.00%**

Total	**223**	**73**	**434**	**730**	**288**	**85**	**427**	**800**	**109.59%**

Table 4(b). Region-wise wells constructed by CGWB during 2023-24

Region	**Tentative Target 2023-24**				**Achievement 01.04.23 to 31.03.2024**				**% Achievement**
	EW	**OW**	**PZ**	**T**	**EW**	**OW**	**PZ**	**T**	
NWHR. Jammu	0	0	15	**15**	0	0	15	**15**	100%
NWR. Chandigarh	6	6	16	**28**	7	7	7	**21**	75%
WR. Jaipur	4	2	44	**50**	5	2	44	**51**	102%
WCR. Ahmedabad	9	6	29	**44**	8	3	33	**44**	100%
NCR. Bhopal	18	6	26	**50**	23	2	25	**50**	100%
NCCR. Raipur	7	3	35	**45**	8	8	30	**46**	102.22%
CR. Nagpur	22	6	50	**78**	24	6	49	**79**	101.28%
NR. Lucknow	24	0	30	**54**	13	12	31	**56**	103.70%
MER. Patna	19	0	15	**34**	9	7	19	**35**	102.94%
ER. Kolkata	5	5	18	**28**	9	7	12	**28**	100%
NER. Guwahati	11	0	11	**22**	9	0	8	**17**	77.27%
SER. Bhubaneshwar	24	10	16	**50**	45	8	2	**55**	110%
SR. Hyderabad	28	12	36	**76**	67	13	62	**142**	186.84%
SWR. Bangalore	8	4	36	**48**	3	4	27	**34**	70.83%
SECR. Chennai	33	10	25	**68**	46	3	31	**80**	117.64%
KR. Trivandrum	0	0	24	**24**	11	3	18	**32**	133.33%
UR. Dehradun	2	0	0	**2**	1	0	0	**1**	50%
NHR. Dharamshala	3	3	8	**14**	0	0	14	**14**	100%
Total	**223**	**73**	**434**	**730**	**288**	**85**	**427**	**800**	**109.58%**

Table 5: State/UT wise cumulative Boreholes Drilled by CGWB till 31st March 2024

S No.	State /UT	EW	OW	PZ	T	EW	OW	PZ	SH	DW	Total	TOTAL (I+II)
		(I) Through Outsourcing (Contractual)				(II) Through Departmental Rigs						
A.	**STATES**											
1	Andhra Pradesh	456	122	0	**578**	995	445	353	9	4	**1806**	**2384**
2	Arunachal Pradesh	0	0	0	**0**	46	10	0	1	1	**58**	**58**
3	Assam	17	20	0	**37**	473	214	78	16	42	**823**	**860**
4	Bihar	151	39	0	**190**	333	199	88	10	514	**1144**	**1334**
5	Chhattisgarh	414	15	105	**534**	906	296	224	0	28	**1454**	**1988**
6	Goa	13	1	0	**14**	58	18	14	0	31	**121**	**135**
7	Gujarat	392	116	0	**508**	1196	557	555	27	255	**2590**	**3098**
8	Haryana	75	55	80	**210**	415	278	251	23	170	**1137**	**1347**
9	Himachal Pradesh	0	0	0	**0**	272	49	38	1	0	**360**	**360**
10	Jammu & Kashmir	21	0	0	**21**	503	124	65	8	114	**814**	**835**
11	Jharkhand	275	28	0	**303**	507	258	80	4	71	**920**	**1223**
12	Karnataka	804	146	0	**950**	1654	746	435	7	5	**2847**	**3797**
13	Kerala	10	0	0	**10**	616	217	264	16	13	**1126**	**1136**
14	Madhya Pradesh	725	57	80	**862**	1474	752	231	8	149	**2614**	**3476**
15	Maharashtra	92	2	88	**182**	1848	584	261	2	166	**2861**	**3043**
16	Manipur	0	0	0	**0**	29	14	1	0	2	**46**	**46**
17	Meghalaya	0	0	0	**0**	122	40	4	2	8	**176**	**176**
18	Mizoram	0	0	0	**0**	3	3	0	0	0	**6**	**6**
19	Nagaland	0	0	0	**0**	15	6	1	0	3	**25**	**25**
20	Orissa	575	40	67	**682**	1814	461	177	21	191	**2664**	**3346**
21	Punjab	121	105	0	**226**	235	218	124	20	14	**611**	**837**
22	Rajasthan	809	195	0	**1004**	1525	568	642	93	591	**3419**	**4423**
23	Sikkim	0	0	0	**0**	31	9	0	0	0	**40**	**40**
24	Tamil Nadu	425	206	179	**810**	1475	454	449	13	93	**2484**	**3294**
25	Tripura	0	0	0	**0**	64	31	1	5	22	**123**	**123**
26	Telangana	373	78	0	**451**	837	544	618	5	27	**2031**	**2482**
27	Uttarakhand	28	12	0	**40**	77	7	4	1	129	**218**	**258**
28	Uttar Pradesh	694	274	0	**968**	1169	798	258	40	501	**2766**	**3734**
29	West Bengal	263	80	100	**443**	616	317	210	12	82	**1237**	**1680**
TOTAL(A)		**6733**	**1591**	**699**	**9023**	**19308**	**8217**	**5426**	**344**	**3226**	**36521**	**45544**

B.	UNION TERRITORIES											
1	Andaman & Nicobar	0	0	0	**0**	46	13	0	1	0	**60**	**60**
2	Chandigarh	0	0	0	**0**	9	17	14	2	15	**57**	**57**
3	Dadra & NagarHaveli	0	0	0	**0**	14	1	0	0	0	**15**	**15**
4	Delhi	0	0	0	**0**	149	64	196	13	380	**802**	**802**
5	Daman & Diu	0	0	0	**0**	0	0	7	0	0	**7**	**7**
6	Pondicherry	0	0	0	**0**	30	20	8	5	14	**77**	**77**
Total(B)		**0**	**0**	**0**	**0**	**248**	**115**	**225**	**21**	**409**	**1018**	**1018**
GRAND TOTAL (A+B)		**6733**	**1591**	**699**	**9023**	**19556**	**8332**	**5651**	**365**	**3635**	**37539**	**46562**

3.4.2 Pumping Test of Exploratory Well

After construction, a well's specific capacity is increased to achieve optimum well life and stop sand from pouring into the well. In order to develop effective well design, yield capability evaluation, and well spacing criteria, pumping experiments are then carried out to evaluate aquifer properties, such as transmissivity and storage coefficient, as well as well parameters, such as specific capacity and well efficiency. Total of 143 tubewells were tested for their aquifer parameters during the year 2023-24 (table 6).

Table 6: Status of Pumping Test of Wells by CGWB in 2023-24

DIVISION	STATE	No. of wells tested for aquifer parameters	Balance No. of wells to be tested (Backlog)
I.Ahmedabad	Gujarat	2	**44**
II.Ambala	Haryana	1	**3**
	Punjab	1	**8**
	Delhi	0	**0**
	Chandigarh	0	**0**
III.Varanasi	Uttar Pradesh	13	**105**
IV.Chennai	Tamilnadu	4	**24**
	Kerala	0	**0**
V.Ranchi	Bihar	2	**18**
	Jharkhand	10	**50**
VI.Nagpur	Maharashtra	16	**11**
VII.Guwahati	Assam	11	**31**
	Meghalya	3	**2**
	Tripura	0	**2**
	Arunachal Pradesh	0	**3**
VIII.Jammu	Jammu & Kashmir	1	**8**
IX.Hyderabad	Andhra Pradesh	5	**0**
	Telangana	4	**7**

X.Bhubneshwar	Odisha	11	**16**
XI.Jodhpur	Rajasthan	8	**18**
XII.Bhopal	Madhya Pradesh	8	**14**
XIII.Raipur	Chhattisgarh	10	**24**
XIV. Bengaluru	Karnataka	8	**2**
	Kerala	0	**3**
XV.Kolkata	West Bangal	13	**12**
XVI.Bareilly	Uttar Pradesh	12	**30**
	Uttarakhand	0	**4**
XVII. Dharamshala	Himachal Pradesh	0	**0**
	TOTAL	**143**	**439**

3.4.3 High Yielding Wells

Ground Water Exploration is carried to study the sub-surface hydrogeological set up of the aquifer to evaluate aquifer parameters. During 2023-24, CGWB under its scientific exploratory drilling programme has explored high yielding aquifers in various parts of the country based on hydrogeological studies coupled with remote sensing and geophysical techniques. Such studies will help in identifying similar ground water sources in other parts of the state having similar hydrogeological conditions and in guiding the state agencies to adopt the follow up action with regard to ground water development for drinking water supply and meeting other demands. High Yielding Wells explored during 2023-24 are indicated in table 7 below. During the year 2023-24, the Board has constructed 133 high yielding wells as detailed below:

Table 7: Region wise details of high yielding wells constructed during 2023-24

Sr no	Region	Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
1	**NWR Chandigarh**	None	Considering discharge >15 lps as high yielding			
2	**NCR Bhopal**	Jaithal Tek EW	Ujjain	23.1707	75.4931	4.5
3		Bakaniya EW	Ujjain	23.2669	75.8502	3.3
4		Bakaniya OW	Ujjain	23.2669	75.8502	7.5
5	**NCCR Raipur**	Tirga EW	Durg	21.084	81.168	5.1
6		Dumardih	Durg	21.1311	81.3913	4.54
7		Mahmara	Durg	21.180141	81.249577	3.252
8		Anjora (Dhaba)	Durg	21.279939	81.242223	10
9		Basin	Durg	21.292302	81.356814	3.87
10		Nankatti	Durg	21.334964	81.336897	4.69
11		Bhoyna	Dhamtari	20.663714	81.587962	28.16

Sr no	Region	Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
12		Bhoyna	Dhamtari	20.663714	81.587962	18.39
13		Sankara	Dhamtari	20.754748	81.621664	15.17
14		Sankara	Dhamtari	20.754748	81.621664	12.5
15		Kasawahi	Dhamtari	20.627235	81.491342	10.066
16		Kasawahi	Dhamtari	20.627235	81.491342	3.25
17		Aroud	Dhamtari	20.666336	81.637204	3.25
18		Balodgahan	Balod	20.648235	81.465996	14.3
19		Balodgahan	Balod	20.64816	81.466054	14.3
20		Jamaruwa EW	Balod	22.441319	82.644432	15
21		Jamaruwa OW	Balod	20.68542	81.271937	3.2
22		Dhaneli EW	Balod	20.68542	81.271937	3.51
23		Indira nagar stadium	Korba	20.74459	81.409018	7
24		Chikhali	Raipur	21.3635092	81.618365	10
25	**UR Dehradun**	Baldiyaparao, Ramnagar	Nainital	29.404732	79.129653	Artesian well discharge of 4.42 lps
26	**NR Lucknow**	Siyara Santha	Sant Kabir Nagar	26.724507	83.040716	17.9
27		Nath Nagar Block	Sant Kabir Nagar	26.638182	83.028432	34.45
28		Hainsar Bazar Block Office	Sant Kabir Nagar	26.5459	83.047	34.43
29		Bedauli	Siddharthnagar	27.412717	82.883285	18.43
30		BTC College,Narkatha	Siddharthnagar	27.192777	82.917609	17.9
31		Devpura(Paharipurwa) [Hardrock]	Mirzapur	24.872069	82.747773	7.68
32		Ghuwas (Mukkha) [Hardrock]	Sonbhadra	24.733476	82.700172	13.48
33		Bedaur Primary Vidyalaya Campus [Hardrock]	Mirzapur	25.02607	82.85683	4.42
34		Tenduhar [Hardrock]	Sonbhadra	24.675969	82.557461	6.8
35		Kanhari [Hardrock]	Sonbhadra	24.749292	82.61066	4.42

Sr no	Region	Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
36		Vishunpuri Colony,Palia	Lakhimpur Kheri	28.541113	80.49139	27.13
37		Adlishpur, Dahurahara	Lakhimpur Kheri	27.90817	81.13248	34.56
38		Shobhapur,Pilana	Baghpat	28.9897	77.36345	17.98
39		Titrodha,Binauli	Baghpat	29.04106	77.375981	22
40		Shahpur Banganga, Binauli	Baghpat	29.138537	77.447556	12.5
41		CCS Rajkiya Mahavidyalaya, Chapprauli	Baghpat	29.205397	77.165547	25.8
42		CSA Campus, Kalyanpur	Kanpur Nagar	26.489532	80.306616	12.6
43	**WCR Ahmedabad**	Chokari-Pz	Vadodara	22.22095	72.93143	14
44		Kujad-EW	Ahmedabad	23.0033	72.74783	12
45		Yashvantpura-PZ	Mehsana	23.06333	72.31684	7.4
46		Kujad-OW	Ahmedabad	23.0033	72.74783	7
47		Latipura-OW	Vadodara	22.22373	73.05943	34
48		Patan Vav-Pz	Rajkot	21.6438	70.26812	3.44
49		Faddang -Pz	Rajkot	22.25536	71.01489	5.29
50		Jhilariya-Pz	Rajkot	22.37756	70.52656	5
51		Khakhra-Pz	Jamnagar	22.51772	70.60922	4.3
52		Nikava-Pz	Jamnagar	22.19702	70.54202	6.5
53		Latipura-OW	Vadodara	22.22373	73.05943	28.29
54	**SR Hyderabad**	Sivarampally Pz-I	Rangareddy	17.327545	78.432985	7.43
55		Sivarampally Pz-II	Rangareddy	17.327543	78.433078	10.76
56		Narasannapeta	Srikakulam	18.425256	84.053914	5.9
57		Kothwalguda	Rangareddy	17.2878806	78.3688755	3.3
58		Kacharam	Rangareddy	17.243317	78.303471	4.13
59		Kacharam	Rangareddy	17.243315	78.303473	4.13
60		Chinna Golkonda	Rangareddy	17.208781	78.393252	3.74
61		Rajendranagar	Rangareddy	17.34288	78.41033	4.25
62		Nagireddyguda	Rangareddy	17.297611	78.32755	9.8

Sr no	Region	Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
63		Nagireddyguda	Rangareddy	17.297613	78.32754	9.8
64		Manneguda EW	Rangareddy	17.251608	78.58508	4.25
65		Manneguda OW	Rangareddy	17.251719	78.585209	3.74
66		Regadi Doswada EW	Rangareddy	17.220942	78.094474	3.74
67		Regadi Doswada OW	Rangareddy	17.22084	78.09442	3.74
68		Pendurthi EW	Visakhapatnam	17.811	83.2085	4.25
69		Pendurthi OW	Visakhapatnam	17.811	83.2085	4.26
70		Kancharapalem EW	Visakhapatnam	17.7387	83.2696	6.7
71		Kancharapalem OW	Visakhapatnam	17.7387	83.2696	6.7
72		Islampeta EW	Visakhapatnam	17.5942	83.1558	3.74
73		Islampeta OW	Visakhapatnam	17.5942	83.1558	3.74
74	**SECR chennai**	Pollikalipalayam	Tiruppur	11.03518	77.392599	7
75		Sivanmalai	Tiruppur	11.037143	77.53172	3
76		Kalanjikattupudur	Tiruppur	10.967811	77.54201	8
77		Kombaithozhu	Theni	9.7128	77.4421	18.8
78		Varasanadu	Theni	9.7272	77.5309	16.3
79		Varasanadu OW	Theni	9.7272	77.5309	18.8
80		Thumakundu	Theni	9.68229	77.48302	3.3
81		Athangaraipatti	Theni	9.8877	77.4987	3.3
82		Muthalamparai	Theni	9.80033	77.59812	16.3
83		Attayampatty	Salem	11.5358	78.03882	14.1
84		Attayampatty OW	Salem	11.5358	78.03882	6.8
85		Chennagiri	Salem	11.536434	78.06214	25
86		Chennagiri OW	Salem	11.536434	78.06214	12
87		Pichampalayam	Salem	11.54393	78.0716	5.53
88		Kongupatty	Salem	11.856457	77.990708	18.9
89		Ramapuram	Salem	11.591383	78.010961	4.36
90		Chinnatirupathy	Salem	11.833425	78.028782	5.53

Sr no	Region	Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
91		Thindamangalam	Salem	11.742978	78.00819	3.34
92		Pannapatty	Salem	11.742978	78.00819	8.4
93		Pulavari	Salem	11.749971	78.106372	12
94	**SWR Bangalore**	Chandrappa Circle, Govt. High School EW	Bengaluru Urban	12.912551	77.376358	7.45
95		Chandrappa Circle, Govt. High School OW	Bengaluru Urban	12.9124279	77.376178	9.9
96		Kadachikanahalli, Govt. Lower Primary School	Chikkaballapur	13.462814	77.575339	6.8
97		PHC Sasalu	Bangalore Rural	13.404152	77.385259	12.76
98		PHC Arodi	Bangalore Rural	13.483826	77.405248	9.03
99		Padesur (EW), Govt. Primary School, Navalgund Taluk	Dharwad District	15.5562	75.3235	4.4
100		Padesur (OW), Govt. Primary School, Navalgund Taluk	Dharwad District	15.556	75.3232	5.5
101		APMC (EW), APMC Premises, Navalgund Taluk	Dharwad District	15.5498	75.3548	3.3
102		Tadhal (EW), Govt. Primary School, Navalgund Taluk	Dharwad District	15.6672	75.4478	8.3
103	**SER Bhubaneshawar**	Hajipur	Jagatsinghpur	20.3487	86.03628	18
104		Tasarada	Mayurbhanj	21.838047	86.754341	15
105		Tasarada	Mayurbhanj	21.838047	86.754341	10
106		Khuntapal	Mayurbhanj	21.8921134	86.9423712	10
107		Chancha	Mayurbhanj	21.91196	86.765144	7
108		Kalanga	Khurda	20.1926215	85.2344999	4.36
109		Dubri	Jajpur	21.003231	85.983006	11.23
110		Dari	Mayurbhanj	21.8746	85.9668	4.58
111		Chadheibhol (EW)	Mayurbhanj	21.819967	85.829765	12

Sr no	Region	Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
112		Chadheibhol(OW)	Mayurbhanj	21.819967	85.829765	13.5
113		Bisoi	Mayurbhanj	22.160724	86.410107	4.5
114		Jharhajamuda	Mayurbhanj	21.8317	86.801533	7
115		Bautiberha(EW)	Mayurbhanj	22.145376	86.349289	9.5
116		Bautiberha(OW)	Mayurbhanj	22.145376	86.349289	10
117		Golamal	Sambalpur	21.649668	84.09433	7.9
118		Golamal(OW)	Sambalpur	21.649668	84.09433	7.9
119		Manada	Mayurbhanj	22.079023	86.234395	7.5
120		Sagunabasa	Mayurbhanj			8
121		Jharmunda	Sambalpur	21.667342	84.023041	16.4
122		Boulasahi	Nayagarh	20.05143	85.19021	3.353
123		Sanbisol	Mayurbhanj	21.52886	86.550697	11.048
124	**ER Kolkata**	Chanak Vivekananda Bani Mandir High School	Murshidabad	24.231537	88.096487	14.50
125		Alladi Ishwar Chandra MSK	Paschim Bardhaman	23.807015	86.875726	10.12
126		Nabagram Gram Panchayat, Chanak	Murshidabad	24.216364	88.111421	10.63
127		Bahadurpur Rural Hospital	Paschim Bardhaman	23.6893	87.1427	10.12
128	**KR Trivanthpuram**	Tholur Panchayath	Thrissur	10.57556	76.12889	10.12
129		Kolazhi Panchayath	Thrissur	10.57805	76.19917	9.24
130		Vallachira Panchayath	Thrissur	10.44111	76.23028	3.34
131		Vengola Panchayath	Ernakulam	10.08360	76.47224	21.72
132		Onamkulam-Vengola Panchayath	Ernakulam	10.07592	76.46495	19.00
133		Thuravur Panchayath	Ernakulam	10.23185	76.43724	14.13
134		Chengalai Panchayath	Kannur	12.0444	75.4905	6.8

Sr no	Region	Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
135		Padiyoor Panchayath	Kannur	12.0002	75.6222	10.96
136		Pudussery Panchayath	Palakkad	10.8572	76.8283	6.83
137	**SUO Ranchi**	Phulwariya, Domchanch (EW)	Koderma	24.46346	85.73931	7.8
138		Demotand, (OW)	Hazaribagh	23.917432	85.3466	4.1
139		Salaidih, (EW)	Koderma	24.47257	85.64261	4.1
140		Salaidih, (OW)	Koderma	24.47257	85.64261	7.1
141	**MER, Patna**	Bhuiyanbigha,Block-Madanpur	Aurangabad	24.720288	84.608037	4.64
142		Sarawak, Block-Rafiganj	Aurangabad	24.779772	84.647905	6.6
143		Jhikauriya, Block-Narhat	Nawada	24.81831	85.423532	5.25
144		Rupauli Sr. Secondary School, Block-Sarairanjan	Samastipur	25.775596	85.766172	18.93
145	**CR, Nagpur**	Abloli village, Guhaghar block (EW)	Ratnagiri	17.336323	73.346	25.05
146		Abloli village, Guhaghar block (OW)	Ratnagiri	17.336323	73.346	25.05
147		Khapa village, Saoner block (PZ)	Nagpur	21.424629	78.833405	3.17
148		Junapani village, Katol Block (EW)	Nagpur	21.152358	78.152358	12.81
149		Jatamkhora village, Saoner Block (EW)	Nagpur	21.475187	78.810231	3.17
150		Jatamkhora village, Saoner Block (OW)	Nagpur	21.475187	78.810231	7.76
151		Kothulana village, Saoner Block (EW)	Nagpur	21.419871	79.037022	5.94
152		Sarra village, Saoner block, (PZ)	Nagpur	21.571424	79.003246	14.88
153		Hivare village, Jath block (OW)	Sangli	17.101194	75.057277	4.43

Sr no	Region	Village / location of High discharge well	District	Lat	Long	Discharge in (LPS)
154		Jat village, Jath Block (OW)	Sangli	17.048436	75.207874	4.43
155		Hivare village, Jath block (EW)	Sangli	17.101179	75.057386	6.81
156		Nangole village, Kawatemahal block (PZ)	Sangli	17.032335	74.92134	5.94
157		Timerwadi teshing village (PZ)	Sangli	17.004089	74.810817	3.17
158		Benapur village (PZ)	Sangli	17.262724	74.743541	4.43
159		Jat village, Jath Block (EW)	Sangli	17.048494	75.207883	4.43
160		Palsai village, Wada block (PZ)	Palghar	19.602093	73.028055	3.17
161		Salvad village (PZ)	Palghar	19.809329	72.720712	12.19
162		Rayate village, Kalyan block (PZ)	Thane	19.2592	73.2268	12.18

3.5 Ground Water Resource Assessment

3.5.1 Introduction

The Dynamic ground water resources (as in 2023) of the entire country have been assessed jointly by CGWB and State Ground Water Departments under the supervision of the State level Committees. The dynamic ground water resources are also known as Annual Ground Water Recharge, since it gets recharged every year from rainfall and other sources (secondary sources) such as applied irrigation water, surface water bodies, water conservation structures, etc.

As a part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction are assessed for each assessment unit (block/taluka/mandal/tehsils/firka etc.). The 'Stage of Ground Water Extraction' is then computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe (<= 70 %), Semi-Critical (>70 % and<=90 %), Critical (>90 % and <=100%) and Over-Exploited (>100 %).

A Central Level Expert Group (CLEG) was constituted by the Department of Water Resources, RD & GR, Ministry of Jal Shakti for over-all supervision of the reassessment of ground water

resources in the entire country with the reference year 2022. The groundwater resources assessment at the State/UT Levels was carried out jointly by Central Ground Water Board and State Nodal/Ground Water Departments under the guidance ofrespective State Level Committees (SLC) with over-all supervision of CLEG. The Report on "National Compilation of Dynamic Ground Water Resources, 2023" was released by Honourable Minister of Jal Shakti on 1st December, 2023. Ground Water Resource Assessment - 2023 (GWRA-2023) has been carried out through a software/web based application "India-Groundwater Resource Estimation System (IN-GRES)" developed by CGWB through IIT-Hyderabad (http://ingres.iith.ac.in). The pictorial categorization of the assessment units of the entire country is shown in fig. 1 and its corresponding details are shown in table 1 below.

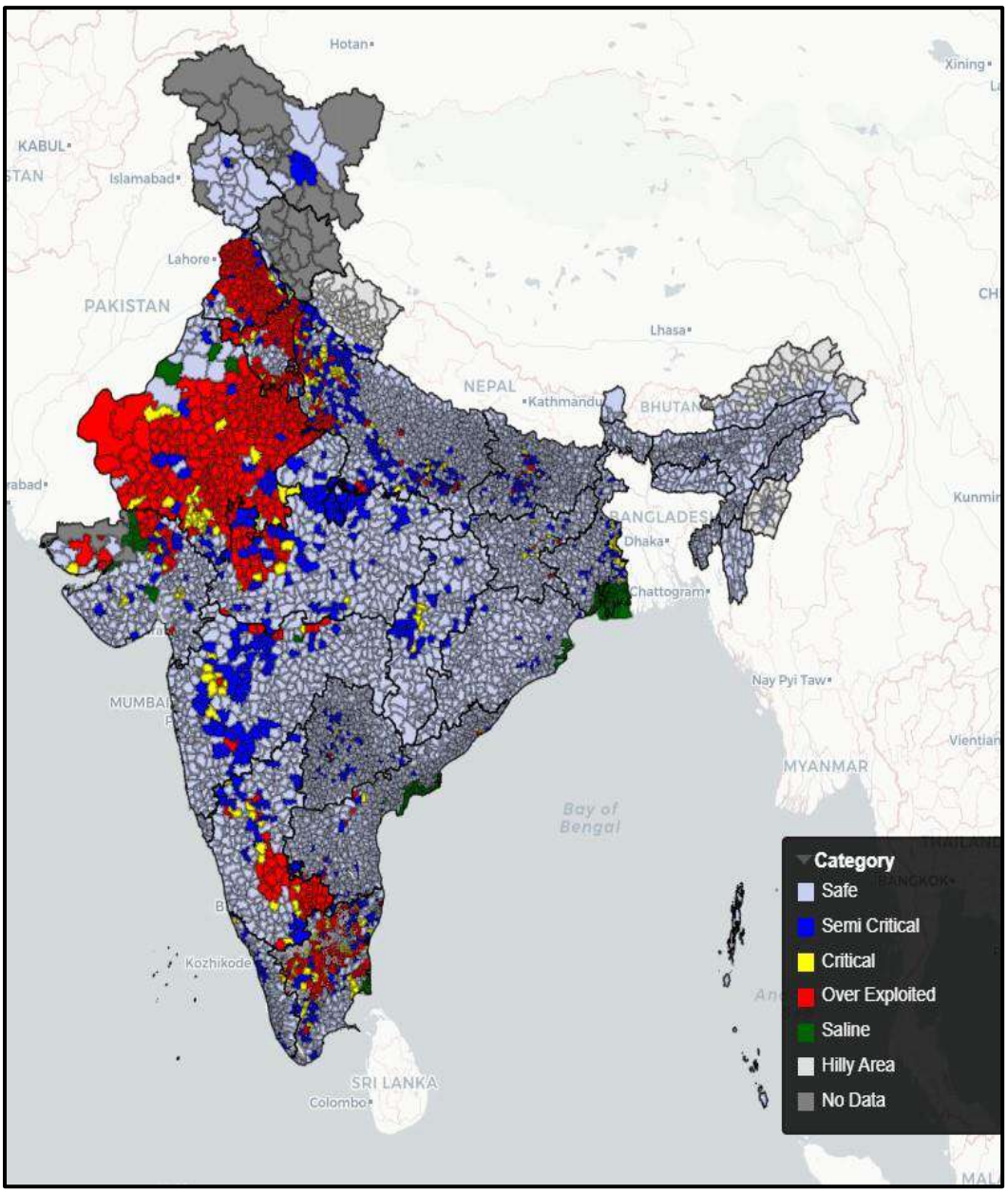


Fig. 1: Categorization of Assessment Units

Table 1: State/UT-wise Categorization of Assessment Units, as per GWRA- 2023

CATEGORIZATION OF BLOCKS/ MANDALS/ TALUKAS IN INDIA (2023)												
S.No.	**State/ Union Territories**	**Total No. of Assessed Units**	**Safe**		**Semi-Critical**		**Critical**		**Over-Exploited**		**Saline**	
	States		**Nos.**	**%**	**Nos.**	**%**	**Nos.**	**%**	**Nos.**	**%**	**Nos.**	**%**
1	**Andhra Pradesh**	667	597	89.5	18	2.7	3	0.5	10	1.5	39	5.85
2	**Arunachal Pradesh**	42	42	100								
3	**Assam**	245	244	99.59	1	0.41						
4	**Bihar**	535	467	87.29	53	9.91	7	1.31	8	1.5		
5	**Chhattisgarh**	146	119	81.51	22	15.07	5	3.42				
6	**Goa**	12	12	100								
7	**Gujarat**	252	189	75	20	7.94	8	3.17	23	9.13	12	4.76
8	**Haryana**	143	35	24.48	9	6.29	11	7.69	88	61.54		
9	**Himachal Pradesh**	10	10	100								
10	**Jharkhand**	263	241	91.63	11	4.18	6	2.28	5	1.9		
11	**Karnataka**	234	146	62.39	32	13.68	12	5.13	44	18.8		
12	**Kerala**	152	119	78.29	30	19.74	3	1.97				
13	**Madhya Pradesh**	317	226	71.29	60	18.93	5	1.58	26	8.2		
14	**Maharashtra**	353	277	78.47	57	16.15	9	2.55	9	2.55	1	0.28
15	**Manipur**	9	9	100.00								
16	**Meghalaya**	39	39	100								
17	**Mizoram**	26	26	100								
18	**Nagaland**	52	52	100								
19	**Odisha**	314	299	95.22	9	2.87					6	1.91
20	**Punjab**	153	20	13.07	13	8.5	3	1.96	117	76.47		
21	**Rajasthan**	302	38	12.58	22	7.28	23	7.62	216	71.52	3	0.99
22	**Sikkim**	6	6	100								
23	**Tamil Nadu**	313	125	39.94	56	17.89	27	8.63	100	31.95	5	1.60
24	**Telangana**	612	530	86.60	61	9.97	10	1.63	11	1.80		
25	**Tripura**	59	59	100								
26	**Uttar Pradesh**	836	559	66.87	172	20.57	43	5.14	62	7.42		
27	**Uttarakhand**	18	14	77.78	4	22.22						
28	**West Bengal**	345	241	69.86	32	9.28	12	3.48			60	17.39
29	**Andaman And Nicobar**	9	9	100								
30	**Chandigarh**	1			1	100						
31	**Dadra and Nagar Haveli and**	3							3	100		

	Daman and Diu											
32	**Delhi**	34	5	14.71	4	11.76	12	35.29	13	38.24		
33	**Jammu And Kashmir**	20	19	95	1	5						
34	**Ladakh**	18	12	66.67	6	33.33						
35	**Lakshadweep**	5	4	80	1	20						
36	**Puducherry**	8	3	37.50	3	37.50			1	12.50	1	12.50
	Grand Total	**6553**	**4793**	**73.14**	**698**	**10.65**	**199**	**3.04**	**736**	**11.23**	**127**	**1.94**

Note-
Blocks- Arunachal Pradesh, Assam,Bihar, Chhatisgarh, Haryana, Jharkhand, Kerala, Madhya Pradesh, Manipur, Meghalaya, Mizoram,Nagaland, Odisha, Punjab, Rajasthan, Tripura, Uttar Pradesh, Uttarakhand, West Bengal,Andaman& Nicobar Island, Ladakh,Lakshadweep
Taluks- Goa, Gujarat, Karnataka, Maharashtra,Tamil Nadu, Puducherry
Mandals- Andhra Pradesh, Telangana
District- Sikkim, Dadra & Nagar Haveli, Daman & Diu, Jammu & Kashmir
Valley- Himachal Pradesh
UT- Chandigarh
Tehsil- Delhi

3.6 National Aquifer Mapping and Management Programme (NAQUIM)

3.6.1 Introduction

Central Ground Water Board (CGWB) is implementing National Aquifer Mapping and Management program (NAQUIM), which envisages mapping of aquifers (water bearing formations), their characterization and development of Aquifer Management Plans to facilitate sustainable management of ground water resources. NAQUIM was initiated in 2012 as a part of the GWMR plan scheme with the objectives to delineate and characterize the aquifers and develop plans for ground water management. Out of 33 lakh km^2 of the entire country, entire mappable area of 25 lakh km^2 has been covered under this programme. NAQUIM outputs are shared with various stakeholders including the District Authorities. Public Interaction Programmes (PIP) are also being conducted in various parts of the country to promote outreach on groundwater management.

3.6.2 Objectives

The National Aquifer Mapping and Management Programme (NAQUIM) was initiated and implemented by Central Ground Water Board under the scheme of Ground Water Management & Regulation during XII plan (2012-17). The Programme was designed as a follow-up of the recommendations of four Working Groups of the erstwhile Planning Commission namely "Sustainable Ground Water Development and Management", "Minor Irrigation and Watershed

Development" , "Urban and Industrial Water Supply and Sanitation "and "Data Base Development and Management".

With an aim to support effective management of ground water resources in the country, the NAQUIM programme was taken up with the following objectives

- Delineation of aquifers in terms of their horizontal and vertical extent.
- Characterization of aquifers in terms of their hydrogeological properties, resource availability and quality
- Preparation of thematic aquifer maps
- Developing management strategies to ensure sustainability of ground water resources.

3.6.3 NAQUIM 2.0

NAQUIM 2.0 was initiated in 2023 with broad objectives of:

i) providing information in higher granularity with a focus on increasing density of dynamic data like ground water level, ground water quality etc.

ii) providing issue based scientific inputs for ground water management upto Panchayat level,

iii) providing printed maps to the users and

iv) putting in place a strategy to ensure implementation of the recommended strategies.

The NAQUIM 2.0 studies are designed to focus on 11 priority areas. In the year 2023-24, a total of 76 studies were conducted under NAQUIM 2.0 covering 8 of these priority areas, which include Auto-flow areas, Coastal areas, Industrial areas, Mining Areas, Over Exploited – Critical – Semi Critical (OCS) areas, Poor Water Quality (WQ) areas, Springshed areas, and Urban Agglomerate areas. These 76 studies cover an area of 44,000 sq. km spanning across 30 States/UTs.

Table 1: Priority area wise NAQUIM 2.0 studies taken up in AAP 2023-24

Sr No	State	Regions	No of Studies	Priority Area type	District/Block
1	Maharashtra	**CR**	**4**	OCS	Sangli (Jath block)
2	Maharashtra	CR		OCS	Nagpur (Katol & Saoner Block)
3	Maharashtra	CR		Poor WQ	Chandrapur (Warora block)
4	Maharashtra	CR		Urban Agglomerate	Mumbai
5	SUO DELHI	**Delhi**	**2**	Poor WQ	Okhla landfill
6	SUO DELHI	Delhi		Poor WQ	Bhalsawa
7	Andaman & Nicobar UT	**ER**	**4**	Urban Agglomerate	South Andaman
8	Sikkim	ER		Springshed	Gangtok
9	West Bengal	ER		Industrial Areas	Purba Medinipur

10	West Bengal	ER		Poor Availability	Purulia
11	Kerala	**KR**	**5**	Industrial Areas	Palakkad
12	Kerala	KR		OCS	Kasargod
13	Kerala	KR		Springshed	Idukki
14	Kerala	KR		Urban Agglomerate	Trivandrum
15	Lakshadweep UT	KR		Poor WQ	Minicoy and Amni
16	Bihar	**MER**	**4**	Poor Availability	Study 1. Mohanpur and adjacent block of Gaya District.
17	Bihar	MER		Poor Availability	Study 2.Meskaur and adjacent block of Nawada District
18	Bihar	MER		Poor WQ	Study 3.Madanpur and adjacent blocks of Aurangabad district
19	Jharkhand	MER		OCS	Koderma, Domchanch and Jainagar Blocks of Koderma district
20	Chhattisgarh	**NCCR**	**5**	Industrial Areas	Korba
21	Chhattisgarh	NCCR		OCS	Balod
22	Chhattisgarh	NCCR		OCS	Dhamtari
23	Chhattisgarh	NCCR		OCS	Durg
24	Chhattisgarh	NCCR		Urban Agglomerate	Raipur urban
25	Madhya Pradesh	**NCR**	**4**	Industrial Areas	Ujjain
26	Madhya Pradesh	NCR		OCS	Ratlam (Alot block)
27	Madhya Pradesh	NCR		Springshed	Dindori, Anuppur
28	Madhya Pradesh	NCR		Urban Agglomerate	Bhopal
29	Arunachal Pradesh	**NER**	**6**	Urban Agglomerate	Itanagar Capital Complex, Papum Pare
30	Assam	NER		Industrial Areas	Bamunimaidan and North Guwahati
31	Assam	NER		Poor Availability	Guwahati City, Kamrup Metro,
32	Assam	NER		Springshed	Cachar and Hailakandi Districts
33	Meghalaya	NER		Urban Agglomerate	Greater Shillong, East Khasi
34	Tripura	NER		Springshed	Dhalai
35	HP	**NHR**	**1**	Springshed	Kangra
36	Uttar Pradesh	**NR**	**4**	Industrial Areas	Unnao

37	Uttar Pradesh	NR		OCS	Baghpat
38	Uttar Pradesh	NR		Poor WQ	Siddharth Nagar
39	Uttar Pradesh	NR		Urban Agglomerate	Kanpur Nagar
40	Jammu & Kashmir UT	**NWHR**	**4**	Springshed	Ramban
41	Jammu & Kashmir UT	NWHR		Springshed	Banihal
42	Jammu & Kashmir UT	NWHR		Urban Agglomerate	Srinagar Urban Area
43	Laddakh UT	NWHR		Springshed	Kargil District
44	Haryana	**NWR**	**4**	Industrial Areas	Gurugram
45	Haryana	NWR		OCS	Samalkha and Bapoli Blocks of Panipat District
46	Punjab	NWR		Deeper Aquifer	Sangrur and Bhiwanigarh Blocks of Sangrur District
47	Punjab	NWR		Poor WQ	Ludhiana City
48	Odisha	**SER**	**5**	Coastal	Balasore Sadar block including Chandipur
49	Odisha	SER		Industrial Areas	Part of Jharsuguda and Sambalpur District
50	Odisha	SER		Mining Area	Sukinda
51	Odisha	SER		OCS	Khordha and Nayagarh
52	Odisha	SER		Urban Agglomerate	Sambalpur
53	Andhra Pradesh	**SR**	**9**	Urban Agglomerate	Vishakhapatnam, Anakkapalli
54	Andhra Pradesh	SR		Urban Agglomerate	Guntur, Vijayavada, Amaravati
55	Tamil Nadu	SR		Mining Area	Ariyalur
56	Tamil Nadu	SR		OCS	Thiruppur
57	Tamil Nadu	SR		OCS	Theni
58	Tamil Nadu	SR		Poor WQ	Salem
59	Telangana	SR		Poor Availability	Vikarabad,Rangareddy
60	Telangana	SR		Urban Agglomerate	Hyderabad, Parts of Rangareddy, Medchal
61	Telangana	SR		Urban Agglomerate	Rangareddy (Parts)
62	Goa	**SWR**	**5**	Poor Availability	South Goa
63	Karnataka	SWR		OCS	Bengaluru Rural
64	Karnataka	SWR		Poor WQ	Dharwad
65	Karnataka	SWR		OCS	Ramnagar
66	Karnataka	SWR		Urban Agglomerate	Bengaluru city

67	Uttarakhand	**UR**	**1**	Autoflow	Udhamsingh Nagar (Jaspur, Gadarpur, Bazpur and Kashipur Block)
68	Gujarat	**WCR**	**5**	Coastal	Navsari
69	Gujarat	WCR		OCS	Banskantha
70	Gujarat	WCR		Poor WQ	Vadodara
71	Gujarat	WCR		Urban Agglomerate	Ahmedabad &Daskaroi
72	Gujarat	WCR		Urban Agglomerate	Gandhinagar
73	Rajasthan	**WR**	**4**	Industrial Areas	Pali
74	Rajasthan	WR		Industrial Areas	Neemrana
75	Rajasthan	WR		Urban Agglomerate	Jaipur
76	Rajasthan	WR		Urban Agglomerate	Jodhpur
		Total	**76**		

Out of these 76 studies taken-up, 69 studies have been successfully completed during the year 2023-24, while remaining 7 studies will contine in 2024-25. The completed 69 studies include 8 priority areas covering 38, 429 sq. km. across the country. The details of priority area and State wise NAQUIM 2.0 studies completed are given in table 2 & 3.

Table 2: Priority area wise NAQUIM 2.0 studies completed during 2023-24

Priority Area	Area (Sq km)	No. of Primary Theme
Auto-flow areas	1030	1
Coastal areas	710	2
Industrial areas	1830	9
Mining areas	1170	2
OCS areas	17539	23
Poor WQ areas	4660	10
Springshed areas	3790	6
Urban Agglomerate areas	7700	16
Total	**38429**	**69**

Table 3: State wise NAQUIM 2.0 studies completed during 2023-24

	State/UT	No. of Studies	Area (in km^2)
1.	Andaman & Nicobar UT	1	40
2.	Andhra Pradesh	2	1660
3.	Arunachal Pradesh	1	490
4.	Assam	3	830
5.	Bihar	3	2040
6.	Chhattisgarh	5	2480
7.	Goa	1	170
8.	Gujarat	5	3430
9.	Haryana	2	670
10.	HP	1	2100
11.	Jammu & Kashmir UT	3	900
12.	Jharkhand	1	800
13.	Karnataka	3	2020
14.	Kerala	2	340
15.	Laddakh UT	1	250
16.	Lakshadweep UT	1	10
17.	Madhya Pradesh	4	3090
18.	Maharashtra	3	4450
19.	Meghalaya	1	180
20.	Odisha	4	1470
21.	Punjab	2	960
22.	Rajasthan	4	1210
23.	Sikkim	1	20
24.	SUO DELHI	2	20
25.	Tamil Nadu	4	2870
26.	Telangana	1	560
27.	Uttar Pradesh	4	2410
28.	Uttarakhand	1	1030
29.	West Bengal	2	2100
	Total	**68**	**38600**

Mid term review Workshops under NAQUIM 2.0: Further, to review the progress of AAP studies, mid-term review workshops were organised for 11 priority areas assigned in AAP 2023-24. Four workshops (table 4) have been organized in different Regional offices of CGWB from April 2023 to March 2024 (fig. 1 & 2).

Table 4: Details of mid-term review workshops organized under NAQUIM 2.0

Sr. No	Priority Area wise Workshop for Mid term review	Number of Studies	Name of Region for Workshop	Date
1	Urban Agglomerates	19	CHQ Faridabad	25 & 26.09.2023
2	OCS + Poor Availability	23(17+6)	SWR Bengaluru	12 & 13.09.2023
3	Springshed+Autoflow	10(9+1)	UR Dehradun	18.10.2023
4.	Poor WQ + Industrial Cluster+Mining Area +Coastal studies	24(10+10+2+2)	WR Jaipur	30 & 31.10.2023

Fig. 1:Workshop on OCS & Poor Availability Areas at SWR Bengaluru

Workshop on Spring Study & Autoflow Areas at UR Dehradun

Fig. 2: Workshop on Poor Water Quality, Industrial Cluster, Mining Area, & Coastal studies at WR Jaipur

3.7 Ground Water Conservation and Artifiicial Recharge

3.7.1 Introduction

Groundwater conservation and Artificial recharge (AR) are critical components in the sustainable management of water resources. The practice of conserving groundwater involves implementing strategies to reduce water wastage, protect aquifers from contamination, and promote the efficient use of this resource. The integration of groundwater conservation and artificial recharge practices is essential for ensuring long-term water security. By adopting a holistic approach that combines policy measures, community engagement, and technological innovations, stakeholders can effectively manage groundwater resources. Furthermore, collaborative efforts among government agencies, local communities, and environmental organizations can lead to the development of sustainable practices that safeguard groundwater resources for future generations, ultimately contributing to the resilience of ecosystems and human societies alike.

3.7.2 Convergence with MGNERGS in Water Stressed areas

CGWB is providing technical guidance in site selection & design of artificial recharge structures and capacity building of MGNREGS officials in selected 09 Blocks of 08 States as a pilot study (table 1). The work has been taken up under the MGNREGS fund. So far, the work has been completed in Andhra Pradesh, Telangana & Tamil Nadu (one block each) and is in progress in the rest of the blocks of the states. Glimpses of the AR structures constructed are given in Fig 1 & 2.

Table 1: Details of water stressed areas taken up under MGNREGS

S.No.	State	District	Block	Target (No. of Structures)	Status of Work (No. of structures constructed)
1	Andhra Pradesh	Prakasam	Peddaravedu	1253	Project Completed
2	Telangana	Nallagonda	Munugode	2174	Project Completed
3	Karnataka	Chickballalpur	Gowribidanur	103	36
4	Tamil Nadu	Krishnagiri	Mathur	20	Project Completed
5	Rajasthan	Bhilwara	Madalgarh	56	47
6	Rajasthan	Sirohi	Reodhar	126	81
7	Haryana	Kurukshetra	Babain	15	13
8	Madhya Pradesh	Ujjain	Badnagar	436	353
9	Maharashtra	Amravati	Warud	1798	422

Fig 1: The Pond constructed in Babain Block, Kurukshetra District, Haryana

Fig 2: Artificial Recharge Structures at Chickballalpur District, Karnataka

3.7.3 Implementation of Master Plan for Artificial Recharge 2020

- CGWB have prepared a conceptual document on "Master Plan for Artificial Recharge to Ground Water in India" in 2020 to work out the feasibility of various structures for the different terrain conditions of the country and estimated costs, the document provides a broad outline of the project and expected investments which include artificial recharge in both rural and urban areas. A total 14,175,144 structures at an estimated cost of Rs 1.33 lakh Cr have been envisaged for rainwater harvesting and artificial recharge in the country.
- One district in each state has been identified for implementation of Master Plan for Artificial Recharge-2020 on pilot basis. The implementation of Master Plan for Artificial recharge has been initiated in 33 States/UTs which are Andhra Pradesh, Assam, Arunachal Pradesh, Bihar, Chhattisgarh, Chandigarh, Delhi, Gujarat, Haryana, Himachal Pradesh, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Punjab, Puducherry, Rajasthan, Sikkim, Tamil Nadu, Telangana, Tripura, Uttar Pradesh, Uttarakhand, West Bengal, UT of Andaman and Nicobar, Jammu&Kashmir, Ladakh. The photographs of the AR structures constructed are provided in Fig. 3 to 5.

- As NAQUIM 1.0 has already been completed in March 2023, the recommendations under NAQUIM 1.0 have to be incorporated in Master Plan for Artificial Recharge-2020. The said updation of Artificial Recharge-2020 Master Plan has been completed.

Fig. 3: Artificial Recharge Structures at South Andaman District, Andaman & Nicobar

Fig.4: Artificial Recharge Structures at Kasargod District, Kerela

Fig.5: Artificial Recharge Structures at Nuapada District, Odisha

3.7.4 Groundwater Augmentation through Artificial Recharge in water-stressed areas of Rajasthan (2021- continuing activity)

During 2021-22, CGWB has taken up artificial recharge for groundwater augmentation in certain water stressed areas of Jaisalmer, Jodhpur & Sikar districts of Rajasthan.

- **Phase-I:** Construction of Concrete Gravity Dam at Bastawa Mata (Block Balesar) and Zoned Earth Fill Dam with clay Core at Indroka (Block Mandore) in Jodhpur district, Rajasthan with the objective of artificial recharge to groundwater aquifer systems. The physical and financial details of the dams are tabulated in table 2.
  - The construction of Zoned Earth Fill Dam with Clay Core at Indroka (Block Mandore) in Jodhpur district, Rajasthan has been completed and inaugurated by the Hon'ble Minister of Jal Shakti on 16.03.2024 (Fig. 6).
  - The concreting work of Concrete Gravity Dam at Bastawa Mata (Block Balesar) in Jodhpur district, Rajasthan has been completed and the ancillary works are in progress. The photographs of the field visit made by Chairman, CGWB, Member (N&W), Member (East) and Officers of CGWB to check the status of Concrete Gravity Dam at Bastawa Mata.
  -

Table 2: Details of AR Structure to be constructed in Phase I

S.No.	Location Details	Type of Artificial Recharge Structure	Fund Allocated
1	Indroka, Mandore, Jodhpur	Zoned Earth Fill Dam with Clay Core	Rs 15.21 Cr
2	Bastawa Mata, Balesar, Jodhpur	Concrete Gravity Dam	Rs. 19.42 Cr
		Total	Rs.34.63 Cr

Fig 6: Inauguration of Indroka Dam by Hon'ble Minister of Jal Shakt

- **Phase-II:** Under this phase, it is proposed to construct Stone Masonary Check Dams (MCD), Anicuts, Concrete Check Dams (CCD & Recharge shafts) at 101 sites in parts of Balesar, Shekhla, Lohawat, Tinwari and Phalodi blocks of Jodhpur district and Sankra block of Jaisalmer and Shrimadhopur block of Sikar district of Rajasthan. The cost of the project is Rs 74.32 Cr.
  - Total 80 Water Harvesting Structures (WHS) have been constructed, out of which 42 have been handed over to WRD, Govt. of Rajasthan, the work of 02 WHS have been partially completed and the construction of remaining 19 WHS has been foreclosed due to local problems. The details of the total WHS structures constructed so far are provided in table.3.
  - Tripartite MoU for handing over of structures has been signed by CGWB, WAPCOS & CEO Zila Parishad Jodhpur on 13.03.2024 in respect of ARS of Jodhpur district under Phase- II & III.

Table 3: Details of the structures constructed under Phase II

Description	Packages								Total
	I		II				III	IV	
	Jodhpur		Jodhpur				Jaisalmer	Sikar	
	Balesar	Sekhala	Tinwari	Sekhala	Phalodi	Lohawat	Sankra	Srimadhopur	
Proposed Structures	56	1	6	9	1	17	4	7	**101**
Work Completed	42	1	6	9	0	16	-	6	**80**

- **Phase III (Extended Phase II):** Due to deterioration in groundwater resources in Rajasthan, the Artificial Recharge Project under Phase –II has been extended as Phase III (Extended Phase-II) in Jodhpur, Jaisalmer, Sikar, Jhunjhunu and Alwar districts of Rajasthan. Further, the cumulative recharge impact may be seen at these places. It is proposed to construct Masonry Check Dams (MCD), Anicuts, Concrete Check Dams (CCD), Model Talab etc. at 53 sites. The estimated cost of the project is 61.82 Cr.

  A total of 27 WHS has been constructed and the construction of remaining WHS is under progress. The details of the total WHS structures constructed so far are provided in table 4.

Table 4: Details of AR structures constructed under Phase II (extended)

Description	Jodhpur District	Jaisalmer District	Alwar District	Jhunjhunu District	Barmer District	TOTAL
Proposed Structures	22	24	1	2	4	53
Work Completed	15	5	1	0	6	27

3.7.5 AR structures in Kala Amb valley, Sirmaur, Himachal Pradesh

Kala Amb Valley in Sirmaur district of Himachal Pradesh falls under ground water dependent area and the ground water is mainly utilized for drinking and industrial development. It has been observed during the past decade that the large withdrawal of ground water in the industrial sector has resulted in depletion of groundwater level in the area. It has also been observed that during the monsoons, runoff quickly flows out of the area with less natural recharge to ground water due to sloping areas. As surplus runoff volume in streams and rivulets are available, appropriate structures across these streams at selective locations may facilitate the harvesting of water which in turn enhances the ground water recharge in downstream areas.

- The Ministry has accorded approval for construction of total 19 artificial recharge structures (07 Concrete Check Dams and 12 recharge Wells) vide letter no. No. T-15014/4/2021-GW section, dated 22.11.2023. The sanctioned cost of the project is 19.92 Cr. The location wise details of of AR structures are to be constructed is provided in table 5.
- The project is under initial stage wherein various meetings have been conducted and committees were formed regarding drafting and finalizing of tender documents. The draft tender document has been prepared and submitted for approval. The photographs of the Field visit by the Officers of CGWB & Jal Shakti Vibhag (JSV), Govt. of Himachal Pradesh regarding verification of the designs/drawings provided in the DPR submitted by WAPCOS is given in Fig. 7.

Fig 7: Field visit by the Officers of CGWB & JSV, Himachal Pradesh Govt. regarding verification of the designs/drawings provided in the DPR submitted by WAPCOS

Table 5: Details of AR structures to be constructed in Kala Amb Valley

S.No.	AR Notation	Longitude	Latitude	Location	District	Block	AR Structure Type
Check Dams							
1	CD-1	77.19638	30.54533	Pothia Kala	Sirmaur	Nahan	Check Dam
3	CD-3	77.21412	30.54288	Tada Baroti Khala	Sirmaur	Nahan	Check Dam

4	CD-4 (a)	77.20134	30.53755	Trilokpur	Sirmaur	Nahan	Check Dam
5	CD-4 (b)	77.19993	30.53344	Trilokpur	Sirmaur	Nahan	Check Dam
7	CD-6	77.17339	30.54914	Arandwala Village	Sirmaur	Nahan	Check Dam
8	CD-7	77.2705	30.50493	Near Konthrogh Bridge	Sirmaur	Nahan	Check Dam
9	CD-8	77.30777	30.52966	D/s of Bhoothpur Bridge	Sirmaur	Nahan	Check Dam
Recharge Wells							
1	RW-1(3)	77.19353	30.535917	Trilokpur Nadi (Kheri)	Sirmaur	Nahan	Recharge Well-(3 Nos)
2	RW-2(6)	77.199619	30.535917	Trilokpur Khad (Budion)	Sirmaur	Nahan	Recharge Well-(6 Nos)
4	RW-4(3)	77.24507	30.52114	Sailani (Forest Deptt.)	Sirmaur	Nahan	Recharge Well-(3 Nos)
Total	**12**						**12**

3.7.6 District Recharge Plan Maps

District Recharge Plans (DRP) were to be prepared for the 255-water stressed district of the country which was identified during Jal Shakti Abhiyan (JSA) 2019 which will be shared to the State Govterment department/implementing agencies and District authorities. The District Recharge Plan will be prepared on the scale of 1:50,000.

Out of the total 255 DRPs identified, 187 DRPs have been finalized and remaining are under finalization. The latest progress of DRPs finaliezed is given in tTable 6.

Table 6: Details of the finalized District Recharge Plan

Name of the State	Target	Achievement
Andaman & Nicobar	1	0
Andhra Pradesh	9	8
Arunachal Pradesh	1	0
Assam	1	0
Bihar	12	1
Chhattisgarh	2	2
Dadra & Nagar Havelli	1	0
Daman & Diu	1	0
Delhi	10	10
Goa	1	0

Gujarat	5	0
Haryana	19	19
Himachal Pradesh	4	0
Jammu & Kashmir	1	0
Jharkhand	2	0
Karnataka	18	18
Kerala	2	1
Lakshadweep	1	0
Madhya Pradesh	11	11
Maharashtra	8	8
Manipur	1	0
Meghalaya	1	0
Mizoram	1	0
Nagaland	1	1
Odisha	1	0
Puducherry	1	1
Punjab	20	5
Rajasthan	29	22
Sikkim	1	0
Tamil Nadu	27	27
Telangana	24	24
Tripura	1	0
Uttar Pradesh	35	28
Uttarakhand	1	0
West Bengal	1	1
Grand total	**255**	**187**

3.7.7 Source Sustainability Interventions

A few areas have been identified jointly by the DoWR, RD & GR and DWS for a pilot study leading to Source Sustainability Interventions (SSI) under Jal Jeevan Mission (JJM). The objective of the initiative was to develop a set of operational guidelines and impart training to state government agencies for planning and interventions for sustainability of groundwater based drinking water sources. The details of the states and districts taken up for SSI are as mentioned in table 7.

Table 7: Details of the states and districts taken up for SSI

State	District	Blocks	Targeted Villages
Karnataka	Chikkaballapura	6	1393
Madhya Pradesh	Chhatarpur	3	375

Rajasthan	Sikar	3	222
Tamil Nadu	Ramanathapuram	11	426
Total Villages			**2407**

In this regard, CGWB officers have visited all the villages and collected basic information assessed demand and supply side, and monsoon runoff carried out risk assessment villages with various parameters and prepared the DPR on village-wise source sustainability intervention and compiled at district level. Based on the outcome of these studies a detailed "SoP on Sustainability of Groundwater Resources" has been prepared, which was inaugurated by the Hon'ble President of India.

Further, CGWB officers have also imparted trainings on Source Sustainability Interventions to the respective State Govt. officials for carrying out this type of similar works. The duration of each training was 03 days with total nos. of 25 – 30 officers in one batch. The Trainings programmes have been conducted from June 2023 to March 2024. Till March 2024 total 49 trainings have been conducted in 29 States and UTs with the total of about 1300 participants. The statuses of trainings conducted under SSI are given as below in table 8.

Table 8: SSI trainings carried out during June 2023 to March 2024

S.No.	Name of the State	No of Training Organized	No. of Participants
1	Andaman &Nicobar	1	6
2	Andhra Pradesh	1	36
3	Arunachal Pradesh	1	25
4	Assam	2	55
5	Bihar	3	94
6	Delhi	1	21
7	Goa	1	40
8	Gujarat	3	85
9	Haryana	2	44
10	Himachal Pradesh	1	20
11	Jammu &Kashmir	2	66
12	Jharkhand	2	51
13	Karnataka	2	75
14	Kerala	1	28
15	Lakshadweep	1	30
16	Madhya Pradesh	4	110
17	Maharashtra	3	77

S.No.	Name of the State	No of Training Organized	No. of Participants
18	Meghalaya	1	33
19	Nagaland	1	39
20	Punjab	2	44
21	Puducherry	1	4
22	Rajasthan	3	59
23	Sikkim	1	12
24	Tamil Nadu	3	89
25	Telengana	1	25
26	Tripura	1	40
27	Uttarakhand	1	25
28	Uttar Pradesh	1	33
29	West Bengal	2	49
Total		**49**	**1315**

3.8 Regulation and Control of Ground Water Extraction by Central Ground Water Authoriity (CGWA)

3.8.1 Introduction

Central Ground Water Authority, constituted under Section 3 (3) of the Environment (Protection) Act, 1986 has been regulating groundwater development in the country The Authority regulates in 19 States/ UTs (15 States/ UTs). Have their own Acts and 2 States are regulating under some. The Authority issues No Objection Certificate for groundwater abstraction.

- Exercise of powers under section 5 of the Environment (Protection) Act, 1986 for issuing directions and taking such measures in respect of all the matters referred in sub-section (2) of section 3 of the said Act.
- To resort to penal provisions contained in sections 15 to 21 of the said Act.
- To regulate and control, the management and development of groundwater in the country and to issue necessary regulatory directions for the purpose.
- Exercise of powers under section 4 of the Environment (Protection) Act, 1986 for the appointment of officers.
- Important activities of CGWA during the period mentioned are given below:

3.8.2 Guidelines for Ground Water Regulation

Ministry of Jal Shakti notified the revised guidelines on 24.09.2020 and for regulation and control of ground water development in the country. These guidelines have Pan India applicability. Further the guidelines have been amended on 29.03.2023. The salient features of guidelines are as follows:

- Seeking of NOC is exempted for individual domestic consumers (in both rural and urban areas) drawing groundwater for drinking/ domestic use, Rural drinking water supply schemes, Armed Forces and Central Armed Police Forces Establishments, Agricultural users and Micro and Small Enterprises drawing <10 KLD of groundwater, All industries/ mining projects/ infrastructure projects drawing ground water only for drinking/ domestic purposes up to 5 Cum /day in all assessment units, Residential Apartments and Group Housing Societies: (a) For drinking water and domestic uses, drawing ground water up to 20 m3/day subject to the conditions mentioned in Para 2.0 of the guidelines. (b) Dwelling units for Economically Weaker Sections (EWS) under Government schemes.
- No new industries except those Micro, small, medium enterprises MSME based on ground water are permitted in Over-exploited assessment units.
- Expansion of existing industries except MSME located in Over-exploited assessment units involving increase in quantum of groundwater abstraction is not permitted
- Industries drawing >100 KLD of ground water in OE, Critical and Semi-critical assessment units are required to submit impact assessment reports along with the application.
- Industries drawing > 100 KLD of ground water in OE, Critical and Semi-critical assessment units are required biennial (once in two years) water audit and reduce fresh groundwater consumption by 20% over the next three years.
- No NOC is granted for groundwater extraction for water parks, theme parks, amusement parks in OE assessment units. Commercial infrastructure projects requiring ground water for drinking /domestic use shall also be covered under this category.
- All ground water users seeking NOC are required to pay ground water abstraction charges in Critical, Semi-critical and Safe assessment units and ground water restoration charges in OE assessment units. The charges so collected are to be utilized by States/ UTs on specific water conservation measures, specific supply/ demand side interventions.
- No charges are to be paid by users drawing saline ground water.
- Users drawing groundwater illegally are liable to pay Environmental Compensation.
- Penal provision has been kept for ensuring compliance of various conditions of NOC.
- Bulk water suppliers have also been brought in the ambit of NOC.

- Installation of Sewage Treatment Plants is mandatory for all new residential apartments/ Group Housing Societies where ground water requirement is more than 20 m3 /day. STP treated water is to be used by such users for flushing, green belt etc.
- All users seeking NOC are required to install tamper proof flow meters along with telemetry on all the ground water abstraction structures.
- In over-exploited assessment units, use of ground water for construction activity is permitted only if no treated sewage water is available within 10 km radius of the site.

3.8.3 Processing of Applications for Grant/ Renewal of No Objection Certificate (NOC) for Ground Water Withdrawal

CGWA evaluated applications from Industries/ Infrastructure Units / Mining Projects for grant of NOC for ground water withdrawal as per provisions of the Notified guidelines dated 24.09.2020. A total **of 2054** nos. of new NOCs were **issued** and **1833** nos. of NOCs **renewed** during the 2023-24. In addition, **exemption** was given in respect of **6966** Nos. **new** applications and **147 Nos renewal** cases. Total **11,000** NOC's/exemptions were issued during the year 2023-24.

3.8.4 Monitoring of Compliance of Conditions Stipulated in the NOC

CGWA has developed an online self-compliance format for the project proponent to upload the data as per compliance conditions defined in the NOC and the same is being monitored by the Regional Offices for ensuring the compliance conditions by the project proponent. Random site inspection is also carried out during the year to check the compliance of NOC issued by C.G.W.A. Mobile application is also under development for this purpose with the provision of verification of compliance by CGWA officers.

Show-cause notices are also being issued to units on a regular basis, which have been found not to have fully complied with the NOC conditions. Orders for sealing of bore/tube wells and/ or disconnection of electricity supply through the concerned DCs/ DMs were also issued in respect of units, which did not give satisfactory replies to the show-cause notice.

While processing applications for renewal of NOC, compliance of NOC conditions by the proponents was ensured. Penalty provisions are kept for non-compliance conditions of NOC issued. Provision of Rs. 1 lakh under Section 15 of Environment (Protection) Act of 1986 was imposed on the violators whose NOC was issued prior to 24.09.2020, i.e prior to issuance of guidelines i.

Separate penalty provisions are kept as per different compliance conditions in the guidelines issued after 24.09.2024 as per table 16.1. Renewal NOCs were issued after the compliance was completed and penalty was deposited by such offenders to the concerned Regional Office of CGWB/CGWA, New Delhi.

3.8.5 Important Meetings for Regulation of Groundwater

a) Meeting of All States/ UTs by the Secretary, DoWR

The Secretary, DoWR, Shri Pankaj Kumar took meeting of all States/ UTs on 05.04.2023 to review status of enactment of GW Act, constitution of SGWA, aligning their existing guidelines in conformity with MoJS guidelines, mandatory RWH conditions etc. The Secretary, DoWR issued necessary directions to expedite the above.

b) Meetings for Implementation of Directions issued by Hon'ble NGT in OA No. 94/2021 and related MA 16-18/ 2023 and EA 41/2023 (Haider Ali vs UoI & Ors) regarding misuse of groundwater by Cricket Stadiums

The JS (A, IC & GW) took meeting of on 11.04.2023 and issued directions for compliance of Hon'ble NGT directions in the matter of misuse of groundwater by Cricket Stadiums.

The Chairman, CGWA took meeting of SAI, BCCI and MoYAS on 19.04.2023 and 17.05.2023 to review the status of compliance of Hon'ble NGT directions.

The Secretary, DoWR, Shri Pankaj Kumar took meeting of BCCI/SAI/MoYAS and State Cricket Associations of 26 Cricket Stadiums (as per list provided by BCCI) on 02.06.2023 and 17.07.2023 and issued strict directions for compliance of Hon'ble NGT directions. A follow-up

c) 48th meeting of the Board of Central Ground Water Authority

48th meeting of the Central Ground Water Authority (CGWA) was held on **24.08.2023** under the chairmanship of Shri A K Agrawal, Chairman, CGWB.

- Board ratified the decisions/ Public Notices issued by CGWA between 47th and 48th meeting as regard to following: extension of last date; revival of rejected applications; revision of charges wef 1st January every year as per latest GWRE category; condonation of penalty with retrospective effect in case of on non-installation of piezometer; EC imposition wef 24.09.2020 on late submission; imposition of EC on saline groundwater extraction; levying of penalty alongside EC on late submission of renewal. Regarding addition of rainwater quantum collectied in mining pit, same was agreed by Board, with the modification that charges will be levied on the basis of annual quantum.
- Amendments to extant guidelines were proposed to the board in order to remove conflicts/ discrepancies in guidelines, as also to introduce provisions for certain scenarios experienced during the processing of applications. Concurrence of Board was sought on these proposed amendments for further submission to Ministry.
- Following proposed amendments were concurred: 1. Modification of the heading of Table 15.3 to ECRGW for industrial/ commercial infrastructure projects; 2. IAR/ Modelling report submission as per SOP to be issued time to time; 3. In case of delay in submission of renewal application even by one day, EC will be

applicable wef the next date of expiry till the date of submission along with penalty towards late submission; 4. Exempted category of 'MSE extracting less than 10 KLD' to 'MSE extracting up to 10 KLD'; 5. Provision to obtain NOC in order to maintain groundwater arrangement in case of operational or natural disasters; 6. Clause of 20% reduction in groundwater requirement by industries may be modified in line with agreed points in Water Audit Policy; 7. Late submission fee of Rs 1,00,000/- to be added to the list of penalties; 8. Provision of extension of validity to be replaced with renewal; 9. Annexure regarding indicative list of site-specific infrastructure projects with two separate lists, one for OE areas (site-specific) and the other for remaining categories; 10. Add a clause at appropriate place in guidelines mandating PPs to adhere to Hon'ble Supreme Court guidelines.

- Regarding other proposed amendments, Board was of view that additional consideration and research are required.
- All the members agreed regarding provisions pertaining to Package Drinking Water (PDW) is to be applicable to the units such as Aerated/ Carbonated/ Alcoholic Beverages and accordingly, heading of charges table for PDW is to be modified.
- It was decided by the members that groundwater NOCs would be granted in accordance with the specified quantum mentioned in CTO, CTE, and CCA.
- Interim Mechanism (SOP) for adjusting Annual Charges as per Actual Consumption was ratified by the Board.
- Revised criteria/ SOP for delisting of accredited GW Consultants was approved by the Board. The accredited GW consultant will be delisted for one year on rejection of three reports.

d) 49th CGWA Meeting of the Board of Central Ground Water Authority

- 49th meeting of the Central Ground Water Authority (CGWA) was held on 19.02.2024. The meeting was held both online and in person under the Chairmanship of Shri Sunil Kumar Ambast, Chairman, CGWB.
- The decision to authorize Region Directors for approval of applications up to 500 KLD was ratified by the Board members.
- In continuation to 48th Meeting some more amendments to extant guidelines were placed before in order to remove conflicts/ discrepancies in guidelines, as also to introduce provisions for certain scenarios experienced during the processing of applications. Concurrence of Board was sought on these proposed amendments for further submission to Ministry.
- Out of total 19 proposed amendments placed before the Board, 16 were concurred to. These amendments relate to 1. Clarification on exempted MSE (to include industries, Infra and Mining); 2. Exclude PDWs from exempted MSE; 3. Exemption to infrastructure facilities situated in the defense establishment; 4. Exemption up to 10 KLD for temporary TWs/ BWs in remote areas for exploration

activities/ temporary camps; 5. Infrastructure projects in remote areas using groundwater up to 10 KLD; 6. Mandatory requirement for zero withdrawal projects to obtain exemption from CGWA; 7. Mandatory requirement for other exempted categories (except Agriculture) to obtain exemption from CGWA; 8. Modification in condition for use of groundwater for construction activity in OE category; 9. Mandatory requirement for exempted MSEs to install digital flowmeter; 10. Permission to bring groundwater from outside the project premises; 11. Add Water Audit condition for Infrastructure projects; 12. Reduction in charges for change in User id and other corrections; 13. Ceiling of GW quantum for new MSMEs in OE area, as well as expansion of MSMEs in OE area; 14. Addition of permission to new Infra MSMEs in OE areas and expansion of MSMEs in OE areas; 15. Criteria for rebate in subsequent renewals; 16. To introduce mandatory condition for installation of digital flow-meter by exempted categories.

- Some other proposed amendments were referred to a committee for their recommendation.
- The Board concurred to proposal for introducing charges in saline category and also to correct EC value for saline water in the glossary so as to have conformity with Saline Water guidelines.
- Proposal to revise Bulk Water Supply guidelines to allow tanker supply for industrial use was agreed to. Further, it was agreed to revise limit and purpose of extraction as per area category.
- For non-assessed blocks, it was agreed to take weighted average of surrounding blocks to decide the category of un-assessed block.
- The Board agreed on initiating proposals to declare CGWB as CGWA, in line with Hon'ble Supreme Court directions, and also to constitute SGWAs under EPA, 1986.

3.8.6 On Site Inspection by CGWB

On-site inspections were carried out by the Regional Offices of CGWB to check the compliance of NOCs accorded by CGWA before recommending the renewal applications to CGWA, New Delhi. Necessary show- cause notices were issued to the project proponents who have not complied with the conditions of the NOC issued by CGWA.

CGWB officers carried out inspections of **26 Cricket Stadiums** (as per list provided by BCCI) in connection with **Hon'ble NGT matter** in OA No. 94/2021 and related MA 16-18/ 2023 and EA 41/2023 (Haider Ali vs UoI & Ors) **regarding misuse of groundwater by Cricket Stadiums**. Show-cause notices were issued to non-complaint stadiums, which had not obtained NOC from CGWA/ SGWA.

An Inter-departmental **Committee** under the chairmanship of **Member Secretary, CGWA** constituted in **Hon'ble NGT matter** in OA No. 198/2023 and related MA 98/2023 (Harpal Singh Rana & Anr vs State of Uttarakhand & Ors) regarding **wastage/misuse of water in Metro ProjectsinspectedMetro Rail projects** in Delhi, Jaipur and Mumbai and submitted its report Hon'ble NGT.

3.8.7 Development of Advanced NOCAP Software

The development of Application form (user interface) for fresh applications for Industry, Mining, Infrastructure & Bulk Water Supply projects are completed. The development of functionalities for charge calculator and check eligibility criteria are also developed and completed. The eight-level processing of application in NOCAP portal has been reduced to five stages i.e. Evaluation, EAC, Recommendation & Approval Stage (1-3 parts) for smooth processing and early disposal of applications. The development of these stages for internal application processing in respect of fresh applications is also completed. Additionally, a Query module also developed for one-to-one communication between applicant and CGWA user for any clarification and update in application or documents.

3.8.8 O.M. regarding relaxation in Environmental Compensation for MSME, relaxation in EC on Saline Water and relaxation in charges for User Id etc

In order to bring more and more projects into the ambit of groundwater regulation by obtaining NOC from CGWA, an OM has been issued on **12.03.2024** providing relaxation in Environmental Compensation (**EC**) for one year period. The target beneficiaries are MSME extracting groundwater up to 100 KLD. The OM also provides relaxation in EC on saline ground water extraction by keeping earlier Public Notice in suspension for one year. In addition, charges for User ID change and other minor corrections have also been reduced for one year vide above OM.

3.8.9 Issuance of SOP for Water Management in Metro Rail Projects

In compliance to Hon'ble NGT directions in OA No. 198/2023 and related MA 98/2023 (Harpal Singh Rana & Anr vs State of Uttarakhand & Ors) regarding **wastage/misuse of water in Metro Projects**, CGWA prepared an **SOP** for '**Water Management in Metro Projects**' and circulated it to **13Metro Corporations –** Kochi Metro Rail Ltd; Jaipur Metro Rail Corporation; Delhi Metro Rail Corporation; Mumbai Metro Rail Corporation; Hydedrabad Metro Rail Corporation; Bengaluru Metro Rail Corporation; UP Metro Rail Corporation; Gujarat Metro Rail Corporation; Chennai Metro Rail Corporation; Haryana Metro Rail Corporation (Panchkula); MP Metro Rail Corporation; Patna Metro Rail Corporation; Kochi Metro Rail Corporation.

3.9 Technical Assistance to State Governments

3.9.1 Short Term Water Supply Investigations

Table 1: Region wise list of short-term investigations carried out in AAP 2023-24

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
1	**NWR Chandigarh**	Feasibility of two (2Nos.) bore-wells in Technical area of Air Force Station, Pathankot, Punjab	Air Force Station, Pathankot, Punjab
2		Feasibility of construction of tubewell in Air Force Station, Ambala Cantt, Ambala, Haryana.	Air Force Station, Ambala, Haryana
3		Feasibility of Construction of tubewell at Old Bamial,Block-Bamial, Pathankot, Punjab	BSF Cantt, Pathankot, Punjab
4		Feasibility of Construction of tubewell at Bana Enclave Mamum Cantt. ,Pathankot, Punjab	Mamum Cantt, Pathankot, Punjab
5		Feasibility Study for Construction of Rain Water Harvesting, Militery Engineering Services,Patiala Cantt, Punjab	MES, Patiala Cantt, Patiala,Punjab
6		Feasibility Study for Construction of Rain Water Harvesting, CSIR, Sector-39, Chandigarh	CSIR, Sector-39, Chandigarh
7		Feasibility Study for Construction of Rain Water Harvesting, Militery Engineering Services, Chandi mandir, Chanidigarh	MES, Chandi Mandir, Chandigarh
8		Feasibility for construction of Tubewell, at Domestic Area 3 BRD Air Force, Chandigarh	3 BRD Air Force, Chandigarh
9	**NCR Bhopal**	Short Term Hydrogeological Investigation for siting of feasible locations for construction of deep borewell in BSF sector HQ campus, village Fatehpur, MP	BSF, Bhopal
10		Short Term Hydrogeological Investigation for siting of feasible locations for construction of deep borewell in BSF sector HQ campus, village Fatehpur Dobra, MP	BSF, Bhopal
11	**NCCR Raipur**	Short Term Hydrogeological Investigation for siting of 3 feasible locations for construction of deep borewell in the Crpf campus, 65 Btn Transit Camp, Raipur	CRPF, Raipur
12	**NHR Dharamshala**	Carried out short term water supply investigation in various ITBP camps along indo-China boarder district Kinnaur, Lahaul and Spiti	ITBP
13		Carried out short term water supply investigation for Army area in Khab, Distt. Kinnaur	Army
14		Carried out short term water supply investigation for Army area in Averi, Distt. Kinnaur	Army
15		Carried out short term water supply investigation for Army area in Powari,	Army

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
		Distt. Kinnaur	
16		Carried out short term water supply investigation for Army area in Pooh, Distt. Kinnaur	Army
17		Carried out short term water supply investigation for Army area in Jakhri, Distt. Shimla	Army
18		Carried out short term water supply investigation in AIIMS, Bilaspur	AIIMS, Bilaspur
19		45 No. of DPRs under Jal Jeevan Mission (JJM) received from CHQ Faridabad to carry out Hydrogeological investigation of proposed Tube wells sites for feasibility and source sustainability.	Jal Shakti Vibhag, Himachal Pradesh
20	**NWHR Jammu**	Groundwater feasibility survey carried out at 150 General Hospital, Rajouri	MES,Rajouri
21		Groundwater Feasibility survey carried out at different locations under Jal Jeevan Mission in Samba district	PHE,Samba
22		Groundwater Feasibility survey carried out at Military Hospital,Kargil	MES,Kargil
23		Groundwater feasibility survey carried out at 63 Engineers,41 FAD, and 3 IDSR at Karu	MES,Karu
24		Groundwater feasibility survey carried out at TBS-255 ABOU,34 AP,22 MECH DAT and 81 BDE at Tangtse	MES,Karu
25		Groundwater feasibility survey carried out at 2 X OTM at 7 VIKAS,LEH	MES,7 VIKAS LEH
26		Groundwater feasibility survey carried out at premises of Chest Disease Hospital, Jammu	Executive Engineer,Hospital & Central Heating Division
27		Groundwater feasibility survey carried out at Victor Force Military Station,Awantipur	Station Headquarter,Awantipur
28		Groundwater feasibility survey carried out at Signal DAT(9 Post),MT Park(Tyakshi), Rockfall site at Prata Pur	GE,Prata Pur
29		Groundwater feasibility survey carried out at Air Force Station,Thoise	GE(AF),Thoise
30		Groundwater feasibility survey carried out at Siachen Base Camp	GE(AF),Thoise
31		Groundwater feasibility survey carried out at Kiari Army station	MES,Karu
32		Groundwater feasibility survey carried out at Chumathang Army station	MES,Karu
33		Groundwater feasibility survey carried out at Nyoma Army station	MES,Karu

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
34		Groundwater feasibility survey carried out at Nanga Village	BADP Project officer,19 JAK RIF
35		Groundwater feasibility survey carried out at Indian Institute of Management,Jammu	IIM,Administrative officer
36		Groundwater feasibility survey carried out at Directorate of Prisons office at Amphalla, Jammu	Technical officer,Director General of Prisons
37		Groundwater feasibility survey carried out at STC,BSF,Udhampur	INSPECTOR GENERAL,STC BSF,UDHAMPUR
38		Groundwater feasibility survey carried out at AF,Station,Awantipur	GE,AF Station,Awantipur
39		Groundwater feasibility survey carried out at Tregham Garrison and Pathahiri Garrison	GE,Kupwara
40		Groundwater feasibility survey carried out at Air Force DETT,Kargil	GE(AF),Leh
41		Groundwater feasibility survey carried out at Military Station,Khundroo,Anantnag,J&k	MES
42		Groundwater feasibility survey carried out for Tubewells at Khanmoh Military station	MES
43		Groundwater feasibility survey carried out at Khundru Military Station,Anantnag	MES
44		Groundwater feasibility survey carried out at 15 CARE & MT UNIT Air force Leh	GE(AF) Leh
45		Groundwater feasibility survey carried out at TACTICAL BPI Air Force Leh	GE(AF) Leh
46		Groundwater feasibility survey carried out at Military station,Pallanwala	MES
47		Groundwater feasibility survey carried out at Tippu lines,Kaluchak Military Station	MES
48		Groundwater feasibility survey carried out at DETT VIJAYPUR,Samba	MES
49	**UR Dehradun**	The Feasibility for Construction of Abstraction Structure for Ground Water at DIBER Field Station, Pithoragarh, Uttarakhand	DIBER, Pithoragarh
50		The Feasibility for Construction of Abstraction Structure for Ground Water at Military station Chakrata in Dehradun district, Uttarakhand	Military Station, Chakrata
51		REPORT ON THE FEASIBILITY FOR CONSTRUCTION OF A TUBEWELL AT RESERVE BANK OF INDIA CAMPUS, IT PARK DEHRADUN DISTRICT, UTTARAKHAND	Reserve Bank of India, Dehradun
52		REPORT ON THE FEASIBILITY FOR CONSTRUCTING A TUBEWELL IN	Kendriya Vidyalaya, Raiwala Cantonment,

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
		THE PREMISES OF KENDRIYA VIDYALAY, RAIWALA CANTT., DEHRADUN DISTRICT, UTTARAKHAND	Dehradun
53		REPORT ON THE FEASIBILITY FOR CONSTRUCTING A TUBEWELL IN THE PREMISES OF PCDA (AIR FORCE) RAJPUR ROAD, DEHRADUN DISTRICT, UTTARAKHAND	PCDA (Airforce), Rajpur Road, Dehradun
54		The Feasibility for Construction of Abstraction Structure for Ground Water at Military station Ranikhet in Almora district, Uttarakhand	Military Station, Ranikhet
55	**NR Lucknow**	Ground survey for borewell at AF STN Gorakhpur against CA No GE(AF)/GKP/E-67 of 2023-24:Special Repair/Replacement of Borewell at PHNo. 02 at AF STN Gorakhpur	Military Engineering Services, Gorakhpur
56		Survey of Borewell in Ordinance Clothing Factory, Shahjahanpur Estate	Ordinance Clothing Factory, Shahjahanpur Estate
57		Ground survey for borewell at AF STN Gorakhpur	AF Station Gorakhpur
58		CA No. CELZ/LKO/T-28 OF 2022-23: Provn. Of Tech. Acc. & Ailled Infra for Command Hospital at Lucknow	Command Hospital,Lucknow
59		Spl Repair to Borewell pump and Misc. items at MES Pump House No. 01 in place of old borewell at 810 CETC Purkazi under Para 35 DWP 2020 for Garrison Engineer, Military Engineer Services, Roorkee	Garrison Engineer, Military Engineer Services, Roorkee
60		Provn. of Technical and OTM Accn for Military Hospital Meerut under Para 56 of DWP 2020	Military Hospital Meerut
61		Rainwater Harvesting Pit for provision of deficient other than married accomodation including guard room at HQ Madhya UP Sub-area camp at Lucknow Cantt.	Lucknow Cantt.
62		Rainwater Harvesting Pit for provision of Children School at Lucknow Cantt.	Lucknow Cantt.
63		Provn. of Technical Accn and OTM Accn for Military Hospital Meerut (AMWP-2021-22/LIST V/SER NO 01)	Military Hospital Meerut
64		Provision of Tech. Acc. and Ailled infrastructure for Command Hospital at Lucknow by Garrison Engineer, MSE, Lucknow	Garrison Engineer, MSE, Lucknow
65		Construction of Regional Network Centre	Indian Army, Telibagh

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
		(RNC) for Unified Network Management System (UNMS) under NFS Project for Indian Army, Telibagh Lucknow (Seeking suggestion for new deep borewell)	Lucknow
66		Provision for Adm. Block, Cook house, Veh. Sheds and single JCO's Acc for 6 JCO's at Lucknow	Lucknow Cantonment, Lucknow
67		Site Selection for construction of Roof Top Harvesting Structure for HQ 71 MTN BDE at Lucknow Cantonment Area Lucknow, UP- Recommendation of Recharge well diameter	Lucknow Cantonment Area Lucknow
68		Inspection of two cricket stadium in UP against the use of Groundwater for maintainace of Cricket Play Ground- Haider Ali vs Union of India- 1.Ekana Stadium, Lucknow 2.Green Park Stadium, Kanpur	Haider Ali vs Union of India
69		Survey of Site selection for boring of new borewell near PAO Pump house under GE E/M Lucknow	Garrison Engineer, Lucknow
70		Provision for Adm. Block, Cook house, Veh. Sheds and single JCO's Acc for single JCO's at Lucknow	Lucknow Cantonment, Lucknow
71	**NER Guwahati**	A brief report on feasiblity for extarction of groundwater at ICAR, KVK , Hailakandi, Assam	ICAR,HAILAKANDI, ASSAM
72		A brief report on feasiblity for extarction of groundwater at ICAR, KVK , Kolasib, Mizoram	ICAR,KOLASIB, MIZORAM
73		Water Supply Investigation for Hq. Eastern Air Command, Upper Shillong and Air Force Station, Laitkor Peak, Meghalaya	Indian Air Force, EAC, Shillong
74		Water Supply Investigation for Umroi Military Station, Barapani, Meghalaya	Indian Army, Barapani, Meghalaya
75		Water Supply Investigation for Special Bureau, Lumbatbgen, Nongthymmai, Shillong, Meghalaya	Special Bureau, Ministry of Home Affairs, Shillong
76		Water Supply Investigation for SHQ, BSF, Mawpat (Dumpling), Shillong, Meghalaya	Border Security Force, Mawpat, Shillong
77		Feasibility report for construction of borewell at U Lumsohpetbneng, Ri Bhoi, Meghalaya	PHED, Govt. of meghalaya
78		Water Supply Investigation at Umladkhur Village, West Jaintia Hills District	District Administration, West Jaintia Hills District
79		STWSI at Laskein, West Jaintia Hills, Meghalaya	District Administration, West Jaintia Hills District

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
80		STWSI at Sohmynting, West Jaintia Hills, Meghalaya	District Administration, West Jaintia Hills District
81		Feasibility survey for construction of GW structures at IOCL, Noonmati, Guwahati, Assam	Indian Oil Corporation Limited (IOCL), Noonmati, Guwahati, Assam
82		Feasibility survey for construction of GW structures at 54 BN ITBP, Bhalukpong, Sonitpur, Assam	54 BN, Indo-Tibetan Border Police Force (ITBP), Sonitpur Assam
83		Feasibility survey for construction of GW structures at BSNL campus, Dispur, Guwahati, Assam	BSNLcampus, Dispur, Guwahati, Sonitpur, Assam
84		Feasibility survey for construction of GW structures at ICAR, Guwahati, Assam	ICAR, Guwahati, Assam
85		A Brief Report On Hydrogeological Investigation At The Premises Of Assam Police Reserve, Rupnagar, Guwahati, Assam	Assam Police
86		A Brief Report On Hydrogeological Investigation At The Premises Of Assam Police Reserve Near 10th Apbn, Kahilipara, Guwahati, Assam	Assam Police
87		A Brief Report On Construction Of A Ground Water Abstraction Structure In The Premises Of Frontier Hq, Border Security Force, Patgaon, Kamrup(M) District, Assam.	Border Security Force
88		Construction Of Groundwater Abstraction Structure In The Premises Of Army, Gerua, Tamulpur Under MES, Rangia Kamrup (R) District, Assam	Army, Gerua, Tamulpur Under MES, Rangia Kamrup (R) District, Assam
89		Construction Of A Groundwater Abstraction Structure At Bholapara Village Piped Water Supply Scheme Under Jal Jivan Mission, Rampur Block, Kamrup District, Assam	JJM, PHED, Govt. of Assam
90		Report On Hydrogeological Investigation At Narangi Military Station, Kamrup (M) District, Assam	Narangi Military Station, Kamrup
91		Note On Hydrogeological Investigation At The Premises Of Married Accomodation, Shantipur Military Area, Guwahati, Assam	Shantipur Military Area, Guwahati
92		Note On Feasibility For Construction Of A Deep Tubewell At The Proposed Site In The Premises Of Ganeshbari Camp Mohanbari Military Area, Dibrugarh District, Assam.	Mohanbari Military Area, Dibrugarh

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
93		Note On Feasibility For Construction Of A Deep Tubewell At The Proposed Site In The Premises Of Dinjan Military Station, Tinsukia District, Assam.	Dinjan Military Station, Tinsukia
94		Hydrogeological Investigation At Lakshmibai National Institute Of Physical Education (Govt. Of India, Ministry Of Youth Affairs & Sports), North East Regional Centre, Tepesia Sports Complex, Sonapur, Guwahati, Assam Kamrup (M) District, Assam	Tepesia Sports Complex, Sonapur,
95		Report On Hydrogeological Investigation In The Defence Area, At Suboti Camp, North Lakhimpur, Lakhimpur District, Assam	Defence Area, North Lakhimpur,
96		A Brief Report On Hydrogeological Investigation At The Premises Of Assam Legislative Assembly, Dispur Guwahati, Assam	Assam Legislative Assembly, Dispur
97		A Brief Report On Hydrogeological Investigation At The Premises Of Jyoti Chitraban, Kahilipara, Guwahati, Assam	Dept. of Cultural Affairs, Govt. of Assam
98		A Brief Report On Hydrogeological Investigation At ISPW Station, Directorate Of Coordination Police Wireless, Moha, Latakata, Kamrup(M), Guwahati, Assam	Directorate of Coordination Police
99		Report On Hydrogeological Investigation At Yuba Niwas Sanstha, Narengi, Guwahati, Assam	District Administration, Kamrup (M) district, Assam
100		A Brief Report On Hydrogeological Investigation At The Premises Of Matri Kalyan Old Age Home, Hengrabari, Guwahati, Assam	PHED, Govt. of Assam
101		Note On Hydrogeological Investigation At Surabala Bordoloi Suworoni PWSS, Geeta Nagar, Guwahati, Assam	PHED, Govt. of Assam
102		Note On Hydrogeological Investigation At Ghoramara Satra, Adarani Path, Cabin Chowk Pwss, Geeta Nagar, Guwahati, Assam	PHED, Govt. of Assam
103		A Brief Report On Hydrogeological Investigation At The Premises Of Assam Legislative Assembly, Dispur Guwahati, Assam	PWD, Govt. of Assam
104		Short Term Hydrogeological Investigation For Assessment Of Feasibility For Groundwater Extraction In The Premises	PHED, Govt. of Assam

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
		Of Multistoried Mla Hostel Under Construction At Dispur, Guwahati, Assam	
105	**WCR Ahmedabad**	Hydrogeological investigation for water supply to Naval Air Enclave, Porbandar, Taluka Porbandar, District Porbandar	The Garrison Engineer (Navy) Military Engineering Service, Porbandar
106	**SR Hyderabad**	Water Supply Investigation at Air Force Academy, Dundigal, Dundigal Gandimaisamma Mandal, Medchal-Malkajgiri District (Old Medak District), Telangana State	Ministry of Defence, Government of India
107		Water Supply Investigation at Employees' Provident Fund Organisation (EPFO) Quarters, Brahman Wadi, Begumpet, Hyderabad, Telangana State	Ministry of Labour and Employment, Govt of India
108		Water Supply Investigation at CRIDA Research Farm, Hayathnagar, Mandal Hayathnagar, District Ranga Reddy, Telangana state	Department of Agricultural Research and Education (DARE), Ministry of Agriculture and Farmers Welfare, Government of India
109		Water Supply Investigation at CRIDA Research Farm, Gunegal, Mandal Yacharam, District Ranga Reddy, Telangana state	Department of Agricultural Research and Education (DARE), Ministry of Agriculture and Farmers Welfare, Government of India
110		Water Supply Investigation at CRIDA head quarters at Santosh Nagar, Mandal Saidabad, District Hyderabad, Telangana state	Department of Agricultural Research and Education (DARE), Ministry of Agriculture and Farmers Welfare, Government of India
111		Water Supply Investigation at Jawahar Navodaya Vidyalaya, at Gopanapally, Nallagandla, Mandal Seri Lingampally, District Ranga Reddy, Telangana State	Ministry of Human Resource Development, Government of India
112	**SECR chennai**	Supply of water from Chinnakulam Eri to Pottanam panchayat	Pottanam Panchayat, Senthamangalam Block, Namakkal district
113		Report on Hydrogeological & Geophysical surveys carried out in the premises of OCF Estate , Avadi, Chennai, Tamil Nadu	Ministry of Defence
114		Report on Ground water investigation at Head Post office, Kancheepuram district,	Ministry of Communication

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
		Tamil Nadu	
115	**SWR Bengaluru**	STWSI at Nofra I, Vasco-da-Gama, Mormugao Taluk, South Goa District, Goa State	Nofra I, Vasco-da-Gama
116		STWSI at Karnataka State Sericulture Research & Development Institute, Thalaghattapura, Bangalore South Taluk, Bangalore Urban District	Karnataka State Sericulture Research & Development Institute, Bangalore
117		STWSI at Centre for Air Borne Systems, DRDO, Belur, Bangalore, Karnataka	CABS, DRDO
118		STWSI at Coastal Gaurd Land, Chicolna, South Goa District, Goa	Coastal Gaurd Land, Chicolna
119		STWSI at DRDO- Microwave Tube Research and Development Centre Township Site, Jalahalli, Yelahanka Taluk, Bangalore Urban District	DRDO
120		STWSI at ALISDA Tech Complex, Jalahalli Camp Road, Yeshwantpur Post, Bangalore	ALISDA Tech Complex
121	**SER Bhubaneshawar**	STWSI for feasibility of site for construction of Tubewell inLC-II Area of ITR, Chandipur, Balasore	DRDO
122		STWSI for feasibility of site for construction of Tubewell in LC-III Area of ITR, Chandipur, Balasore,Odisha	DRDO
123	**ER Kolkata**	STWSI at Indian Reserve Battalion, Port Blair, Andaman & Nicobar Islands	IRBn
124		STWSI at Military Station, Salugara, Jalpaiguri District, West Bengal	MES
125		STWSI at Military Forward Station, Bagrakote, Jalpaiguri District, West Bengal	MES
126		STWSI at Military Forward Station, Binnaguri, Jalpaiguri District, West Bengal	MES
127		STWSI at Military Station, Sukna, Darjeeling District, West Bengal	MES
128		STWSI at Military Station, Panagarh, Paschim Bardhaman District, West Bengal	MES
129		STWSI at Vidyasagar University, Paschim Medinipur District, West Bengal	Vidyasagar University, Govt. of West Bengal
130		STWSI at Sikkim Manipal Institute of Technology	SMIT, Sikkim Manipal University
131	**KR Trivanthpuram**	Short term Water Supply Investigation at Air Force Station,Ministry Of Defence, Government Of India, At Akkulam, Trivandrum district, Kerala State	Air force
132	**SUO Delhi**	Feasibility report on artificial recharge to Ground Water in Delhi Cantt Area for solving the surface water logging problems	MES

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
133		Hydrogeological investigation report for construction of tube well at A1 Defence land, Shakurbasti, Delhi	MES
134		Hydrogeological investigation report for construction of 02 tube well at Salaria Officers Enclave, Sec-21, Dwarka, Delhi	MES
135		Hydrogeological investigation report for construction of 02 tube well at Delhi Cantt, under GE(U) Water supply, Delhi	MES
136		Hydrogeological investigation report for construction of 04 tube well at CEDZ at Mehram Nagar, Delhi Cantt.	MES
137		Hydrogeological investigation report for construction of 01 tube well at Sekhon Vihar Md Accomodation complex area at Airforce Station, Palam.	MES
138		Hydrogeological investigation report for construction of 02 tube wells at Centre for United Nations Peace Keeping (CUNPUK) at New Delhi (SPL WK) AMWP 2017-18, Shankar Vihar, Delhi Cantt.	MES
139		Inspection of Hiagshi Autism School for the available recharge structures	Higashi Autism School
140		Hydrogeological investigation report for construction of 04 tube wells at Jailerwala Bagh, Ashok Vihar, Delhi.	Delhi Development Authority
141		Hydrogeological investigation report for construction of 01 tube well at IHQ of MOD(Army) Camp, Delhi Cantt Phase-I of two phases.	MES
142		Hydrogeological investigation report for construction of 02 tube wells at Baird Place, Delhi Cantt.	MES
143		Hydrogeological investigation report for construction of 04 tube wells at CVD Lines, Delhi Cantt.	MES
144		Hydrogeological investigation report at 23 acres complex & 12 acres complex, Delhi Cantt, New Delhi	MES
145		Rainwater Harvesting structure at FIE, Pocket-A, Okhla Phase-II, New Delhi-110020	DSIIDC
146		Hydrogeological Investigation for constructure of 23 recharge structures along with the recharge wells at GPRA complex, New Moti Bagh, New Delhi	NBCC
147		Ground Water recharge system for greenhouse facility at CPCT, IARI PUSA, Delhi	PUSA Institute

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
148		Technical advise/recommendation/comments in connection with permission of borewells in the notified area of North West District	DM (North West District)
149		Technical advise/recommendation/comments in connection with permission of borewells in the notified area of North West District	DM (North West District)
150		Technical advise/recommendation/comments in connection with permission of borewells in the notified area of North West District	DM (North West District)
151		Clarification regarding high water table areas in Sadar Bazar, Chandni Chowk, Matia Mahal and Ballimaran constituencies	DJB
152		Information about the water table in area of Mayur Vihar Phase-I & Mayur Vihar Ext. in Patparganj Constituency along Noida Link Road	DJB
153		Clarification regarding high water table areas in Seelampur AC-65, Ghonda AC-66 & Karawal Nagar AC-70 constituencies	DJB
154		Clarification regarding high water table area in Assembly Constituencies Shalimar Bagh, Shakur Basti, Tri Nagar, Wazirpur and Madipur	DJB
155		Clarification regarding the high water table areas for extraction of Ground Water in Patparganj, Laxmi Nagar, Vishwas Nagar, Krishna Nagar and Gandhi Nagar under ACE (M)-1	DJB
156	**SUO, Ranchi**	Report on the Feasibility for Construction of Abstraction Structure for Ground Water at Army Cantonment, Dipatoli, Ranchi, Jharkhand	Army Cantonment, Dipatoli, Ranchi
157		Report on the Feasibility for Construction of Abstraction Structure for Ground Water at 106 Bn RAF (CRPF) Campus, Sundarnagar, Jamshedpur, Jharkhand	106 Bn RAF (CRPF) Campus, Sundarnagar, Jamshedpur
158	**MER,Patna**	Feasibility study for identification of suitable sites for Ground Water Abstraction at 205, CoBRA Regiment, Barachatti, Gaya District, Bihar	CoBRA Commando Battalion
159		Feasibility study for identification of suitable sites for Ground Water Abstraction at Danapur Cantt Area, Patna District, Bihar	Danapur,MES,Patna

Sr no	Region	Title of Short Term Water Supply Investigation	Agency for which study was taken up
160	**CR, Nagpur**	Groundwater availability survey for IIIT Pune campus	IIIT, Pune

3.10 Important events

- A Workshop on NAQUIM 2.0 was held on 14th April, 2023 at CSMRS, New Delhi which was attended by Regional Directors of CGWB and senior officers from regions.
- Union Minister of Jal Shakti, Shri Gajendra Singh Shekhawat released the Dynamic Ground Water Resource Assessment (DGWRA) Report for the entire country for the year 2023 on 01.12.2023.

Fig. 1. Releasing of DGWRA Report 2023 for the entrire country by Union Minister of Jal Shakti, Sh. Gajendra Singh Shekhawat

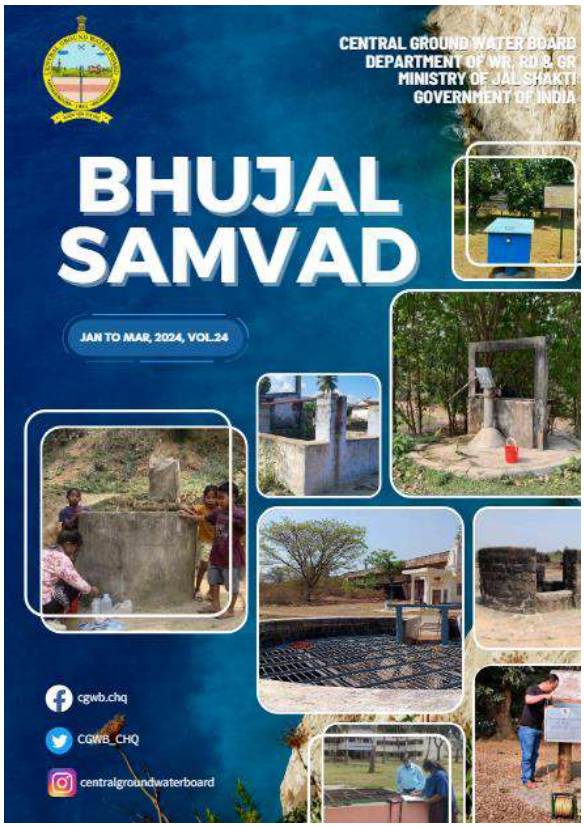
- Sh.Gajendra Singh Shekhawat, Union Minister of Jal Shakti inaugurated Dam at Indroka, district Jodhpur, Rajasthan under AR Project Rajasthan (Phase-I) in the august presence of Jogaram Patel, Minister of Parliamentary Affairs & Justice, Govt. of Rajasthan on 16.03.2024. Dr. Sunil Kumar Ambast, Chairman, CGWB, various dignitaries and officials from Ministry of Jal Shakti graced the occasion. The project was executed and funded by the CGWB under GWM&R Scheme for the construction of the Rainwater Harvesting Structures.

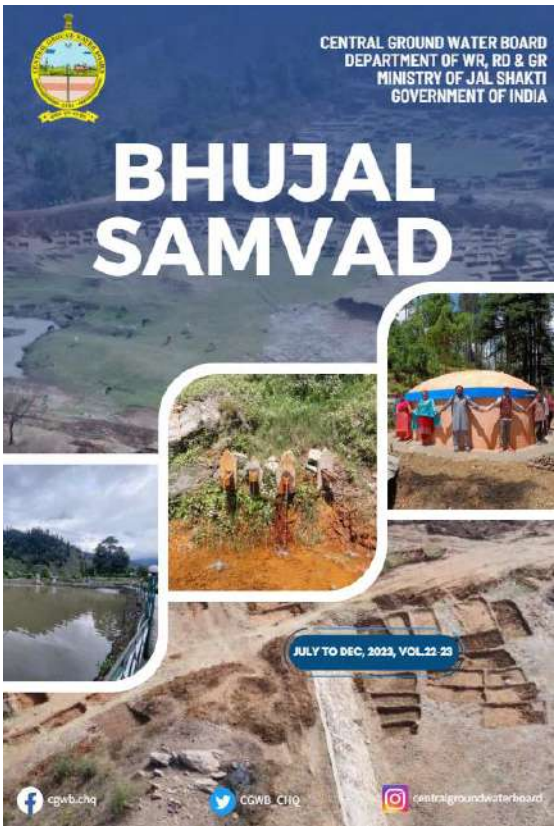
3.10.1 Bhujal Samwad

Bhujal Samvad, a quarterly publication by the Central Ground Water Board, was launched in January 2018. Since its inception, it has published 24 editions up to March 2024, all accessible through the official CGWB website. The magazine features distinct sections including the Cover Story, In Focus, NAQUIM Success Stories, Reports, Bhujal Samvad Talk Series, Pathshala, and Social Media Highlights and Collectables.

The 'Cover Story' section aims to highlight significant endeavours undertaken by the Board. 'In Focus' provides a quarterly overview of CGWB's primary activities and achievements. The 'Report' and 'Pathshala' segments educate readers on various hydrogeological subjects.

Bhujal Samvad is continuously evolving, and a new section called the 'Bhujal Samvad Talk Series' has been introduced. This section provides a brief overview of insightful talks delivered by eminent speakers from both within and outside the Board.





3.11 Public Interaction Programmes (PIP)

Under Azadi ka Amrit Mahotsva, Public Interaction Programmes were organized to sensitize people in respect of NAQUIM outcomes and water related aspects in all States. Details are as follows in table 1.

Table 1: List of PIPs conducted during the year (2023-24)

S. No.	State /UT	Region	Annual Target (2023-24) for PIP	PIPs done upto 31.03.2024	No of Total Participants	No of Female participants
1	Madhya Pradesh	NCR	**8**	8	754	349
2	Himachal Pradesh	NHR	**5**	5	658	250
1	Jammu and Kashmir	NWHR	**7**	6	609	289
4	Haryana	NWR	**5**	5	628	240
5	Punjab & Chandigarh	NWR	**6**	6	583	238
6	Gujarat	WCR	**8**	8	824	257
7	Rajasthan	WR	**8**	8	680	230
8	Uttar Pradesh	NR	**8**	8	1177	564
9	Delhi	SUO	**2**	2	389	191
10	Uttarakhand	UR	**5**	5	421	173
11	West Bengal	ER	**11**	11	879	420
12	Bihar	MER	**5**	4	478	209
13	Jharkhand	MER	**5**	4	340	151
14	Chhattisgarh	NCCR	**8**	7	843	256
15	NER	NER	**10**	9	1659	867
16	Odisha	SER	**8**	7	781	322
17	Maharashtra	CR	**8**	8	722	317
18	Kerala	KR	**5**	5	84	42
19	Tamil Nadu	SECR	**10**	10	1135	314
20	Andhra Pradesh	SR	**5**	6	714	90
21	Telangana	SR	**5**	6	740	161
22	Karnataka	SWR	**8**	8	789	417
	Total		**150**	**146**	**15887**	**6347**

4. Public Investment Board (PIB) Project for Strengthening NAQUIM's Data Generation Infrastructure

4.1 Introduction of Public Investment Board (PIB)

The activities proposed under the National Aquifer Mapping and Management Project were originally proposed as part of the Central Sector Scheme "Ground Water Management and Regulation" under Component I (Aquifer Mapping and Management Programme, NAQUIM) of the Scheme for implementation during the period 2021-26. As per the recommendations of the EFC meeting of GWMR scheme held on 31st August 2021, proposals for three sets of activities of time bound nature were prepared and presented before the Public Investment Board (PIB) Committeee and approved as NAQUIM(PIB) Project under the GWMR scheme. The NAQUIM(PIB) project has three components, namely: (i) Automation of infrastructure for water level monitoring through construction of Piezometers and installation of Digital Water Level Recorders (DWLRs) in the country. (ii) High resolution aquifer mapping through heli-borne geophysical surveys and (iii) Data generation for aquifer mapping through well construction with an approved outlay of Rs.805 crore.

4.1.1 Creation of Infrastructure for high density automated monitoring of water levels

It is proposed to create infrastructure for automated monitoring of water levels to generate high frequency water level data in areas falling under assessment units that have been categorised under Over-exploited (OE), Critical and Semi-Critical category and other priority areas like large urban agglomerations, industrial clusters, mining areas, coastal areas, seismic zones IV and V and other such assessment units where adequate monitoring networks are not available. It is proposed to install Piezometers with Digital Water Level Recorders and telemetry in such areas for generation of high frequency water level data. The exercise would be taken up in convergence with the National Hydrology Project (NHP) and Atal Bhujal Yojana. Considering existing piezometers with CGWB and State Governments and those proposed under NHP and Atal Bhujal Yojana, additional requirement of piezometers has been estimated as 9,000. The actual requirement may vary depending upon field conditions like availability of suitable sites, security issues etc. Out of the 9000 Piezometers, 2000 piezometers are planned to be constructed through in-house resources of CGWB under the Ground Water Management and Regulation Scheme. Construction of remaining 7000 piezometers and installation of 7000 DWLR with telemetry is envisaged to be implemented through outsourcing under this project at an estimated expenditure of Rs 529 Cr. High frequency water level data would form a critical input to water budgeting and planning for ground water interventions at the Panchayat level in such areas besides providing inputs in planning for

source sustainability under Jal Jeevan Mission and participatory ground water management under Atal Bhujal Yojana.

4.1.2 High resolution Mapping of aquifers through heli-borne geophysical survey in nearly 3 lakh sq km area in arid parts of NW India

It is proposed to take up high resolution mapping of aquifers on 1:10,000 scale in ~3 lakh sq km area covering the arid parts of the States of Rajasthan, Gujarat, Haryana, Punjab and Himachal Pradesh using heliborne geophysical studies. The heli-borne study is aimed at providing 3D Geophysical model of the area, geophysical thematic maps at horizontal and vertical planes, Aquifer Geometry in 2D and 3D, Aquifer system with fresh and saline zones and suitable sites for artificial recharge. Spatial distribution of paleochannel network, if any, and its linkage with aquifer system would also be studied. The high resolution maps would help in formulating site specific plans for improving groundwater situation in the water stressed regions. An outlay of Rs 151 crores is envisaged towards conducting heliborne survey along with associated ground geophysical surveys covering 3 lakh sq km during 2022-25.

4.1.3 Data Generation for Aquifer Mapping through construction of 1135 wells in parts of 11 States of the country

National Aquifer mapping and Management Programme was initiated under the Ground Water Management and Regulation Scheme during the XII Plan (2012-17) period. Under the programme, an area of nearly 24.8 lakh sq km of the country has been identified for mapping out of which an area of 21.15 lakh sq km has already been covered upto March 2022. To meet the data generation requirement for completing aquifer mapping on 1:50,000 scale by March 2023, drilling of 1135 wells through outsourcing has been envisaged in parts of 11 States of the country. This activity is proposed to be taken up through outsourcing. The data generated through outsourcing will be used for preparation of aquifer maps and management plans. An outlay of Rs 125 Crore has been envisaged towards this activity.

4.2 Project Activities and Outputs

The outputs of this project will serve as inputs for preparing of aquifer maps and management plans under the ground water management and regulation scheme. This information, in turn will be shared with state government departments for planning sustainable ground water management. Activity wise major outputs are given in the table 1 below.

Table 1: Activity wise major outputs of PIB

S.I. No	Item	Measurable Outcomes
1	Construction of 7000 Piezometers and installation of DWLRs with telemetry in OE, Critical, Semi-Critical assessment units and other priority areas	Creation of infrastructure for automated monitoring of water levels in water stressed assessment units. High frequency data would also provide scientific inputs for driving corrective action by State/Central agencies based on ground water level monitoring system. High density and high frequency water level data would form a critical input to water budgeting and planning for ground water interventions at the Panchayat level in such areas and in improved assessment of groundwater resources.
2	Heli-borne surveys for high resolution mapping in ~3 lakh sq km area in arid parts of NW India	High Resolution mapping of aquifers on 1:10,000 scale in arid parts of the country will help formulate better ground water management plans. Will help provide Pin-pointed sites and depth zones for taking up appropriate management interventions for water harvesting and recharge in water stressed areas.
3	Data Generation for Aquifer Mapping through construction of Wells in 11 States	Construction of 1135 wells for data generation for completing remaining works of Aquifer Mapping on 1:50,000 scale. Improved information about aquifer disposition, quantification of available ground water resources and assessment of groundwater quality.

4.3 Estimated Cost

Total estimated cost of the project is Rs 805 Cr (Rupees Eight Hundred and five crores) for the entire project period i. 2022-23 to 2025-26. Year wise and activity wise proposed outlay for the project are given in the table 2.

Table 2: Activity wise estimated cost of PIB project

Activity	Total Physical targets	Financial outlay (Rs in Cr)				
		2022-23	2023-24	2024-25	2025-26	Total
Construction of Piezometers	7000 no of piezometers	25	150	150	59	**384**
Installation of DWLRs with telemetry	7,000 no of DWLRs	-	21	62	62	**145**
Heli-borne surveys for high resolution mapping in 3 lakh sq km area in arid parts of NW India	~3 lakh sq km	45	45	61	-	**151**
Data Generation for Aquifer Mapping through construction of Wells in parts of 11 States	1135 number of wells	55	70	-	-	**125**
	Total	**125**	**286**	**273**	**121**	**805**

5. Rajiv Gandhi National Ground Water Training and Research Institue (RGNGWTRI) - non Scheme

5.1 Introduction

Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI) is the training wing of CGWB**,** under the aegis of Department of Water Resources, River Development & Ganga Rejuvenation, Ministry of Jal Shakti, Govt. of India with the vision as a "Centre of Excellence for Training and Research for Sustainable Development of Ground Water Resources of the Country"

Since its Inception in 1997, as India's first exclusive Training Institute in Groundwater studies to improve the capacity building of groundwater professionals for sustainable groundwater development & management with participatory approach, a total of 1, 18,255 participants, including ground water professionals as well as stakeholders at the grassroots level, participated in the 1755 programs conducted by the Institute.

The institute mainly conducts training courses for Officers of Central Ground Water Board (CGWB). The subject matters in the training courses cover various stages of their professional careers starting from the Induction, Mid-career and Senior Management levels. In addition to this, officers from other Central and State Government Organizations, Non-Governmental Organizations (NGO) as well as University Faculty Members and Students associated with the realm of ground water are also beneficiaries of RGNGWTRI training programs.

5.2 Three - tiered Training Programme

Since 2012 (XII Plan) RGNGWTRI has embarked upon a three-tier training system (Tier I- National Level, Tier II-State/ District Tier- III: Block Level) to improve the capacity building of groundwater professionals for sustainable groundwater development & management involving participatory approach to create a pool of trained resource persons who would be working towards sustainability of ground water resources across the country with community participation.

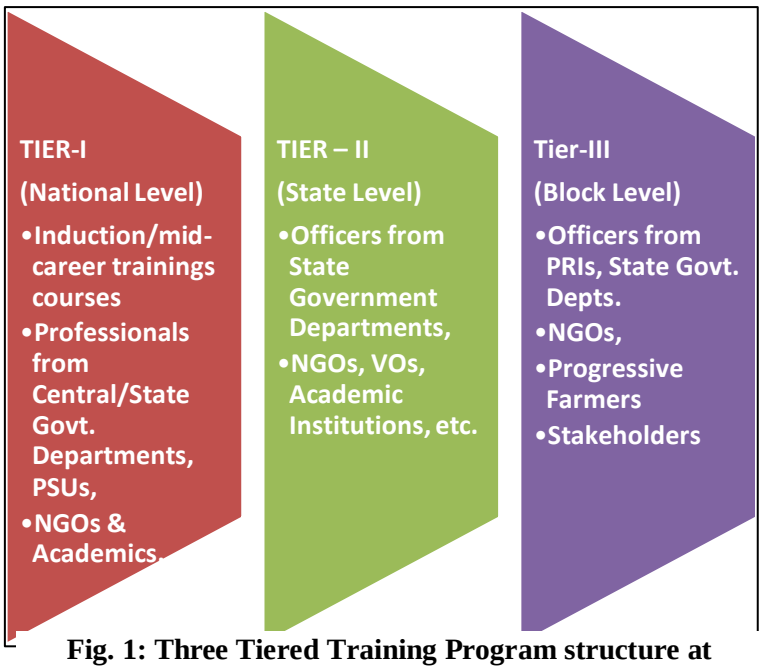


Fig. 1: Three Tiered Training Program structure at RGNGWTRI

The Tier I trainings are conducted at National Level and cover three verticals of capacity building *viz.*, domain specific trainings, behavioral and functional competency improving training programs. During 2023-24, 59 tier I trainings were organized by RGNGWTRI, against the target of 50, with a total of 966 participation (including 160 female participants). The tier I training also included:

- 29 mandatory trainings for Scientific, Ministerial and Engineering cadres of CGWB,
- 3 demand driven trainings for Implementing Agencies and state government agencies,
- 3 Trainings under National Hydrology Project,
- 2 Water Auditor Certification training and
- 6 trainings for synergy of the three Training institutes, RGNGWTRI, NERIWALM & NWA ofr Ministry of Jal Shakti.

RGNGWTRI as a part of Tier I training progammes organizes Induction Level Trainng Cources (ILTC) for newly recruited officers of CGWB. In the year 2023, RGNGWTRI has introduced the newly designed ILTC of one year duration. The first batch under this newly designed 1 year ILTC programme started in October 2023.

Fig. 2: Sh. Satish Kumar, Member (HQ), CGWB interacting with newly recruited officers of CGWB, during ILTC

Tier II trainings programs are conducted at the State level for capacity building. These trainings are organized through CGWB Regional Offices. During the AAP 2023-2024, the Tier II State Level training programs were held on the theme “Local Ground Water Management Issues”. A total of 20 training courses against the target of 20 were organized in the entire country from April 2023 to March 2024, in which 555 trainees have participated including 156 women participants.

During the AAP 2023-2024, 49 Block level (Tier-III) trainings on "Local Ground Water-Related Issues and Participatory Ground Water Management" were organized to build capacity at the village/ block level. All 49 trainings were organized in physical/offline mode.

Summary of the Targets and Achiements are as given below in table 2.

Table 2: Summary of Targets & Achievements of training courses in AAP 2023-2024

Training Courses	Targets Training	Targets Trainees	Achievements (Nos.)	No. of Participants	Male	Female
TIER–I (National Level) at RGNGWTRI	**50**	1000	59	966	806	160
TIER–II (StateLevel)*	**20**	400	20	555	399	156
TIER–III (Block Level)*	**50**	6000	49	6002	3522	2480
	120	**7400**	**128**	**7523**	**4727**	**2796**

*Tier–II and Tier III training courses are conducted through all the Regional Offices of CGWB.

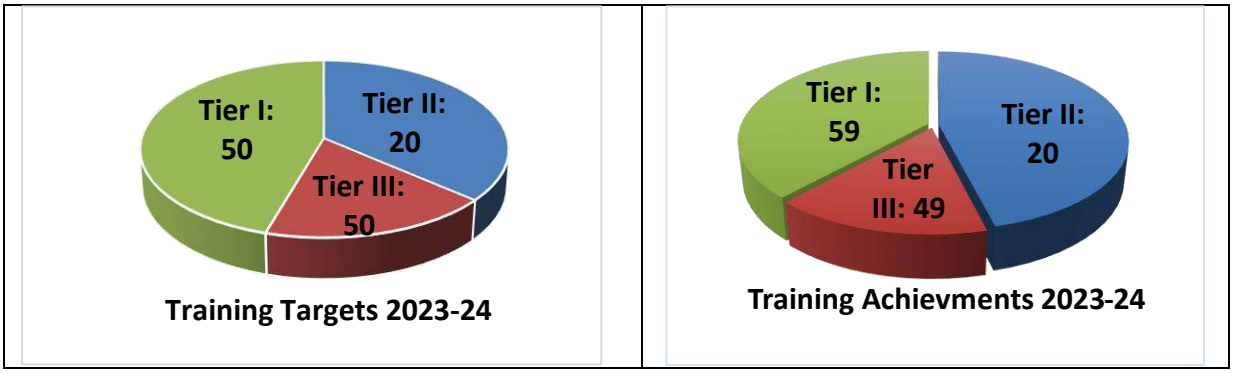


Fig 3: TrainingTargets and Achievments in 2023-24

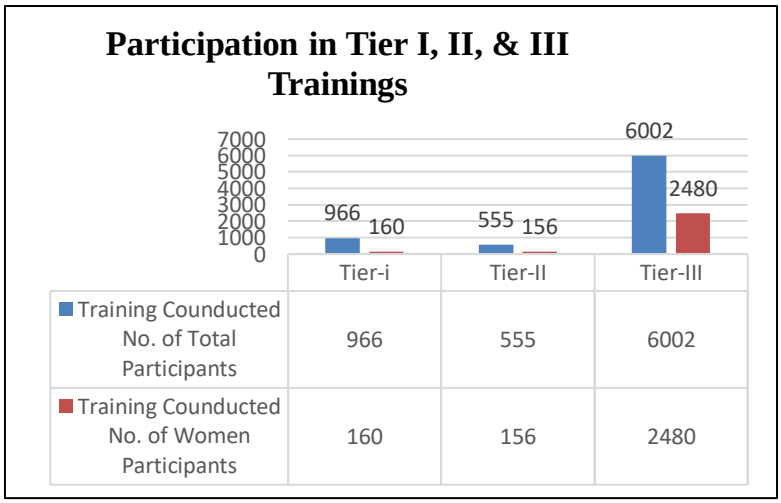


Fig 4: Participation in Trainings in 2023-24

5.3 BHUJAL News

Bhujal News is a quarterly Journal of CGWB which is being published by Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI), Raipur from the year 2020.

The works in CGWB are mostly field-based and the outputs are measured by the reports based on these field studies. Bhujal News serves as a medium to highlight CGWB's contribution in the field of ground water.

RGNGWTRI, Raipur has published NAQUIM Special Volume - showcasing the outcomes of the NAQUIM 1.0 in the Bhujal News, Volume 33 (2023-24, No 1-4) in the year 2024. It includes 25 papers that covers Aquifer mapping in various hydrogeological environments, alongwith Geophysical, and Chemical aspects of ground water in different aquifer systems.

5.4 Accreditation of Ground water Professionals

Accreditation is a process of instilling assurance and confidence amongst all stakeholders to submit reports to acceptable standards. As per the guidelines of Central Ground Water Authority (CGWA), applicants who seek permission for Ground Water extraction hire the services of accredited individuals or institutions for conducting hydrogeological survey and prepare a report on the groundwater scenario and the impact of the proposed extraction on the ground water reservoir. RGNGWTRI is entrusted with the screening and scrutiny of applications received from individuals/ Institutions for accreditation, conducting examinations, and issuance of certificates. RGNGWTRI has conducted four rounds of Accreditation till 2023-24. The details of the 'Accredited individuals/ Institutes' are as below:

- Total number of Groundwater professionals accredited are : 66
- Total number of Institutions accredited are: 30

5.5 Training Programme for Water Auditors Certification

RGNGWTRI has completed two batches of training on Water Auditors Certification and a total of 33 water auditor certification has been awarded during 2023-24. The trainee participants in both batches were from FICCI, CII, Industries and former Scientists and Engineers from organizations in water sector.

Fig. 5: Water Auditor Certification training first batch, April 2023

5.6 Other Important Trainings programs

- 2 days Training program on synergy among all the Training Institutes of Ministry of Jal Shakti (MoJS) was organized at at NERIWALM, Tezpur, Assam. The program was presided over by Ms. Debashree Mukherjee, Special Secretary, DoWR, RD&GR, MoJS, who delivered a keynote address urging for the necessity of synergy among all training institutions/ centers of the country.

Fig. 6: Ms. Debashree Mukherjee, Special Secretary, DoWR,RD&GR, MoJS presiding over the inaugural session of 2-Days training programmes at NERIWALM, Tezpur, Assam

- **Organised National Workshop on Training Needs Assessment** for Water Resources Development & Management in collaboration with NWA Pune, Maharashtra and NERIWALAM Tezpur on 7. 7.2023 at SCOPE Convention Centre New Delhi.

5.7 Other Achievements

- Six e-Learning modules on Atal Bhujal Yojana were created and uploaded in iGOT Karmayogi portal by RGNGWTRI. A glimpse of the portal is shown below in fig. 7.

Fig 7: Six e-Learning modules on Atal Bhujal Yojana created and uploaded in iGOT Karmayogi portal by RGNGWTRI

- **RGNGWTRI** has been accredited as □□□ □□□□□ under the **Capacity Building Commission's National Standards** assessed by the **National Accreditation Board of Education and Training (NABET)** on 22.11.23. The accreditation was awarded after the completion of the desktop and onsite assessment of the activities based on the eight pillars of capacity building. The accreditation is valid for a period of 2 years.

CAPACITY BUILDING COMMISSION

Certificate of Accreditation

NABET

The training institute **Rajiv Gandhi National Ground Water Training and Research Institute, Raipur** is accredited as **अति उत्तम** under the Capacity Building Commission's National Standards as assessed by National Accreditation Board of Education and Training (NABET).

The Accreditation shall remain in force subject to continued compliance with the National Standards. The accreditation needs to be renewed before the expiry date by the training institution following the due process of assessment.

Accreditation granted on **22/11/2023** is valid until **21/11/2025**

Secretary — Capacity Building Commission

Chairperson — Capacity Building Commission

Certificate No. APP/23/N00084

- **Sanyukt Hindi Kaaryshala (Joint Rajbhasha Workshop):** A joint Hindi workshop was organized on 21.12.2023 under the aegis of RGNGWTRI, Raipur. More than 30 officials from 6 Government of India offices participated in the workshop. The workshop was chaired by THE Regional Director, RGNGWTRI, Raipur.

- 10 Students from Lucknow University, Uttar Pradesh took M.Sc. dissertation under the aegis of RGNGWTRI in collaboration with CGWB, NR, Lucknow during 2023-24.

6. National Hydrology Project (NHP)

6.1 Introduction

National Hydrology Project (NHP**)** is a continuation of Hydrology Project (HP)- I and HP- II, a central sector scheme of the Department of Water Resources, RD&GR, Ministry of Jal Shakti, Government of India with a total outlay of Rs. 3679.77 Crore. The total duration of the project is of 08 years starting from 2016-17 to 2023-24, which has now been extended for a period of one-and-half years more from April 2024 to Sep 2025. Out of the total outlay of Rs. 3679.77 Crore, 50% of the amount has to be repaid by Central Government to the Word Bank (WB) and the remaining 50% would be Central Assistance. The project will be implemented by 48 agencies involving 9 Central and 34 States agencies, 02 UTs and 03 River Basins. Central Ground Water Board (CGWB) is one of the Central implementing agencies under the National Hydrology Project.

6.2 Objective of NHP

To improve the extent, quality, accessibility of water resources information, decision support system for floods basin level resource assessment/ planning strengthen the capacity of targeted water resources professionals and management institutions in India.

The components of NHP are:

- Component A: Water Resources Monitoring systems
- Component B: Water Resources information systems
- Component C: Water Resources Operation and Planning systems
- Component D: Institutional Capacity Enhancement

NHP will improve and expand hydrology data and information systems, strengthen water resources operation and planning systems, and enhance institutional capacity for water resources management. The project will strengthen the information base and institutional capacity for effective decision making in water resources planning and operational management at the basin level across the country using the latest available technology and tools. NHP will also help in contributing to the Digital India initiative by integrating water resources information across state and central agencies.

6.3 Role of CGWB in NHP

CGWB being the Nodal Agency for Groundwater in the country, NHP cell in all the Regional Offices have been created for providing Technical Guidance & Support to States in the implementation of NHP.

The proposed major activities of CGWB under NHP

- **Real time monitoring Water Quality in coastal aquifers of Tamil Nadu & UT of Puducherry** through construction of 60 piezometers & installation of 60 DWLRs.
- **Establishment of Center of Excellence for groundwater modeling** for providing technical support to the Central/ State Agencies in Ground Water Flow Model Studies.
- **Real time monitoring of Water Level through Installation of 5200 additional DWLRs (Phase-1 & 2) in the existing Piezometers** located in the Overexploited, Critical and Semi-Critical Blocks of the Country.
- **Capacity Building** of Officers from CGWB, State / other Central implementing agencies through National and International level trainings every year.
- **Awareness Campaigns** are conducted at six places every year on state specific groundwater issues.
- Organizing **National Level Workshops** on need based Topics.

The Status of various activities of CGWB under NHP along with photographs are as below:

- **Real-time Monitoring of Water Quality through 60 DWLRs with EC Probe:** The activity construction of 60 Piezometers in coastal aquifers of Tamil Nadu & UT of Puducherry and Installation of 60 DWLRs for Real time monitoring of Water Level & Water Quality has been completed in the month of Oct-2021 and July-2022, respectively. Data from all the 60 Stations are started sharing with Water Information Management System (WIMS) server of National Water Informatics Center (NWIC), Department of Water Resources, River Development and Ganga Rejuvenation. Based on the preliminary study of DWLR data obtained for the period of August 2022 to April 2023 SECR, Chennai has submitted an interim report on the real time water quality analysis.
- Electrical conductivity of more than 30000 µS/cm has been observed in few Piezometers located in Ramanathanpuram, Tirunelveli, Tuticorin districts. In order to study the high level of salinity in ground water, a detailed study has been taken up by SECR, Chennai by way of collecting more number of water samples in and around all 60 piezometers during pre-monsoon and post-monsoon period.

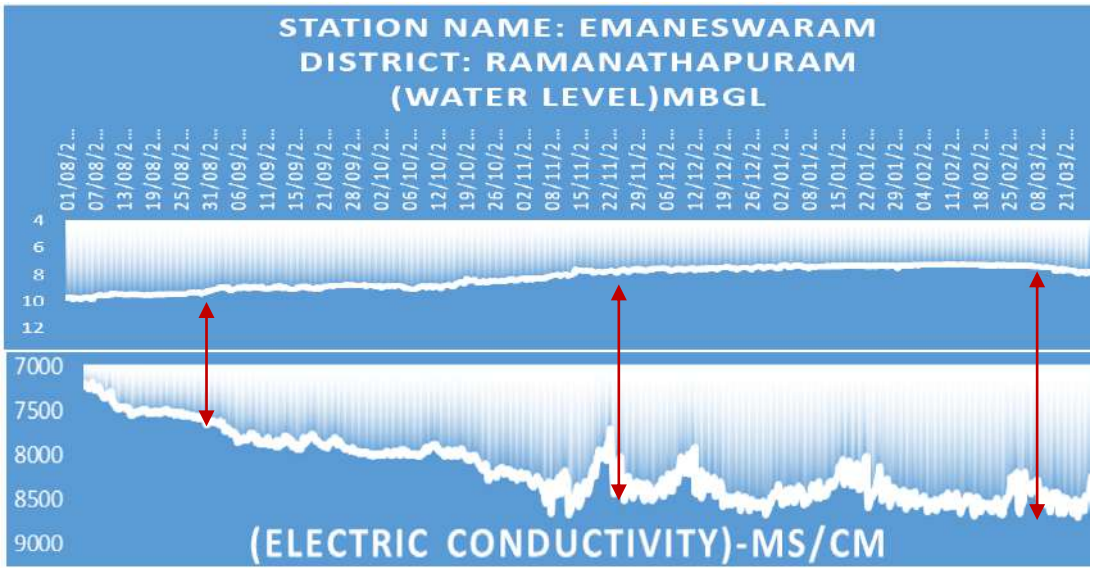


Fig- 1.: Combined chart of Hydrograph and Electrical conductivity of Emaneshwaram, Ramanathapuram District of Tamil Nadu

- **Installation, commissioning and Data Integration of 5200 DWLRs:** Installation, commissioning and Data Integration of 5200 DWLRs procured under Phase-1&2 are its various stages of completion. The state-wise summary of these DWLRs are given in table 1 below.

Table 1: Installation, commissioning and Data Integration of 5200 DWLRs procured under Phase-1&2

Sl. No	Regional Office In-charge	State / UT	3400 DWLRs	1800 DWLRs	Total DWLRs Awarded	Installed & commissioned	No. of Stations Integrated with WIMS Server of NWIC
1	2	3	4	5	6 = 4+5	7	8
1	CR, Nagpur	Maharashtra	148	29	177	177	169
2	KR, Trivandrum	Kerala	226		226	226	220
3	MER, Patna	Bihar	7	16	23	23	21
4	NCCR, Raipur	Chhattisgarh	124	144	268	269	177
5	NCR, Bhopal	Madhya Pradesh	291	100	391	391	361
6	NER, Guwahati	Arunachal Pradesh	4		4	4	4
7	NER, Guwahati	Assam	24	38	62	62	54
8	NER, Guwahati	Meghalaya	8	8	16	13	8
9	NER, Guwahati	Nagaland	5		5	1	1
10	NER, Guwahati	Tripura	7		7	6	6
11	NHR, Dharamshala	Himachal Pradesh		10	10	10	9
12	NR, Lucknow	Uttar Pradesh	34	138	172	172	136
13	NWHR, Jammu	Jammu & Kashmir, UT	14	20	34	34	32
14	NWR, Chandigarh	Chandigarh, UT	19	1	20	20	18

15	NWR, Chandigarh	Haryana	245	5	250	214	205
16	NWR, Chandigarh	Punjab	199	60	259	294	283
17	SECR, Chennai	Puducherry	7		7	7	7
18	SECR, Chennai	Tamil Nadu	468	165	633	633	627
19	SER, Bhubaneswar	Odisha	50	52	102	102	97
20	SR, Hyderabad	Andhra Pradesh	119	161	280	280	278
21	SR, Hyderabad	Telangana	397	120	517	517	509
22	SUO, Delhi	Delhi	51	41	92	92	90
23	SUO, Ranchi	Jharkhand	5	15	20	18	17
24	SWR, Bengaluru	Goa		39	39	39	36
25	SWR, Bengaluru	Karnataka	260	134	394	394	369
26	UR, Dehradun	Uttarakhand	1	11	12	12	12
27	WCR, Ahmedabad	Diu UT	1		1	1	0
28	WCR, Ahmedabad	Gujarat	295	116	411	411	377
29	WR, Rajasthan	Rajasthan	304	258	562	562	439
30	ER, Kolkata	West Bengal	87	119	206	206	155
	Total		**3400**	**1800**	**5200**	**5190***	**4717***

** Installation and commissioning of remaining 10 No DWLRs under progress and likely to be completed in a week of time*

** Due to signal issues 473 stations are not sharing data with NWIC server which are under technical rectification by vendors*

Fig 2: Data Integration and Data View on CGWB dashboard

- **Workshop on State Water Informatics Center (SWIC)** was organized by NWIC along with NHP in New Delhi on 09. Jan.2024. The basic data base of SWIC application, tools developed for data view and analysis, applications, etc. and overall performance of State Water Informatics Centers were assessed by a committee of Senior Level Officers including an Officer from CGWB.

Fig 3: Glimpse of SWIC workshop held in New Delhi on 09.01.2024

- **5th Meeting of the National Level Steering Committee(NLSC)** under National Hydrology Project (NHP) was conducted on 10.Jan.2024 in New Delhi under the Chairpersonship of Secretary, Department of Water Resources, River Development and Ganga Rejuvenation (DoWR, GD&GR). The status of RTDAS Supply, Installation and Data Integration with WIMS server of NWIC was reviewed by Secretary, DoWR, RD&GR. Subsequently, CGWB has emphasized NPMU & World Bank to promote more indigenous vendors (make in India) in the RTDAS service, as majority of the RTDAS integral components such as Lithium Batteries, GPRS modem, Data logger-IC, Real-time clock, Resister, Capacitor, Pressure Sensor, polyurethane, etc. are imported from other countries like China, Singapore, Taiwan, US, Europe which consumes more time, eventually delays the project schedule.

Fig 4: Glimpse of 5th NLSC meeting held in New Delhi

- **International Training**
  CGWB in co-ordination with IHE Delft the Netherlands has organized 03 week customized International Training on Ground Water Modelling for 25 No. of Officers from CGWB (15 Nos), NPMU NHP (02 Nos), and State Government Departments (08 Nos) during Sep 2023.

Fig 5: Group photo of Indian Officers in IHE Delft the Netherlands

- **National Training**
  03 Nos Domains Specific Training for Officers from CGWB, State Government Departments were organized through Rajiv Gandhi National Ground Water Training and Research Institute. There are 22 participants were attended the trainings (State Government – 16 Nos + CGWB – 06 Nos).

Fig 6: The group photo of trainees along with RGI faculties

- **Meeting cum Discussion** with Andhra Pradesh and Telangana State Ground Water Departments (GWD) regarding mechanism adopted for High-Frequency Data Storage, Data view and Data Validation, etc. in Telengana on 12th, Feb 2024. The glimpses of meetings are given below in Fig. 7.

Fig. 7: Meeting with Vassar Lab, Hyderabad **Meeting with Telangana State GWD**

- **Factory Inspection** of M/s Swan Environmental Private Ltd, Hyderabad DWLR Manufacturing Unit in order to access the capacity of vendors in line with various units involved in the assembling/making of integral parts, facilities available in the Lab, testing of DWLRs in Lab conditions, Calibration mechanism and instruments available, IT cell, manpower capacity, etc.

Fig 8: Inspection of DWLR Manufacturing Units at Hyderabad

- **Site inspection DWLRs stations** and feedback to vendors for rectification of technical issues for seamless data transfer to SECR, Chennai RoDC servers and WIMS server of NWIC in state Tamil Nadu and Puducherry between dates 14-19th, Feb 2024. The filed inspection involved screening of DWLR server data of all 03 vendors.

Fig- 9: Meeting with RD, SECR Chennai and Vendors at RoDC Chennai

Fig 10: Glimpse of DWLR site Inspection in Tamil Nadu and UT of Puducherry

Fig 11: Filed Inspection in Andhra Pradesh State &Field view of WL & EC data and Onsite Data Calibration

- **Young Water Professional (YWP) Cohort-II:**

National Hydrology Project (NHP) in collaboration with Australian Water Partnership (AWP), is organising Young Water Professional (YWP) training course led by NHP, National Water Academy (NWA), Pune and Western Sydney University, Australia for the last 2 years with the programme objectives to equip YWPs with the necessary skills, knowledge, behavior and networks that will better enable them to contribute to the development & management of water resources in India and to address the competency and leadership needs and priorities of the water sector in India. Under this activity, CGWB has submitted a list of 15 Officers after careful examination of their work profile and out of which 06 of them are selected from CGWB for the training programme. The YWP Cohort-IItraining is likely to be start from 8th July 2024.

- **DWLR Trainings to Regional Offices:**

As per the contract agreement, the DWLR vendors have to provide hands-on training to CGWB Officers on basics of DWLR, Data calibration in field, Data retrieval, Data Integration with RoDC server of CGWB and WIMS server of NWIC. There are 12 No. of 1-day training on DWLR were organized by different vendors in 12 no. of States such as Telangana, Maharashtra, Karnataka, Kerala, Tamil Nadu, West Bengal, Odisha, Uttar Pradesh, Madhya Pradesh, Rajasthan, Gujarat and Chandigarh, in which more than 200 Officers/Officials from CGWB were participated.

7. Pradhan Mantri Krishi Sinchayi Yojana - Har Khet Ko Pani - Ground Water (PMKSY-HKKP-GW)

7.1 Introduction (PMKSY-HKKP-GW)

The Pradhan Mantri Krishi Sinchayee Yojana - Har Khet Ko Pani - Ground Water (PMKSY-HKKP-GW) initiative serves as a pivotal program aimed at enhancing irrigation facilities across agricultural lands in India. This scheme is designed to ensure that every field has access to adequate water resources, thereby promoting sustainable agricultural practices and improving crop productivity. The scheme, launched by Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti, envisages to provide irrigation facility for small and marginal farmers in areas having sufficient potential for future development of ground water.

The scheme provides financial assistance to States for assured ground water irrigation to small and marginal farmers with priority to SC/ST and women farmers. The funding pattern is in the ratio of 90:10 (Central : State) in case of NE/Hilly areas and 60:40 (C:S) in case of other areas. The scheme is applicable only in areas having stage of ground water development less than 60%, average rainfall more than 750 mm rainfall and having shallow ground water levels (less than 15 m bgl).

7.2 Details of PMKSY in AAP 2023-24

Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti has sanctioned 13 projects under PMKSY-HKKP-GW scheme since 2019 in 10 States namely Assam, Arunachal Pradesh, Gujarat, Nagaland, Manipur, Mizoram, Tripura, Tamil Nadu, Uttar Pradesh, Uttarakhand.

The total cost of these 13 ongoing projects are Rs. 978.00 Crore and Central Assistance is Rs.827.15 Crore of which 764.89 crore has already been released as on 31st March 2023 by the Ministry of Jal Shakti. No fund has been released during the year 2023-24.

During the year 2023-24, 84 wells have been constructed, 1058.3 Ha command area have been created and 624 number of small & marginal farmers have been benefited under the scheme. All the 13 projects have been completed with the help of combined effort of CGWB, DoWR, RD& GR, Ministry of Jal Shakti and concerned State Implementing Agency.

Cumulative physical progress, as on March 2024, 29779 wells have been constructed, 88547 Ha command area have been created and 67909 number of small & marginal farmers have been taken the benefit of PMKSY-HKKP-GW scheme so far.

The details & present status (As on 31st March 2024) of these projects are summarized in the following table 1 & 2:

Table 1: Year wise Fund released under PMKSY-HKKP-GW (till 31 March' 2024)

Sl. No	State	Projects	Project cost in crore	CA Released (As on 31 March'23)	CA released in 2019-20	CA released in 2020-21	CA released in 2021-22	CA released in 2022-23
1	Assam	Phase-I	246.69	439.29	132.87	13	37.8	3.69
2		Phase-II	292.96		0	10	242.29	0
3	Arunachal Pradesh	Phase-I	45.3	79.90	24.46	10	5.985	0
4		Phase-II	44.95		24.15	0	15.295	0
5	Gujarat	Gujarat	119.19	71.44	6	0	28.5	36.94
6	Nagaland	Nagaland	18.15	16.25	9.75	0	5.85	0.65
7	Manipur	Manipur	61.68	54.40	0	33.306	21.09	0
8	Mizoram	Mizoram	16.04	13.86	0	8.66	0	5.20
9	Tamil Nadu	Tamil Nadu	9.13	5.355	0	3.67	1.61	0.08
10	Tripura	Phase-I	13.31	43.63	7.15	0	2.38	0.264
11		Phase-II	48.34		0	0	26.1	7.735
12	Uttarakhand	Uttarakhand	15.89	13.72	0	1.36	12.36	0
13	Uttar Pradesh	Uttar Pradesh	46.37	26.69	16.69	0	0	10.00
	Total		**978.00**	**764.89**	**221.07**	**80.0**	**399.26**	**64.55**

Table 2: Cumulative Status of PMKSY-HKKP-GW Scheme (till 31st March 2024)

Cumulative physicalStatus of PMKSY-HKKP-GW Scheme As on 31st March'24				
S. No	**Projects**	**Wells to be constructed Target/ Achievement Nos.**	**Project Command Target/ Achievement Ha.**	**Beneficiaries Target/ Achievement Nos.**
1	Assam Ph-I	4779 / 4779	19116 / 19116	19643 / 19643
2	Assam Phase-II	4916/ 4916	19664/ 19532	17216/ 17200
3	Arunachal Pradesh Ph-I	473 / 473	1785/ 1785	3350/ 3350
4	Arunachal Pradesh Ph –II	519/ 519	1957/ 1957	3633/ 3633
5	Nagaland	262/ 262	667/ 667	264/ 264
6	Tripura Ph-I	231/ 231	339/ 339	851/ 851
7	Tripura Ph-II	890/ 890	2670/ 2670	1639/ 1639
8	Manipur	550/ 550	2057/ 2057	1445/ 1445

9	Mizoram	209/ 209	553/ 553	411/ 411
10	U.P.	14752/ 14752	36365/ 36365	15252/ 15252
11	Uttarakhand	206/ 206	1030/ 1030	1085/ 1085
12	Gujarat	1826/ 1826	1866/ 1866	1908/ 1908
13	Tamil Nadu	166/ 166	610/ 610	1233/ 1228
	Total	**29779/ 29779**	**88679/ 88547**	**67930/ 67909**

8. Jal Shakti Abhiyaan (JSA)

Jal Shakti Abhiyan (JSA) was launched by Ministry of Jal Shakti in the year 2019 in the 256 water stressed districts of the country. Jal Shakti Abhiyan: Catch the Rain (JSA: CTR) has become an annual feature. In continuation to this, JSA: CTR 2023 was implemented across the country with the theme of "Source Sustainability for Drinking Water" focusing on 150 districts of the country.

CGWB took part in the JSA: CTR campaign through 150 officers who were nominated as Technical Officers (TO) in the Central Team of JSA for various districts across the country along with Central Nodal Officer (CNO).

Glimpses of JSA:

Central Nodal Officer-JSA-CTR concludes 3-day visit to Ramban district

RAMBAN, NOVEMBER 22: Director (Law & Order), Internal Security-I Division, Ministry of home Affairs, Government of India, and Central Nodal Officer (CNO) for Jal Shakti Abhiyaan (JSA)- Catch the Rain (CTR), Nishtha Tiwari, along with Scientist Central Ground water, Chandreyee De, concluded their 3-day visit to the district Ramban.

During the visit, they toured to a number of locations in the district and interacted with officers of the District Administration, students and teachers, PRIs, Pani Samitis, and common masses and got feedback about the implementation of 'Catch the Rain' programme under Jal Shakti Abhiyaan. They inspected GSRs, FHTCs, gravity lines, filter plants, Water Supply Schemes and other infrastructure under the Jal Jeevan Mission besides reviewing various activities of the Implementation Support Agency (ISA).

Conducting detailed assessment of the crucial aspects of the program, the CNO reviewed implementation of the programme in the district. She reviewed the financial and physical progress achieved, so far, in the district and appreciated the efforts of the district administration and other stakeholders in effectively implementing the Jal Shakti Abhiyan in the district.

CNO and Scientist CGW sensitized the locals about the need for water conservation, its judicious use, protection of water sources and rain water harvesting to make safe and sufficient water available in future. CNO asked the PRIs to plan under various developmental schemes for conservation of water and rain water harvesting. She also emphasised on wider publicity and awareness on the importance of rain water harvesting and its techniques.

The Scientist CGW also reviewed the physical progress of Jal Jeevan Mission works through virtual mode. She interacted with different beneficiaries and sensitized them about the initiative 'Catch the Rain' 'Where it falls, When it falls'.

The program aims to raise awareness about the importance of water conservation, promote water harvesting and improve infrastructure for water storage, distribution and ground water recharge, she said.

She asked the concerned officers for maintenance and cleanliness of Amrit Sarovars and other water bodies. She also asked to submit detailed reports regarding water management projects and ensure Geo-tagging of all the water bodies.

Meanwhile, the central team also inspected the Jal Shakti Kendra, Ramban and asked the concerned officer to ensure proper functioning of Jal Shakti Kendra to address timely public queries.

Technical Officer from CGWB visited Ramban District, Jammu along with CNO on a field visit as part of JSA CTR 2023

Online meeting on JSA and Amrit Sarovar 2023 chaired by Hon'ble Principal Secretary, WSSD, Govt. of Maharashtra, Mumbai- attended by officers, from CGWB, CR, Nagpur

Post Monsoon Field Visit under JSA CTR-2023 in Kamrup district, Assam carried out by Shri Nitin Khade, IAS Joint Secretary, D/o Land Resources, MoRD and Officer from CGWB

9. Rajbhasha

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10. Vigilance and RTI

10.1 Vigilance

Out of 4 disciplinary cases that were sent to Ministry for preliminary investigation, 1 case is disposed off. one penalty has been imposed during the year 2023-24. 2 cases are under investigation by the CBI and 3 cases are under preliminary investigation in CGWB.

10.2 RTI

The opening balance of RTI applications of CHQ as on 01.04.2023 was 37 Requests and 08 Appeals. During the year 2023-24, 193 RTI requests and 68 appeals were received. Total RTI requests and RTI appeals disposed of were 230 and 53 respectively. Details are given in table 1 below.

Table 1: Status of RTI Applications for Year 2023-24 At CHQ

Received during the year (including cases transferred from other public authorities)	Opening balance at CHQ as on 01.04.2023	Decisions where requests/appeals rejected
Requests: 230	Requests: 37	Requests: NIL
Appeals: 53	Appeals: 08	Appeals: NIL

11. Budget and expenditure

Table 1: Budget Details for the financial year 2023-24

Name of the Scheme	Revenue			Capital			Total		
	BE	**RE**	**Expenditure**	**BE**	**RE**	**Expenditure**	**BE**	**RE**	**Expenditure**
GWM&R	8805.00	9500.00	8383.00	23350.00	16741.00	11019.00	32155.00	26241.00	19402.00
TSP	750.00	700.00	596.00	800.00	1059.00	424.00	1550.00	1759.00	1020.00
Total	**9555.00**	**10200.00**	**8979.00**	**24150.00**	**17800.00**	**11443.00**	**33705.00**	**28000.00**	**20422.00**
HQ-Establishment	28951.00	28970.00	27886.60	1498.00	1430.00	1142.03	30449.00	30400.00	29028.63
RGI	357.00	290.00	238.80	28.00	10.00	3.64	385.00	300.00	242.44
External Support (NHP)	136.00	44.45	14.94	1704.00	820.00	842.13	1840.00	864.45	857.07
Domestic Support (NHP)	0.00	67.67	0.00	0.00	0.00	3.07	0.00	67.67	3.07
Total (NHP)	**136.00**	**112.12**	**14.94**	**1704.00**	**820.00**	**845.20**	**1840.00**	**932.12**	**860.14**

12. Photo - Gallery

Photo Gallery of the various activities held during the year 2023-24

Welcome of Hon'ble Union Minister of Jal Shakti,
Sh. Gajendra Singh Shekhawat, at Ahmedabad Airport by Officers of WCR Ahmedabad

Sh. Prahlad Singh Patel, Hon'ble Minister of State for Jal Shakti & Food Processing Industries and Officers of CGWB, NCR Bhopal ion International Yoga Day Celebration at Jabalpur, MP

Sh Subodh Yadav, Joint Secretary, (Admin, IC & GW) MoJS interacting with the Regional Directors of CGWB during progress review meeting at CGWB, CHQ, Faridabad.

Dr Sunil Kumar Ambast, Chairman, CGWB delivered a talk in a Seminar on Water Resources in Kerala held , inaugurated by Shri Roshy Agustine. Honble Minister for Water Resources, Govt. of Kerala

Inaugural function of 3-Day training on "Mentoring Skill" was organized at RGNGWTRI, Naya Raipur. Training Session was chaired by Sh. Sourabh Gupta, RD, RGNGWTRI, Raipur, and Dr Ashwin Janaikar, RT-Mentoring, DoPT.

Scientist from CGWB, SUO, Jodhpur conducted Preliminary Yield Test (PYT) and prepared litholog at Indroka Dam (upstream), Jodhpur.

Painting competition and PIP conducted at Hamdard Public School, Sangam Vihar, Delhi. The programme was organized by CGWB, SUO, Delhi to popularize Goals of LiFE initiative for the students in order to develop the sustainable sources for future.

Study visit of Parliamentary Standing Committee carried out by Hon'ble Members of Parliament (Rajya Sabha & Lok Sabha) along with the officials from CGWB, BBMB, State Govt. Punjab, and Haryana on water resources 2022-23 in Chandigarh.

Officers of CGWB, SUO, Ranchi presented and shared NAQUIM report of Hazaribagh district, Jharkhand with DC Hazaribagh, Smt. Nancy Sahay, IAS.

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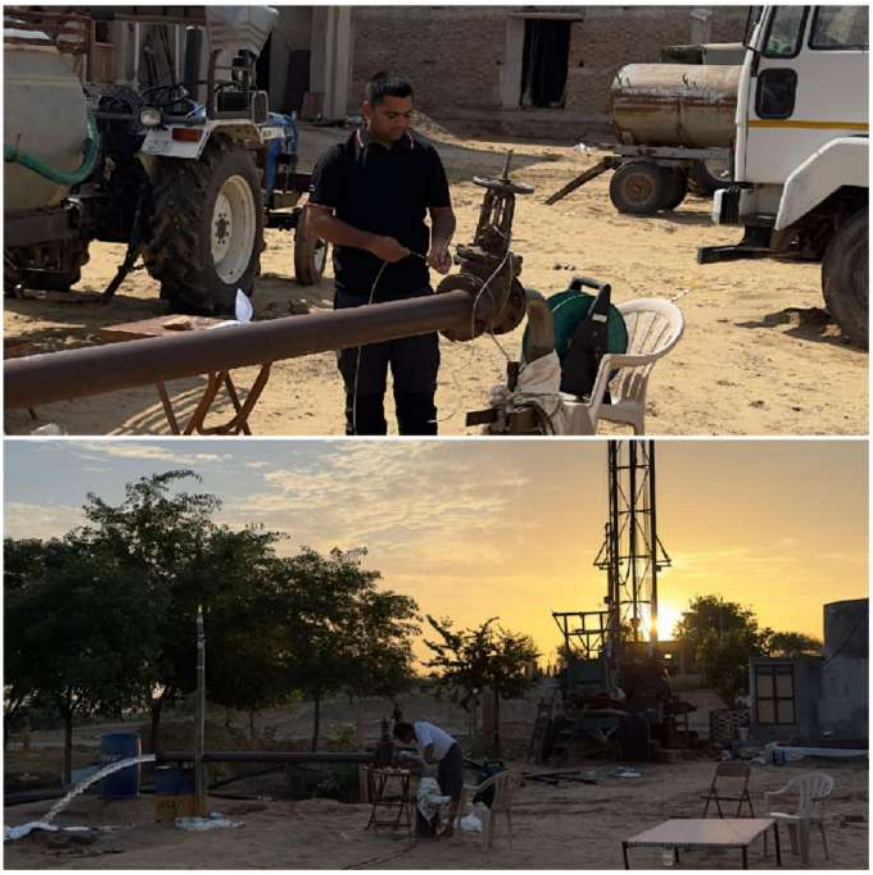
Study visit of Parliamentary Standing Committee on water resources 2022-23 held in Chandigarh. Visit was carried out by hon'ble members of parliament of Rajya Sabha and Lok Sabha in order to develop the sustainable sources for future.

Presentation and sharing of NAQUIM report of Dehradun district with DM, CDO (Dehradun), Sh. Dharmendra Km. Singh, JS, M/o Defence

CGWB, NR Lucknow organized a 3-day Training of Trainers program on "source sustainability of groundwater resources" under the Jal Jeevan Mission. 33 engineers associated with JJM from Jal Nigam and UPGWD across 16 districts attended the training

CGWB, NHR, Dharamshala organised One day Public Interaction Program under the aegis of Mission LiFE near lake of Shri Renuka Ji , Nahan, district Sirmour H.P.



Officer from, WR- CGWB conducting Step Drawdown test (SDT) and Aquifer Performance Test (APT) at Site- Jaitasar Churu District, Rajasthan.

Regional Director, Eastern Region, Kolkata visited Namchi, Sikkim for Ground Verification of recharge structure implementation in the district.

Officers and officials of CGWB, CHQ, Faridabad participated in the Swacchata Pledge as part of Special Campaign 3.0.

WCR Ahmedabad carried out Preliminary Yield Test of Piezometer drilled at Hadmatiya village, Paddhari Taluka, Rajkot District

CGWB pavilion at International Water Conclave, 2024, Shillong

NWR Chandigarh organized a Public Interaction Program on Ground water and its management management at Brahmrishi College of Education

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